



PEDIATRIC MOONSHOT

Pediatric Nephrology Atlas The Complete Challenge

176 Challenges

AI can accelerate diagnosis and treatment for children rurally, locally and globally

“If you know the enemy and know yourself, you need not fear the result of a hundred battles.”
— Sun Tzu, The Art of War

To defeat pediatric kidney disease, we must first define it. The Pediatric Nephrology Atlas catalogs 176 distinct conditions affecting children worldwide, spanning glomerular diseases, congenital anomalies, inherited tubulopathies, hypertension, and rare metabolic and inflammatory disorders. For each condition, the atlas details its biology, U.S. and global incidence, diagnostic workup, and treatment — distinguishing established standard-of-care protocols from emerging investigational therapies. 118 of these conditions carry a hereditary predisposition, classified as Defining, Dominant, or Associated based on the strength of the germline contribution.

Diagnosing pediatric kidney disease is complicated by its heterogeneity and the nonspecific nature of early symptoms. Edema, hypertension, and proteinuria are common presentations shared across dozens of distinct conditions. Many genetic nephropathies progress silently for years before renal function declines. Accurate diagnosis now requires genetic panel testing, renal biopsy with electron microscopy, and biomarker profiling — tools that remain unavailable at most community hospitals and inaccessible in lower-income settings where the global burden of pediatric kidney disease falls heaviest.

Standard-of-care remains anchored in corticosteroids, calcineurin inhibitors, and RAAS blockade — tools largely unchanged since the 1970s. Yet the past four years have delivered more disease-specific approvals than the prior four decades combined, including sparsentan, iptacopan, avacopan, and obinutuzumab. Emerging therapies targeting complement, podocyte biology, and genetic root causes are advancing rapidly through trials.

Yet structural challenges persist. Most pediatric nephrology conditions are rare, trials are slow to enroll, and nearly all approvals are extrapolated from adult data. There are fewer than 1,000 pediatric nephrologists in the United States, concentrated in urban academic centers. A child in rural America with FSGS or a rare tubulopathy may have no specialist within reach, no access to genetic testing, and no pathway to a trial. Data is siloed, rare subtypes stay invisible — problems a distributed, privacy-preserving AI infrastructure is built to solve.

AI can accelerate the diagnosis and treatment of children with kidney diseases. Kidney disease can end with AI.

Josh Tarnoff, CEO, NephCure Kidney International NephCure
Howard Trachtman, MD, Pediatric Nephrologist; formerly Chief of Nephrology at Schneider Children's Hospital
Timothy Chou, PhD, Founder PediatricMoonshot.org

Pediatric Nephrology Atlas

176 Conditions · 17 Categories · SOC vs. Emerging Therapies

Frequency / Epidemiology · Current Diagnostic Methods · Treatment

176

Conditions

17

Categories

118

Hereditary 

6

Columns

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Validated: ABP Pediatric Nephrology Board (March 2026) · RCPCH SPIN · ERKNet · ESPN/IPNA

LEGEND

CONDITION NAME — Hereditary Predisposition Tier

 **Strong Hereditary Predisposition** — monogenic or high-penetrance variants; immunosuppression often ineffective; genetic testing essential

 **Moderate Hereditary Predisposition** — significant genetic component; genetic testing recommended

 **Mild / Partial Hereditary Association** — genetic variants contribute; often multifactorial (no emoji) **No Significant Hereditary Component** — acquired, structural, infectious, or multifactorial etiology

COLUMN COLORS & TREATMENT FORMATTING

Frequency / Epidemiology column — light blue tint; US and global incidence with cited sources

• **Standard of Care** — navy left border + blue header; FDA-approved / guideline-endorsed treatments

▶ **Emerging / Investigational** — amber left border + yellow header; late-stage trials, accelerated approvals, pipeline therapies

Category banner rows — teal full-width dividers between disease categories

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
GLOMERULOPATHIES					
1	Nephrotic Syndrome — Minimal Change Disease (MCD)	Podocyte injury causing massive proteinuria, hypoalbuminemia, edema, and hyperlipidemia; most common cause of nephrotic syndrome in young children	US: ~2/100,000 children/year; ~80% of nephrotic syndrome in young children; peak age 2–6 yrs World: ~2/100,000 children/year globally; ~80% of childhood nephrotic syndrome; higher rates in Asia (3–7/100,000) Sources: Kidney Int 2012;81:1094; NAPRTCS 2011; ISKDC original studies	Urinalysis (heavy proteinuria), serum albumin, lipid panel; kidney biopsy if atypical or steroid-resistant (shows effacement of foot processes on EM)	Standard of Care - Corticosteroids (prednisone) — first-line; remission in ~90% of children - Relapsing/steroid-dependent disease: mycophenolate mofetil, tacrolimus, levamisole (off-label), rituximab - Steroid-resistant MCD: calcineurin inhibitors (tacrolimus preferred over cyclosporine); RAAS blockade for proteinuria management Emerging/Investigational ▶ Anti-nephrin antibodies identified as causative in a subset of MCD/podocytopathies (NEJM 2024) — nephrin-directed therapies under early development ▶ Obinutuzumab (anti-CD20) — Phase 2/3 trial in idiopathic nephrotic syndrome including MCD (Roche MAJESTY/POSTERITY program) ▶ Inebilizumab, ianalumab — B-cell depleting agents in early investigation ▶ SGLT2 inhibitors — adjunctive antiproteinuric role; emerging evidence in pediatric nephrotic syndrome
2	 Focal Segmental Glomerulosclerosis (FSGS) — Primary/Idiopathic	Glomerular scarring affecting some glomeruli focally and segmentally; most common cause of steroid-resistant nephrotic syndrome in children	US: ~1/100,000 children; 20–30% of childhood nephrotic syndrome; highest incidence in Black children (~3×) World: ~1/100,000 children overall; significant racial/ethnic variation; higher in West African and Hispanic populations Sources: NAPRTCS 2011; JASN 2016;27(4); Kitiyakara et al., CJASN 2019	Kidney biopsy (light microscopy, IF, EM); podocin/NPHS2 genetic testing; circulating permeability factor assay; eGFR	Standard of Care - Corticosteroids (prednisone) — first-line for primary/idiopathic FSGS; complete/partial remission in ~40–50% - Calcineurin inhibitors (tacrolimus, cyclosporine) — for steroid-dependent or steroid-resistant disease - Mycophenolate mofetil — steroid-sparing; used in relapsing/refractory disease - RAAS blockade (ACEi/ARB) — antiproteinuric; mainstay for all FSGS regardless of immunosuppression - LDL apheresis — for circulating permeability factor-mediated FSGS refractory to other therapy

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					- Rituximab (off-label) — anti-CD20; increasingly used for refractory steroid-dependent FSGS; evidence from pediatric cohorts Near-Approval / Late-Stage Investigational - Sparsentan (Filspari) — oral dual endothelin A/AT1R antagonist (Travere Therapeutics); sNDA under FDA review for FSGS (PDUFA date April 13, 2026; extended from Jan 2026); Phase 3 DUPLEX trial showed significant proteinuria reduction and lower ESKD rates vs. irbesartan; Phase 2 DUET (ages 8–75) met primary endpoint; EPIIK trial includes pediatric FSGS; would be first FDA-approved FSGS-specific therapy if approved; already FDA/EMA approved for IgAN Investigational - Inaxaplin (VX-147; Vertex Pharmaceuticals) — oral first-in-class APOL1 channel inhibitor; FDA Breakthrough Therapy Designation + Rare Pediatric Disease Designation for APOL1-mediated FSGS; Phase 2a (NEJM 2023) showed 47.6% UPCR reduction; now in Phase 3 AMPLITUDE trial (NCT05312879) including adolescents ages 10–17; interim analysis at Week 48 may support accelerated approval in US - Obinutuzumab — anti-CD20 type II antibody; more complete B-cell depletion than rituximab; case series/trials in steroid-resistant NS - Povetacicept (Alpine Immune Sciences) — APRIL/BAFF dual inhibitor; Phase 2 in proteinuric kidney disease - SGLT2 inhibitors (e.g., dapagliflozin) — renoprotective; growing evidence in proteinuric CKD including pediatric populations; not FSGS-specific but increasingly used adjunctively
3	 FSGS — Genetic/Monogenic	Hereditary podocyte dysfunction causing FSGS; caused by mutations in NPHS1, NPHS2, WT1, TRPC6, INF2, and >50 other genes	US: ~30% of steroid-resistant nephrotic syndrome (SRNS) cases have monogenic cause; higher yield in infants World: ~30% of SRNS globally have monogenic etiology; >60 causative genes identified; higher in consanguineous populations Sources: NEJM 2013;369:1995; Bierzyńska et al. JASN 2017; Trautmann et al. Kidney Int 2018	Comprehensive genetic panel (podocyte genes); kidney biopsy; EM for foot process morphology; family history and inheritance pattern	Standard of Care - RAAS blockade (ACEi/ARB) — antiproteinuric; mainstay regardless of genotype - Supportive care — management of edema, hypertension, hyperlipidemia, nutrition - Immunosuppression generally ineffective — steroids/calcineurin inhibitors not recommended for monogenic FSGS - IMPORTANT EXCEPTION: CoQ10-deficient FSGS (ADCK4, COQ6 mutations) — coenzyme Q10 supplementation may halt or reverse renal decline; critical to identify - Genetic counseling — essential for family planning and living donor selection - Renal transplant planning — excellent outcomes; recurrence risk very low for most monogenic subtypes (<5%)

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					 Investigational (Genotype-Specific) ▶ Inaxaplin (VX-147; Vertex Pharmaceuticals) — for APOL1 high-risk genotype (G1/G2 homozygous or compound heterozygous); AMPLITUDE Phase 3 trial (NCT05312879) now includes adolescents 10–17 years; FDA Breakthrough Therapy + Rare Pediatric Disease Designation; Phase 2a showed ~48% UPCR reduction (NEJM 2023) ▶ Gene therapy approaches — early-stage research for NPHS1 (nephrin) and NPHS2 (podocin) deficiencies ▶ Antisense oligonucleotides / mRNA approaches — preclinical for podocin mutations ▶ Sparsentan — under FDA review for FSGS broadly (PDUFA April 2026); DUET trial included patients with FSGS-associated podocyte gene mutations Emerging Supportive Therapies ▶ SGLT2 inhibitors — renoprotective in proteinuric CKD; adjunctive use growing even in genetic forms
4	 APOL1-Mediated Nephropathy	Kidney disease caused by high-risk APOL1 variants (G1/G2) in individuals of African ancestry; presents as FSGS, collapsing glomerulopathy, or focal global glomerulosclerosis. APOL1 expressed in podocytes causes cell injury under second-hit triggers (infection, interferon, hemodynamic stress). Autosomal recessive — two risk alleles required for high phenotypic risk.	US: ~13% of Black adults carry 2 APOL1 risk alleles; ~35–50% of FSGS in Black children attributable to APOL1 high-risk genotype; >7 million high-risk individuals in the US. World: G1/G2 variants in ~15% of West Africans; rare in non-African populations. Sources: Genovese et al. Science 2010;329:841; Kopp et al. NEJM 2011;365:2183; Kasembeli et al. JASN 2015; Dummer et al. CJASN 2023	APOL1 genotyping (G1: rs73885319 + rs60910145; G2: rs71785313) — 2 risk alleles required for high-risk genotype Kidney biopsy — FSGS or collapsing variant on light microscopy; podocyte foot process effacement on EM Urinalysis — heavy proteinuria, nephrotic range; urine protein/creatinine ratio; eGFR, serum creatinine Exclude secondary causes: HIV, interferon exposure, SLE, sickle cell nephropathy Family history — screen first-degree relatives; autosomal recessive inheritance pattern	 Standard of Care • RAAS blockade (ACEi/ARB) — antiproteinuric; mainstay regardless of immunosuppression status • Corticosteroids — generally ineffective for APOL1-mediated FSGS; avoid unless MCD overlap suspected • Blood pressure control — target <130/80 mmHg; high cardiovascular-renal risk in this population • Avoid second-hit triggers — NSAIDs, contrast, uncontrolled infections, interferon-alpha • Genetic counseling — autosomal recessive; 2 risk alleles required; family screening recommended • Renal transplant — effective for ESKD; APOL1 high-risk donor kidneys associated with shorter graft survival; genotype donors of African ancestry Emerging / Investigational ▶ Inaxaplin (VX-147; Vertex Pharmaceuticals) — oral first-in-class APOL1 ion channel inhibitor; FDA Breakthrough Therapy + Rare Pediatric Disease Designation; Phase 2a (NEJM 2023): 47.6% UPCR reduction vs placebo; Phase 3 AMPLITUDE trial (NCT05312879) includes adolescents ages 10–17; interim Week 48 data may support accelerated approval 2025–2026 ▶ Interferon pathway inhibition — targeting APOL1 second-hit activation via type I IFN axis; anifrolumab under early investigation ▶ APOL1 mRNA / antisense silencing — preclinical; podocyte-targeted suppression of cytotoxic G1/G2 isoform production ▶ SGLT2 inhibitors — renoprotective in proteinuric CKD; adjunctive use growing in APOL1 nephropathy; not APOL1-specific

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5	Membranous Nephropathy — Pediatric (PLA2R and non-PLA2R)	Immune complex deposition in subepithelial space causing nephrotic syndrome; secondary causes more common in children than adults	US: Rare; ~1–5% of pediatric nephrotic syndrome; secondary causes (HBV, SLE) predominate World: Rare in high-income countries (~1–5% of pediatric NS); more prevalent in endemic HBV/HCV regions (Asia, Africa) Sources: JASN 2012;23:650; Debiec et al. NEJM 2004; Ronco & Debiec Nat Rev Nephrol 2012	Kidney biopsy (IgG4 deposits, spike pattern on silver stain); anti-PLA2R antibody; HBV serology; ANA/dsDNA; urine protein	Standard of Care - Treat underlying cause: HBV → antivirals (tenofovir, entecavir); SLE → immunosuppression; malignancy → tumor resection - Idiopathic/PLA2R-positive: watchful waiting (spontaneous remission ~30%); RAAS blockade for proteinuria - Rituximab — now widely considered first-line immunosuppression for PLA2R-positive MN (MENTOR and GEMRITUX trial evidence); replaces cyclophosphamide-based regimens in many centers - Cyclosporine/tacrolimus ± MMF — alternatives for rituximab-refractory or relapsing disease - Anti-PLA2R antibody titer monitoring guides treatment decisions Emerging/Investigational - Felzartamab (anti-CD38 monoclonal antibody) — Phase 2 M-PLACE trial showed efficacy in moderate-to-high-risk PLA2R-positive MN; Phase 3 underway - Daratumumab (anti-CD38) — case series data for refractory PLA2R-positive MN - Ofatumumab — for rituximab-resistant/intolerant patients (case series 2024) - Obinutuzumab — under investigation (MAJESTY trial) - Complement-targeted therapies (iptacopan, pelecopan) — Phase 2 trials; results inconclusive to date - Expanding antigen panel (THSD7A, NELL-1, SEMA3B, EXT1/2) informs precision diagnosis
6	 IgA Nephropathy	Most common primary glomerulonephritis worldwide; IgA immune complex deposition in mesangium causing hematuria and variable proteinuria	US: ~1–2/100,000 children; most common glomerulonephritis in adolescents; ~10–20% of pediatric GN biopsies World: ~1/100,000 globally; highest in East Asia (10–40/100,000 in Japan/China); most common GN worldwide Sources: KDIGO IgAN Guidelines 2021; Wyatt & Julian NEJM 2013; Roberts et al. AJKD 2020	Urinalysis (hematuria ± proteinuria); kidney biopsy (mesangial IgA on IF); Oxford MEST-C score; serum IgA; eGFR and proteinuria trend	Standard of Care - RAAS blockade (ACEi/ARB) — cornerstone; target proteinuria <0.5 g/day - SGLT2 inhibitors — now recommended as part of CKD supportive bundle per 2025 KDIGO IgAN guideline - Systemic corticosteroids — for progressive disease with active inflammation despite optimized supportive care - Mycophenolate mofetil, azathioprine — adjunct immunosuppression <i>FDA-Approved Disease-Specific Therapies (2021–2025):</i> - Targeted-release budesonide (Tarpeyo/Nefecon, Calliditas) — full FDA approval Dec 2023; reduces eGFR decline in adults with primary IgAN - Sparsentan (Filspari, Travere) — dual endothelin A/AT1R antagonist; full FDA approval Sept 2024 for IgAN; Phase 3 DUPLEX trial ongoing for FSGS (PDUFA Apr 2026) - Iptacopan (Fabhalta, Novartis) — Factor B / alternative complement pathway inhibitor; FDA accelerated approval Aug 2024; 38% UPCR reduction vs placebo at 9 months (APPLAUSE-IgAN Phase 3)

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					- Atrasentan (Vanrafia, Novartis) — first selective endothelin A receptor antagonist; FDA approved 2025 for IgAN - Sibeprenlimab (Voyxact, Otsuka) — anti-APRIL monoclonal antibody; FDA accelerated approval Nov 2025; 54% UPCR reduction vs placebo in VISIONARY Phase 3 Emerging/Investigational - Povetacicept (BAFF/APRIL dual inhibitor) — Phase 2/3; RUBY-3 trial showing sustained UPCR and Gd-IgA1 reductions - Felzartamab (anti-CD38) — Phase 2 IGNAZ trial interim results positive - Finerenone (non-steroidal MRA) — Phase 2 data: 25% relative UACR reduction vs placebo - Telitacicept (BAFF/APRIL inhibitor) — Phase 3 ongoing - Mezagitamab (anti-CD38) — Phase 2/3 ongoing - Combination approaches under study per 2025 KDIGO guideline: layered pathway-directed therapy
7	IgA Vasculitis Nephritis (Henoch-Schönlein Purpura)	Systemic small vessel vasculitis with IgA deposits causing palpable purpura, arthritis, abdominal pain, and nephritis in children	US: ~10–20/100,000/year; most common systemic vasculitis in children; nephritis in 20–50%; peak age 4–6 yrs World: ~10–20/100,000/year in Western countries; slightly lower in Asia; renal involvement in 20–50% globally Sources: Gardner-Medwin et al. Lancet 2002; Ozen et al. ARD 2010; EULAR/PRES/PRINTO criteria	Urinalysis; kidney biopsy if significant proteinuria or impaired renal function (mesangial IgA, similar to IgAN); skin biopsy; clinical criteria	 Standard of Care - Supportive care — fluid management, antihypertensives, analgesics for joint/abdominal symptoms - RAAS blockade (ACEi/ARB) — for persistent proteinuria - Corticosteroids (prednisone) — for moderate-severe nephritis; crescentic disease: pulse IV methylprednisolone - Mycophenolate mofetil or azathioprine — for steroid-dependent or frequently relapsing nephritis - Cyclophosphamide — reserved for severe crescentic nephritis (Class III/IV histology) Emerging/Investigational - IgAN-directed therapies may benefit IgA vasculitis nephritis given shared pathophysiology (Gd-IgA1 axis); trials generally have excluded pediatric IgAV - Rituximab — case reports/series for refractory crescentic IgAV nephritis - SGLT2 inhibitors — adjunctive antiproteinuric benefit in proteinuric IgAV nephritis; emerging use 2024 KDIGO IgAV guideline aligns IgAV nephritis management with IgAN framework
8	 MPGN — Complement-mediated	Glomerulonephritis with mesangial and endocapillary hypercellularity; driven by dysregulation of alternative complement pathway	US: Rare; <5% of pediatric glomerulonephritis biopsies; complement-mediated recognized with genetic testing World: Rare globally; <5% of pediatric GN; increasing recognition	Kidney biopsy; complement studies (C3, C4, CH50, AP50); CFH/CFI/C3/CFB genetic panel; anti-CFH antibodies; electron microscopy	 Standard of Care Eculizumab (anti-C5) — for complement dysregulation confirmed by genetic testing or low C3/low C4 pattern; evidence base primarily from aHUS but used off-label for complement-driven MPGN

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			with improved complement genetic testing panels Sources: Smith et al. CJASN 2019; Servais et al. JASN 2007; C3 Glomerulopathy Consensus Report KI 2013		- Ravulizumab — extended dosing alternative to eculizumab - RAAS blockade — antiproteinuric - Plasma infusion/fresh frozen plasma — for CFH or CFHR deficiency (rare) - Mycophenolate mofetil ± steroids — for immune complex-driven component - Meningococcal vaccination — mandatory before complement inhibitor therapy FDA-Approved (2025)  Pegcetacoplan (Empaveli, Apellis) — FDA approved July 2025 for IC-MPGN (patients aged ≥12 years); first approved therapy for this indication; proximal C3 inhibitor Emerging/Investigational  Iptacopan (Factor B inhibitor) — Phase 2/3 data in complement-mediated kidney disease  Danicopan (Factor D inhibitor) — Phase 2 trials  Personalized complement pathway targeting based on genetic complement profiling
9	MPGN — Immune Complex	Glomerulonephritis with MPGN pattern driven by chronic antigen-antibody complex deposition; secondary to infections, autoimmune disease, or monoclonal proteins	US: Uncommon; secondary causes (HBV, HCV, SLE, endocarditis) more common; primary rare in pediatrics World: Uncommon; incidence declining with HBV vaccination; secondary MPGN dominant globally; prevalence varies by infection burden Sources: Sethi & Fervenza NEJM 2012; Appel et al. JASN 2005; KDIGO GN Guidelines 2012	Kidney biopsy (IF showing IgG/IgM/C3); identify underlying cause: ANA, ANCA, complement, HBV/HCV serology, cryoglobulins, SPEP	Standard of Care - Treat underlying cause — antivirals for HCV (DAAs: sofosbuvir-based regimens; often curative); antibiotics for chronic bacterial infection; chemotherapy for lymphoproliferative disease - RAAS blockade — antiproteinuric management - Immunosuppression (mycophenolate, cyclophosphamide, rituximab) — for autoimmune-driven IC-MPGN (SLE, Sjögren) - Cryoglobulinemia-associated: rituximab + antiviral (if HCV+) as primary therapy Emerging/Investigational  Pegcetacoplan (Empaveli) — FDA approved July 2025 for primary IC-MPGN (idiopathic, not secondary); relevant for complement-amplified secondary IC-MPGN under investigation  B-cell depleting agents for cryoglobulinemia — obinutuzumab (superior B-cell depletion vs rituximab) under investigation
10	 C3 Glomerulopathy (C3GN)	Glomerulonephritis with dominant C3 deposition and absent/minimal immunoglobulin, caused by alternative complement pathway dysregulation	US: Very rare; <1/1,000,000; distinguished from DDD by EM findings; genetic and acquired complement disorders World: Very rare globally; <1/1,000,000; distinct from DDD on electron microscopy; both genetic	Kidney biopsy (dominant C3 on IF, absence of Ig); complement pathway workup (C3, C4, factor H/I/B/properdin); genetic panel; anti-CFH antibodies	Standard of Care - RAAS blockade — antiproteinuric and antihypertensive; first-line supportive - Mycophenolate mofetil ± low-dose steroids — for progressive C3G; modest evidence base - Plasma therapy (FFP infusion or plasma exchange) — for CFH deficiency or anti-CFH antibodies

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			and acquired complement dysregulation Sources: Pickering et al. KI 2013; Servais et al. JASN 2012; C3G Consensus Report KI Suppl 2013		• Eculizumab — off-label; limited and variable efficacy in C3G (better evidence for dense deposit disease); response predicted by terminal complement activation markers • Meningococcal vaccination — mandatory if complement inhibitor used • Renal transplant for ESKD (risk of recurrence ~50%) FDA-Approved (July 2025) ▢ Pegcetacoplan (Empaveli, Apellis) — FDA approved July 2025 for C3 glomerulopathy (aged ≥12 years); FIRST ever approved therapy for C3G; proximal C3/C3b inhibitor; subcutaneous twice-weekly self-administration; approval based on Phase 2/3 data showing proteinuria reduction and complement suppression Emerging/Investigational ▢ Iptacopan (Factor B inhibitor, Novartis) — Phase 2/3 ongoing in C3G; APPELHANSEN trial ▢ Danicopan (Factor D inhibitor, Alexion) — Phase 2 trials ongoing ▢ Complement genetic panel — essential for patient stratification and precision therapy selection
11	 Dense Deposit Disease (DDD / MPGN Type II)	Rare MPGN variant with highly electron-dense ribbon-like deposits in GBM; caused by C3 nephritic factor or complement gene mutations	US: Extremely rare; ~2–3 cases/million; median onset childhood/adolescence; C3 nephritic factor in majority World: ~2–3/million globally; most data from European registries; North American Complement-Mediated TMA Registry Sources: Appel et al. JASN 2005; Bomback & Appel KI 2010; Smith et al. CJASN 2014	Kidney biopsy with EM (pathognomonic dense deposits); C3NeF; C3, factor H levels; complement genetic panel; fundoscopy (drusen)	Standard of Care • RAAS blockade — antiproteinuric; mainstay • Plasma infusion for CFH deficiency — replaces absent complement regulatory proteins • Mycophenolate mofetil — for progressive disease; modest evidence • Eculizumab — variable efficacy in DDD (evidence less robust than in aHUS); may benefit patients with documented terminal complement activation • Renal transplant (high recurrence risk ~80–90%; allograft loss common) FDA-Approved (July 2025) ▢ Pegcetacoplan (Empaveli, Apellis) — FDA approved July 2025 for C3G including DDD (aged ≥12 years); FIRST approved disease-specific therapy; proximal C3/C3b complement inhibitor Emerging/Investigational ▢ Iptacopan (Factor B inhibitor) — Phase 2/3 in C3G/DDD ▢ Danicopan (Factor D inhibitor) — Phase 2 trials ▢ C3 nephritic factor (C3NeF) quantification guides complement inhibitor selection ▢ Monoclonal antibody targeting C3 convertase stabilization — investigational

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12	Anti-GBM Disease (Goodpasture) — Pediatric	Autoimmune disease with anti-GBM antibodies attacking type IV collagen in glomerular and alveolar basement membranes; pulmonary-renal syndrome	US: Very rare in children; <0.5/million/year; mostly adolescents; pulmonary involvement less common than adults World: Very rare globally; <0.5/million/year; predominantly adult disease; European multicenter study reports <5% pediatric cases Sources: Cui et al. Medicine 2015; Johnson et al. KI 1985; Levy et al. QJM 2001	Anti-GBM antibody (ELISA/CLIA); ANCA (concurrent positivity in ~30%); kidney biopsy (linear IgG on IF, crescentic GN); chest imaging; pulmonary function	Standard of Care - Emergency plasmapheresis (plasma exchange) — removes circulating anti-GBM antibodies; 14 exchanges over 2–3 weeks; most urgent intervention - Cyclophosphamide + high-dose corticosteroids (pulse IV methylprednisolone → oral prednisone) — concurrent immunosuppression to halt antibody production - Dialysis support — if AKI requires renal replacement therapy - Renal transplant — after antibody clearance (typically ≥6–12 months of undetectable anti-GBM titers) - Lung protection — oxygen supplementation; avoid pulmonary hemorrhage triggers (smoking cessation, treat infections) Emerging/Investigational ▢ Rituximab — for refractory or relapsing anti-GBM disease; case reports/series ▢ Imlifidase (IdeS) — IgG-cleaving enzyme; experimental compassionate use to rapidly remove anti-GBM IgG when plasmapheresis cannot be performed urgently ▢ CAR-T/B-cell depleting approaches — very early stage ▢ Dual anti-GBM + ANCA positivity (~10–30% of cases) — influences treatment duration (longer immunosuppression needed for ANCA component)
13	 Lupus Nephritis — Pediatric (Class I–VI)	Renal involvement in childhood-onset SLE; more severe and rapid progression than adult SLE nephritis; ISN/RPS classification I–VI	US: Pediatric SLE: ~0.5–0.6/100,000/year; nephritis in 50–80% of pediatric SLE; Class III/IV most common World: Pediatric SLE: ~0.3–0.9/100,000/year globally; nephritis in up to 80%; higher prevalence in Asian, Hispanic, Black populations Sources: Tarr et al. ARD 2012; CARRA Registry 2019; ACR/EULAR SLE Classification Criteria 2019	Urinalysis, urine protein:creatinine; ANA, anti-dsDNA, complement (C3/C4); kidney biopsy for class; CBC, renal function; SLEDAI score	Standard of Care - Mycophenolate mofetil (MMF) + hydroxychloroquine + corticosteroids — first-line induction for Class III/IV LN (Euro-Lupus or high-dose MMF regimen) - Cyclophosphamide (NIH high-dose or Euro-Lupus low-dose IV) — alternative induction for severe/proliferative disease; preferred for Class IV with crescents - Voclosporin (Lupkynis, Aurinia) — calcineurin inhibitor; FDA approved Jan 2021 for adult LN; used with MMF + steroids; increasing use in older adolescents (AURORA 1 & 2 trials; KDIGO 2024-endorsed) - Belimumab (Benlysta, GSK) — anti-BLyS/BAFF; FDA approved Dec 2020 for LN; IV or SC; used as add-on to standard of care; adolescent data emerging - RAAS blockade — Class I/II or proteinuric background - Hydroxychloroquine — standard background therapy for all LN classes FDA-Approved (October 2025) ▢ Obinutuzumab (Gazyva, Genentech/Roche) — anti-CD20 type II monoclonal antibody; FDA approved Oct 2025 for active adult LN receiving standard therapy; REGENCY Phase 3 superiority vs placebo + SOC; pediatric trial

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					(POSTERITY) ongoing for adolescents with Class III/IV LN Emerging/Investigational ▣ CAR-T cell therapy (anti-CD19) — early case series showing deep B-cell depletion and LN remission; Phase 1/2 trials ongoing ▣ Anifrolumab (anti-IFN-α receptor) — approved for SLE; potential role in LN under investigation ▣ Finerenone, SGLT2 inhibitors — adjunctive renal protection per KDIGO 2024 CKD bundle ▣ Ustekinumab (IL-12/23), iberdomide — investigational for refractory LN
14	ANCA-associated Vasculitis Nephritis (GPA / MPA)	Necrotizing crescentic GN caused by anti-neutrophil cytoplasmic antibodies; part of systemic small-vessel vasculitis	US: Rare; ~1–2/million/year; GPA more common than MPA in pediatrics; mean age ~12–14 years World: ~1–2/million/year in children; European registries report slightly higher rates; GPA predominates over MPA in pediatric cohorts Sources: Cabral et al. Pediatrics 2009; EULAR/PRES recommendations 2010; ARPAKid Registry	ANCA (PR3/MPO by ELISA); kidney biopsy (pauci-immune crescentic GN); chest CT; ENT evaluation; eGFR; urinalysis	 Standard of Care • Rituximab — preferred induction agent for GPA/MPA over cyclophosphamide in most adults and older children; equivalent or superior remission rates (RAVE/RITUXVAS trials); preferred in young women and relapsing disease • Cyclophosphamide (IV pulse or oral) + high-dose corticosteroids — alternative induction; preferred for severe alveolar hemorrhage or renal failure requiring dialysis • Azathioprine or rituximab — maintenance therapy; rituximab increasingly preferred for maintenance (MAINRITSAN trials) • Plasma exchange (PLEX) — previously standard for severe dialysis-requiring disease; PEXIVAS trial (2020) showed no benefit for renal survival or mortality; now NOT routinely recommended; reserved for concurrent anti-GBM antibodies • Mycophenolate mofetil — alternative maintenance agent FDA-Approved (Oct 2021) ▣ Avacopan (Tavneos, Amgen/ChemoCentryx) — C5a receptor antagonist; FDA approved Oct 2021 for GPA/MPA as adjunct to SOC; ADVOCATE Phase 3: non-inferior remission at 26 wks, superior sustained remission at 52 wks; reduces corticosteroid burden; increasingly incorporated into induction regimens Emerging/Investigational ▣ Obinutuzumab — improved B-cell depletion vs rituximab; Phase 2/3 in ANCA vasculitis ▣ Imlifidase (IdeS) — for refractory ANCA nephritis with high IgG burden; investigational ▣ Complement pathway (C3/C5) — additional targets under study ▣ Pediatric-specific data limited; treatment extrapolated from adult trials with weight-based dosing

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
15	Post-infectious / Post-streptococcal GN	Immune complex GN occurring 1–3 weeks after streptococcal or other bacterial infection; characterized by hematuria, edema, hypertension	US: Common; ~10/100,000/year; peak age 5–12 yrs; most common acute GN in North American children World: Very common in developing countries; ~10/100,000/year in high-income countries; major cause of GN in resource-limited settings Sources: Carapetis et al. Lancet Infect Dis 2005; Rodriguez-Iturbe & Musser JASN 2008; Eison et al. Pediatrics 2011	Urinalysis (hematuria, RBC casts); ASO titer; streptozyme; low C3 (transiently); throat/skin culture; kidney biopsy rarely needed	Standard of Care - Supportive management — primary approach; most cases self-limited - Salt and fluid restriction — for hypertension and edema - Antihypertensives (calcium channel blockers, hydralazine, loop diuretics) — for symptomatic hypertension - Diuretics (furosemide) — for fluid overload and edema - Antibiotics — eradicate residual streptococcal infection (penicillin, amoxicillin); does NOT alter renal course but prevents spread - RAAS blockade — for persistent proteinuria in recovery phase - Dialysis — for severe AKI with oliguria/fluid overload unresponsive to diuretics (rare, <1%) Emerging/Investigational - No disease-specific investigational therapies; prognosis excellent in children (>95% full recovery) - Atypical/non-resolving PSGN — consider complement evaluation (rare overlap with C3G or aHUS) - IgA-dominant PIGN (staphylococcal) — emerging recognition as distinct entity; management similar but may require immunosuppression in refractory cases
16	 Thin Basement Membrane Nephropathy	Diffuse thinning of GBM on EM causing persistent microscopic hematuria; benign in most cases; allelic with Alport syndrome (COL4A3/A4)	US: Common; ~1% of general population; majority of asymptomatic isolated hematuria in children World: ~1% globally; autosomal dominant inheritance; majority of isolated hematuria in school-age children worldwide Sources: Pierides et al. KI 2009; Haas et al. AJKD 2009; KDIGO GN Guidelines 2012	Kidney biopsy with EM (GBM <150–200 nm); COL4A3/A4 genetic testing; urinalysis; family history of hematuria; exclude Alport	Standard of Care - No treatment typically required — benign prognosis in most pediatric cases; isolated hematuria - RAAS blockade (ACEi/ARB) — initiate if proteinuria develops (>300 mg/day) or if blood pressure rises - Annual monitoring — urinalysis, blood pressure, eGFR, urine protein/creatinine ratio - Genetic counseling — essential; COL4A3/COL4A4 heterozygous mutations overlap with autosomal dominant Alport (TBMN-Alport spectrum); family screening recommended - Avoid nephrotoxins — NSAIDs, contrast media, aminoglycosides - Renal biopsy — not always required; EM finding of diffuse thinning (<250 nm); necessary to distinguish from early Alport Emerging/Investigational - Reclassification ongoing — 2025 guidance recognizes TBMN/heterozygous Alport as a spectrum; SGLT2 inhibitors under investigation for COL4A3/4 heterozygous mutations with proteinuria - No novel disease-specific therapies approved or in late-stage trials

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
17	HBV-Associated Membranous Nephropathy — Pediatric	Membranous nephropathy caused by hepatitis B viral antigen-antibody immune complex deposition in subepithelial space	US: Rare in US (low HBV prevalence); mostly in immigrant populations from endemic regions World: Major cause of childhood membranous nephropathy in Asia and Sub-Saharan Africa; declining with HBV vaccination programs Sources: Lin et al. Kidney Int 1990; KDIGO HBV Guidelines 2018; WHO HBV vaccination data	HBsAg, HBeAg, anti-HBc; kidney biopsy (HBsAg/HBeAg deposits on IF); liver function; viral load; anti-PLA2R (usually negative)	Standard of Care - Antiviral therapy — cornerstone; tenofovir disoproxil fumarate (TDF) or tenofovir alafenamide (TAF — preferred in adolescents for renal/bone safety); entecavir as alternative - RAAS blockade — for proteinuria management - Avoid immunosuppression unless under antiviral cover — risk of HBV reactivation; if required (e.g., severe class V), ensure antiviral prophylaxis started ≥ 1 week prior - HBeAg seroconversion — associated with spontaneous remission of nephropathy (~30%); antiviral therapy accelerates seroconversion - Monitor HBsAg, HBeAg, HBV DNA, ALT, renal function, proteinuria throughout Emerging/Investigational - Functional cure strategies for HBV (cccDNA silencing, capsid assembly modulators, RNAi) — if HBV eliminated, nephropathy expected to resolve; trials ongoing - Rituximab — for antiviral-resistant or refractory proteinuric disease under antiviral cover; limited case reports - TAF over TDF preferred in young patients and those with renal or bone concerns — increasing adoption
18	 Antiphospholipid Syndrome Nephropathy	Thrombotic microangiopathy of renal vessels caused by antiphospholipid antibodies; occurs as primary APS or in association with SLE	US: Rare in children; mostly adolescent females; APS in ~20% of pediatric SLE; renal in ~25% of APS World: Rare globally; predominantly adolescent females; APS in pediatric SLE varies 10–35% across registries Sources: Avcin et al. Arthritis Rheum 2008; Levy et al. Lupus 2004; PAPS collaborative data	Antiphospholipid antibodies (aCL, anti- $\beta 2$ GPI, lupus anticoagulant) $\times 2$ positive 12 weeks apart; kidney biopsy (TMA pattern); anti-dsDNA, ANA	Standard of Care - Anticoagulation — warfarin (target INR 2.5–3.5 for arterial thrombosis; 2.0–3.0 for venous); LMWH as bridge or in pregnancy - Hydroxychloroquine — reduces thrombotic events and aPL titers; recommended for all SLE-related APS - RAAS blockade — for hypertension and proteinuria (hypertensive nephrosclerosis component) - Aspirin — adjunctive antiplatelet for arterial events; combined with anticoagulation in high-risk patients - Blood pressure control — stringent targets (<130/80 mmHg) Note: DOACs (rivaroxaban, apixaban) — NOT recommended for triple-positive aPL (LAC + anti-cardiolipin + anti-$\beta 2$GPI); warfarin preferred; DOACs may be used for double/single positivity with venous-only events after shared decision-making Emerging/Investigational - Ecilizumab — for catastrophic APS (CAPS) and refractory TMA; evidence from case series and small cohorts; increasingly accepted in CAPS - Rituximab — for refractory APS or aPL-positive SLE nephritis; emerging evidence - Belimumab — for SLE-APS overlap; BLISS trials included LN patients with aPL positivity

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					▶ Sirolimus (mTOR inhibitor) — for APS nephropathy with chronic vascular injury; investigational; some case series ▶ Complement inhibition — aPL activates complement; anti-C5 strategies under investigation
HEREDITARY / GENETIC KIDNEY DISEASES					
19	 Alport Syndrome — X-linked (COL4A5)	Most common hereditary nephritis; mutations in COL4A5 causing progressive nephritis, sensorineural hearing loss, and ocular abnormalities in males	US: ~1/5,000–10,000; accounts for ~1% of pediatric ESKD; X-linked most common (85% of Alport); males severely affected World: ~1/5,000–10,000 globally; predominant in European-descent populations; ESKD by 25 yrs in 50% of affected males Sources: Kashtan et al. JASN 2013; European Alport Registry; Jais et al. JASN 2003	Urinalysis (hematuria ± proteinuria); audiogram; ophthalmology exam; kidney biopsy (EM: GBM lamellation/thinning); COL4A5 genetic testing	Standard of Care • ACE inhibitors (enalapril, lisinopril) / ARBs (losartan) — initiated at first sign of microalbuminuria/proteinuria; ESCAPE trial and Alport registry data show significant delay to ESKD; 2022 KDIGO Alport guideline recommends ACEi/ARB in ALL males with hematuria + family history regardless of proteinuria level • KDIGO 2022 Alport Guidelines — landmark update; recommends early ACEi/ARB even before proteinuria; tier-based approach: Tier 1 (persistent proteinuria) → mandatory; Tier 2 (microalbuminuria) → strong recommendation; Tier 3 (isolated hematuria, high-risk genotype) → consider • BP control to <50th percentile (or CKD target <75th %ile) — essential for renal preservation • Low-protein diet — adjunct in advanced CKD stages • Kidney transplant — highly effective; no recurrence of Alport nephritis in graft (no type IV collagen made); ESKD median age ~25–35 yr in untreated males; anti-GBM disease in transplant rare (<5%) • Hearing aids — for sensorineural hearing loss; audiological surveillance annually • Ophthalmological monitoring — anterior lenticonus, macular flecks; annual slit-lamp exam Emerging/Investigational ▶ Sparsentan (FILSPARI) — dual ET-A/AT-II antagonist; DUPLEX trial includes Alport patients; ACEi/ARB-refractory proteinuria; investigational ▶ Lademirsen (anti-miR-21) — microRNA-21 inhibitor targeting TGF-β fibrotic pathway; Phase 2 (NCT02855268); Alport-specific; halted 2023 for strategic reasons; mechanism validated ▶ Bardoxolone methyl — Nrf2 activator; improves GFR in CKD; Phase 3 CARDINAL trial in Alport (NCT03019185); completed; modest eGFR improvement ▶ COL4A5 gene therapy — AAV delivery to podocytes/GBM; preclinical (mouse models); long-term goal ▶ Stem cell therapy — mesenchymal stem cells reducing fibrosis; preclinical

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
20	 Alport Syndrome — Autosomal Recessive (COL4A3/A4)	Biallelic mutations in COL4A3 or COL4A4 causing severe nephritis similar to X-linked Alport; affects males and females equally	US: ~15% of Alport syndrome cases in US; typically presents in first decade with rapid ESKD progression World: ~15% of all Alport globally; ESKD in adolescence/young adulthood; consanguineous families at higher risk Sources: Jais et al. JASN 2003; Gross et al. JASN 2012; Kashtan Ped Nephrol 2010	COL4A3/COL4A4 genetic testing; kidney biopsy (EM: GBM thinning and lamellation); audiogram; ophthalmology; family history	Standard of Care - ACE inhibitors / ARBs — same indications as X-linked Alport; KDIGO 2022 guidelines apply equally; early initiation at microalbuminuria stage; AR Alport often has earlier, more severe course than X-linked in males - BP control — CKD targets; combination antihypertensive as needed - Monitoring — proteinuria (monthly initially), eGFR (every 3–6 months), hearing (annually), ophthalmology (annually) - Kidney transplant — curative; no recurrence; ESKD typically earlier than X-linked (median ~20s); living-related donor evaluation requires COL4A3/A4 carrier status testing (heterozygous carriers have favorable outcomes but need long-term follow-up) - KDIGO 2022 note: COL4A3/A4 heterozygous carriers ("thin basement membrane nephropathy spectrum") now recognized as having ~30% lifetime CKD risk; ACEi warranted if proteinuria develops Emerging (same pipeline as X-linked Alport)  Lademirsen, bardoxolone methyl, sparsentan — see X-linked Alport; trials include all COL4 genotypes  COL4A3/A4 gene therapy — autosomal recessive inheritance makes gene therapy more straightforward than X-linked; preclinical
21	 Alport Syndrome — Autosomal Dominant	Heterozygous COL4A3/COL4A4 mutations causing variable nephritis; often milder than AR form; previously called familial hematuria	US: Underdiagnosed; heterozygous COL4A3/A4 mutations in ~1/200 population; significant CKD in 30–40% World: ~1/200 globally carry heterozygous COL4A3/A4 mutations; 30–40% develop proteinuria/CKD; likely most common inherited GN Sources: Gross et al. JASN 2018; Savige et al. AJKD 2016; Deltas et al. Ped Nephrol 2013	COL4A3/COL4A4 genetic testing; urinalysis; family history of hematuria; kidney biopsy (may show GBM thinning or lamellation)	Standard of Care - ACE inhibitors / ARBs — KDIGO 2022: same tier-based approach as other Alport forms; heterozygous COL4A3/A4 mutations; milder course than X-linked or AR; ESKD in ~15–30% of affected individuals (often adult onset) - BP management — standard CKD targets - Genetic counseling — AD inheritance; 50% transmission risk; test at-risk family members - Monitoring — same surveillance as other Alport forms; hearing and ophthalmology involvement less common than X-linked - Kidney transplant — if ESKD reached; excellent outcomes; no recurrence Emerging  Same investigational pipeline as X-linked Alport (lademirsen, bardoxolone methyl, sparsentan)
22	 Congenital Nephrotic Syndrome — Finnish Type (NPHS1)	Mutations in NPHS1 encoding nephrin causing massive proteinuria from birth; originally described in Finnish population	US: ~1/50,000 in US; Finnish prevalence ~1/8,200; presents at birth with massive proteinuria and anasarca	Neonatal urinalysis (massive proteinuria); NPHS1 genetic testing; kidney biopsy (diffuse mesangial sclerosis)	Standard of Care Albumin infusions — frequent (often daily/alternate day) to maintain serum albumin ≥ 20 g/L and minimize complications; mainstay of supportive management

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			World: ~1/50,000 globally; highly prevalent in Finland (~1/8,200); high carrier frequency in Finnish population (~1/200) Sources: Kestila et al. Mol Cell 1998; Holmberg & Jalanko Ped Nephrol 2014; Jalanko et al. Ped Nephrol 2009	pattern); serum albumin; AFP elevated antenatally	- IV albumin + IV furosemide — manages massive edema; goal serum albumin ≥ 15–20 g/L - Nephrectomy (bilateral) — key management decision; unilateral or bilateral; removes massive protein loss; timing individualized; usually performed before transplant or when protein losses unmanageable; bilateral nephrectomy + dialysis bridge to transplant - Nutrition — high-protein diet (2–3 g/kg/day); vitamin D, trace element supplementation; nasogastric/gastrostomy tube feeding commonly required - Thyroid hormone supplementation — thyroxine loss in urine causes hypothyroidism; levothyroxine replacement standard - Anticoagulation — hypercoagulable state; prophylactic low-molecular-weight heparin often used - Prophylactic antibiotics — pneumococcal vaccine + penicillin prophylaxis; high infection risk - Kidney transplant — definitive treatment; NPHS1 mutations → no nephrin in native kidneys; ~25% risk of recurrent nephrotic syndrome post-transplant due to anti-nephrin antibodies; rituximab used for post-transplant recurrence - ACEi/ARBs — reduces proteinuria; used pre-nephrectomy/transplant Emerging ▶ Anti-nephrin antibody detection — NEJM 2024 data demonstrate circulating anti-nephrin antibodies in subset of NPHS1 patients; rituximab targets this mechanism; potential monitoring tool ▶ Gene therapy (AAV-NPHS1) — podocyte-targeted; preclinical; NPHS1 promoter-driven expression challenging due to size
23	 Congenital Nephrotic Syndrome — NPHS2 (podocin)	Autosomal recessive mutations in NPHS2 (podocin) causing steroid-resistant nephrotic syndrome in infancy or early childhood	US: ~10–15% of childhood steroid-resistant nephrotic syndrome in US; onset 3 months to 5 years World: 10–15% of childhood FSGS/SRNS globally; prevalent in Mediterranean, Middle Eastern, and European populations Sources: Boute et al. Nat Genet 2000; Weber et al. Nat Genet 2000; Ruf et al. JASN 2004	NPHS2 genetic testing; kidney biopsy (FSGS or minimal change on LM); steroid trial confirms resistance; EM (foot process effacement)	 Standard of Care - ACE inhibitors / ARBs — partial antiproteinuric effect; less responsive than MCD/primary FSGS; use in all NPHS2 patients with proteinuria - Calcineurin inhibitors (cyclosporine, tacrolimus) — limited efficacy vs. primary FSGS; some NPHS2 genotypes (especially R138Q homozygous) are steroid-resistant AND CNI-resistant; genotype guides expectations - Dietary management — protein supplementation; edema management - Kidney transplant — standard treatment for ESKD; recurrence risk much lower than primary FSGS (NPHS2-associated FSGS rarely recurs unless compound heterozygous with one non-truncating mutation); R229Q variant in one allele increases recurrence risk - Genetic testing — critical before transplant to stratify recurrence risk; donor evaluation

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Steroids — generally ineffective (steroid-resistant NS by definition in most); brief trial acceptable per KDIGO 2021 NS guidelines Emerging - Podocyte stabilization strategies — coenzyme Q10, antioxidants for mitochondrial podocytopathies (indirect benefit) - Anti-nephrin antibody monitoring — subset may have immune overlay; rituximab considered - mRNA therapy for NPHS2 — restoring podocin; preclinical concept
24	 Pierson Syndrome (LAMB2)	Biallelic LAMB2 mutations causing congenital nephrotic syndrome + microcoria (fixed miosis) + neurodevelopmental impairment	US: Extremely rare; <200 cases worldwide including US; presents at birth with NS and ocular findings World: Extremely rare globally; <200 cases reported in literature; severe early-onset NS with microspherophakia and retinal dystrophy Sources: Zenker et al. Nat Genet 2004; Matejas et al. KI 2010; Pierson review Ped Nephrol 2012	LAMB2 genetic testing; ophthalmology (microcoria, retinal abnormalities); kidney biopsy (FSGS/DMS); neonatal urinalysis	 Standard of Care - Renal replacement therapy — ESKD in infancy/early childhood; peritoneal dialysis preferred in infants; nasogastric feeding + intensive nutrition support during dialysis phase - Kidney transplant — ESKD treatment; renal outcomes good post-transplant (LAMB2 not expressed in donor kidney in pathologic form); timing when weight/size permits (typically ≥8–10 kg) - Ophthalmological management — microcoria is pathognomonic; does not worsen post-transplant; manage refractive errors; amblyopia prevention; specialist ophthalmology follow-up lifelong - Neurological monitoring — hypotonia, developmental delay in severe forms; physiotherapy, occupational therapy - Nutritional support — intensive supplementation in nephrotic phase and dialysis; growth promotion - Supportive NS management — albumin infusions, diuretics, anticoagulation as for other congenital NS Emerging - LAMB2 gene therapy — very early concept; basement membrane protein delivery challenges; preclinical
25	 Galloway-Mowat Syndrome	Heterogeneous group of conditions (WDR73, KEOPS complex genes, etc.) causing congenital nephrotic syndrome + microcephaly + neurological disease	US: Very rare; multiple genetic causes; poor prognosis with early ESKD and neurodevelopmental impairment World: Very rare globally; caused by mutations in >20 genes (OSGEP, WDR73, NUP107, etc.); universal poor prognosis Sources: Braun et al. NEJM 2017; Colin et al. JASN 2014; Galloway-Mowat review Ped Nephrol 2019	Genetic panel (WDR73, OSGEP, TP53RK, etc.); brain MRI; kidney biopsy; developmental assessment	 Standard of Care - Renal replacement therapy — ESKD typically in infancy to early childhood; peritoneal dialysis; intensive nutritional support - Kidney transplant — feasible for renal component; extra-renal manifestations (microcephaly, severe intellectual disability, seizures) impact candidacy and outcomes; center-by-center evaluation; controversial given multi-organ severity in many genotypes - Neurological management — antiepileptic drugs for seizures; seizure control challenging; anticonvulsant optimization

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Nutritional support — nasogastric/gastrostomy feeding; protein and caloric supplementation - Physical/occupational/speech therapy — developmental support - Genetic testing — multiple genes (WDR4, TP53RK, TPRKB, OSGEP, LAGE3, GON7, etc.); genotype-phenotype correlation informs prognosis and transplant decision - Genetic counseling — autosomal recessive in most; prenatal diagnosis available Emerging - tRNA modification pathway — multiple GMS genes encode KEOPS complex; translational regulation pathway; research stage - No disease-modifying therapies available
26	 Denys-Drash Syndrome (WT1)	WT1 missense mutations causing diffuse mesangial sclerosis, Wilms tumor predisposition, and 46XY DSD (male pseudohermaphroditism)	US: Rare; ~150 cases reported worldwide; WT1 exon 8/9 mutations; Wilms tumor risk >90% World: Extremely rare globally; ~150 total reported cases; autosomal dominant de novo WT1 mutations; near-universal Wilms tumor risk Sources: Pelletier et al. Cell 1991; Breslow et al. Cancer 2000; Lehnhardt et al. Ped Nephrol 2011	WT1 genetic testing; kidney biopsy (diffuse mesangial sclerosis); gonadal evaluation (karyotype, pelvic ultrasound); Wilms tumor surveillance	 Standard of Care - Nephrectomy — CRITICAL: bilateral prophylactic nephrectomy recommended given near-universal Wilms tumor risk (>90% lifetime); timing debated but most centers recommend before age 2–3 years or at diagnosis; pathological specimens required - Dialysis + kidney transplant — after bilateral nephrectomy; peritoneal dialysis bridge; transplant once oncologically clear; excellent renal outcomes post-transplant (no disease recurrence) - Wilms tumor surveillance — prior to nephrectomy: ultrasound every 3 months; MRI for better characterization - Wilms tumor treatment if present — nephrectomy + chemotherapy (actinomycin D, vincristine ± doxorubicin per COG/SIOP protocols); radiation for high-stage disease - DSD management — gonadal dysgenesis; gonadectomy (streak gonads at high risk of gonadoblastoma); hormonal management; psychosocial counseling; multidisciplinary DSD team - Steroid-resistant NS — supportive; ACEi/ARB for proteinuria; steroid-resistant by definition; dialysis planning Emerging - WT1 targeted therapies — understanding WT1 missense mutation effects on podocyte transcription; no targeted drugs yet - Genotype-specific counseling — WT1 exon 8/9 mutations (zinc finger domain) associated with DDS; exon 9 splice site mutations → Frasier syndrome
27	 Frasier Syndrome (WT1)	WT1 splice site mutations causing slowly progressive FSGS, male	US: Very rare; WT1 intron 9 splice mutations; slower ESKD	WT1 genetic testing (intron 9); kidney biopsy (FSGS); karyotype; gonadal	 Standard of Care

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		pseudohermaphroditism (46XY), and gonadoblastoma risk; milder than Denys-Drash	(adolescence/young adulthood); gonadoblastoma ~30% World: Very rare globally; WT1 intron 9 donor site mutations; distinct from Denys-Drash; streak gonads with gonadoblastoma risk Sources: Barbaux et al. Nat Genet 1997; Pelletier et al. Cell 1991; Lehnhardt et al. Ped Nephrol 2011	ultrasound; urinalysis and renal function monitoring	- Gonadectomy — CRITICAL: streak gonads have high gonadoblastoma risk (30–60%); bilateral gonadectomy recommended at diagnosis; gonadoblastoma can transform to dysgerminoma - Kidney transplant — ESKD typically in adolescence/young adulthood (slower than DDS); no Wilms tumor risk in Frasier (unlike DDS); transplant after nephrectomy; good outcomes - Steroid-resistant NS management — ACEi/ARBs for proteinuria; progressive CKD; dialysis planning - Hormonal management — testosterone or estrogen replacement post-gonadectomy per gender identity; endocrinology comanagement - DSD psychosocial support — multidisciplinary team; gender identity counseling Emerging WT1 splice site research — Frasier caused by +KTS splice isoform imbalance; therapeutic splicing modulation investigated
28	 Nail-Patella Syndrome (LMX1B)	Autosomal dominant LMX1B mutations causing nail dysplasia, absent/hypoplastic patella, iliac horns, glaucoma, and nephropathy in ~30–40%	US: ~1/50,000; nephropathy in 30–40%; ESKD in ~10%; hematuria/proteinuria may appear in childhood World: ~1/50,000 globally; COL5A1/LMX1B mutations; renal involvement in 30–40%; COL4A3/A4 involvement established Sources: Bongers et al. Nat Genet 2002; Sweeney et al. Am J Hum Genet 2003; NPS Consortium data	LMX1B genetic testing; kidney biopsy (EM: 'moth-eaten' GBM with collagen fibrils); X-ray (iliac horns, absent patella); ophthalmology (glaucoma)	Standard of Care - ACE inhibitors / ARBs — first-line for proteinuria/hypertension; ~30–50% of NPS patients develop nephropathy; KDIGO CKD guidelines apply; early initiation at microalbuminuria - BP control — essential; renovascular and glomerular mechanisms - Monitoring — annual renal function, urinalysis, BP; ophthalmology (open-angle glaucoma in 30–50%); orthopedic surveillance (patella dislocation, elbow dysplasia) - Glaucoma treatment — topical beta-blockers, prostaglandin analogues; laser trabeculoplasty; trabeculectomy; regular IOP monitoring - Kidney transplant — for ESKD; NPS does not recur in graft (LMX1B mutation affects host cells, not donor); good outcomes - Orthopedic management — physiotherapy for patellar instability; surgical stabilization if required; skeletal surveillance Emerging LMX1B transcription factor research — podocyte-expressed; understanding downstream targets for therapy
29	 Schimke Immuno- Osseous Dysplasia (SMARCAL1)	SMARCAL1 mutations causing spondyloepiphyseal dysplasia, T-cell immunodeficiency, and progressive steroid-resistant	US: Very rare; ~250 cases worldwide; AR; immunodeficiency and ESKD lead to poor prognosis World: Extremely rare globally; ~250 cases reported; SMARCAL1	SMARCAL1 genetic testing; kidney biopsy (FSGS); skeletal survey; lymphocyte immunophenotyping; T-cell function studies	Standard of Care Emerging/Investigational: Hematopoietic stem cell transplant — experimental

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		FSGS; often fatal in childhood	mutations; T-cell lymphopenia and ESKD uniform findings Sources: Boerkoel et al. Nat Genet 2002; Lücke et al. Ped Nephrol 2012; SIOD International Registry		
30	 CoQ10 Nephropathy (ADCK4 / COQ6)	Mitochondrial coenzyme Q10 biosynthesis defects causing steroid-resistant nephrotic syndrome/FSGS that may respond to CoQ10 supplementation	US: Very rare; ADCK4 and COQ6 mutations; important treatable cause; pediatric onset typical World: Very rare globally; both ADCK4 and COQ6 mutations reported; coenzyme Q10 supplementation may halt renal progression Sources: Ashraf et al. JASN 2013; Heeringa et al. JASN 2011; Lipka et al. Ped Nephrol 2018	Genetic testing (ADCK4, COQ6, COQ2, COQ8B); kidney biopsy (FSGS); urine organic acids; muscle CoQ10 levels; mitochondrial function testing	Standard of Care - Oral CoQ10 supplementation (ubiquinol or ubiquinone) — may dramatically halt or even reverse renal decline if initiated before significant fibrosis; dose 30 mg/kg/day; mechanism: restores mitochondrial electron transport chain function; ADCK4 mutations respond best - RAAS blockade (ACEi/ARB) — antiproteinuric; adjunctive - Renal transplant — if progressive to ESKD; post-transplant outcomes good <i>Note — CRITICAL DIAGNOSTIC PRIORITY: COQ2/COQ6/ADCK4 mutations represent the most treatment-responsive form of genetic FSGS; CoQ10 initiation before dialysis essential for renal recovery</i> Emerging/Investigational - Optimized CoQ10 formulations (nanomicellar, liposomal) — improved bioavailability under study - Gene therapy for ADCK4/PDSS2 — preclinical - Mitochondrial-targeted antioxidants (MitoQ) — investigational - Exogenous CoQ10 analogues — early stage

CILIOPATHIES / CYSTIC DISEASES

31	 Autosomal Recessive Polycystic Kidney Disease (ARPKD — PKHD1)	PKHD1 mutations causing enlarged echogenic kidneys with collecting duct cysts + congenital hepatic fibrosis; presents perinatally or in childhood	US: ~1/20,000 in US; severe forms cause perinatal death from pulmonary hypoplasia; survivors face CKD and portal HTN World: ~1/20,000 globally; PKHD1 mutations; most common recessive PKD; perinatal/neonatal mortality ~30–50% in most severe Sources: Zerres et al. Nat Genet 1994; Bergmann et al. JASN 2005; Guay-Woodford AJKD 2014	Prenatal/neonatal renal ultrasound (enlarged echogenic kidneys); PKHD1 genetic testing; liver ultrasound; liver biopsy (ductal plate malformation); MRI	Standard of Care - Supportive management of renal disease — antihypertensives (ACEi/ARB preferred); proteinuria management; CKD staging and management per KDIGO guidelines - Antihypertensive therapy — critical; RAS blockade first-line; BP target <50th %ile for age - Biliary/hepatic management — congenital hepatic fibrosis (CHF) universal; portal hypertension surveillance; beta-blockers for varices; endoscopy for variceal band ligation; TIPS for refractory portal hypertension; Caroli disease component (ductal plate malformations) - Urinary tract infection management — increased susceptibility; aggressive antibiotic treatment; prophylaxis if recurrent - Dialysis — peritoneal or hemodialysis for ESKD; PD may be limited by hepatosplenomegaly/ascites - Combined kidney-liver transplant — for patients with ESKD + severe portal hypertension/liver disease; timing
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#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					optimization critical; outcomes improving; isolated kidney transplant if liver disease mild - Neonatal management — bilateral enlarged echogenic kidneys; respiratory support for pulmonary hypoplasia in severe neonatal cases; oligohydramnios-related pulmonary complications - Growth support — growth hormone supplementation for growth failure in CKD Emerging/Investigational - Tolvaptan (Jynarque, Otsuka) — V2R antagonist; approved for ADPKD adults; PEDIATRIC TRIAL ONGOING (ORBIT trial, NCT02964599) for ARPKD; inhibits cAMP-driven cystogenesis; not yet approved for ARPKD or pediatric use; monitoring: liver toxicity, aquaretic effects - mTOR inhibitors (everolimus, sirolimus) — reduce cyst growth in animal models; no sustained benefit in human PKD trials; not recommended outside trials - HDAC inhibitors — preclinical for ARPKD - PKHD1 gene therapy — hepatocyte-targeted; preclinical; complex gene (largest human gene)
32	 Autosomal Dominant PKD — Pediatric Onset (ADPKD — PKD1/PKD2)	PKD1/PKD2 mutations causing bilateral renal cysts that typically present in adulthood but increasingly diagnosed in childhood with screening	US: ~1/1,000 overall; pediatric manifestations in ~2–5% including HTN, hematuria, cyst complications World: ~1/1,000 globally (most common inherited renal disease); PKD1 (85%) and PKD2 (15%); pediatric manifestations in 2–5% Sources: Torres et al. Lancet 2007; CRISP Study; ADPKD Consensus Report 2015	Renal ultrasound (cysts); MRI for detailed assessment; PKD1/PKD2 genetic testing; blood pressure monitoring; urinalysis; family history	 Standard of Care - Blood pressure management — ACEi/ARBs first-line; hypertension begins in childhood; ESCAPE trial shows BP target <50th percentile; early intervention reduces renal progression - Monitoring — annual BP, urinalysis, renal function; renal imaging (ultrasound every 3–5 years or per symptoms); avoid CT radiation in children - Avoid nephrotoxins — NSAIDs, contrast agents, aminoglycosides — limit; optimize hydration - High fluid intake — reduces vasopressin levels; delays cystogenesis; 2–3 L/m²/day; water preferred over juice/soda - Pain management — for renal cyst hemorrhage/infection; analgesics; drainage if severe - Psychosocial support — genetic condition; disclosure counseling; impact on child/family - Genetic counseling — PKD1 (chromosome 16) vs PKD2 (chromosome 4); PKD2 milder course; prenatal diagnosis; pre-implantation genetic testing available FDA-Approved (adults; pediatric under study) - Tolvaptan (Jynarque, Otsuka) — FDA approved 2018 for adults with ADPKD at risk of rapid progression; TEMPO 3:4 and REPRISE trials; slows TKV growth ~50%; not approved in pediatrics; hepatotoxicity monitoring required; PEDIATRIC trials planned; liver function tests mandatory

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					every 18 months; aquaretic side effects (nocturia, thirst); contraindicated in liver disease Emerging/Investigational - Tolvaptan pediatric trials — safety and PK studies in children underway; PREVENT-ADPKD trial; potential future approval - mTOR inhibitors — no benefit demonstrated in adults; not recommended - CFTR inhibitors — c-AMP pathway link; investigational - Metformin — reduces cAMP via AMPK; Phase 2 TAME-PKD (NCT02656017); promising; off-label early use in some centers - Tesevatinib (EGFR/Src inhibitor) — Phase 2 data; slows TKV progression - KD025 (ROCK2 inhibitor) — anti-fibrotic; preclinical for PKD
33	 Nephronophthisis — Infantile (NPHP2/INVS)	Most severe NPHP form; INVS mutations causing renal cysts and fibrosis presenting in first year of life with rapid progression to ESKD	US: Very rare; ESKD before age 2; accounts for small fraction of infantile ESKD in US registries World: Very rare globally; INVS/NPHP2 mutations; situs inversus in ~25%; ESKD typically before 18 months Sources: Otto et al. Nat Genet 2003; Tory et al. JASN 2009; ESCAPE Trial data	Renal ultrasound (small echogenic kidneys ± cysts); NPHP2/INVS genetic testing; kidney biopsy (tubular basement membrane disruption, interstitial fibrosis)	Standard of Care - Renal replacement therapy — ESKD in infancy (typically within first year of life); peritoneal dialysis; aggressive nutritional support for growth - Kidney transplant — curative for renal disease; excellent outcomes; NPHP does not recur in graft; timing when size permits (≥8–10 kg); infantile NPHP has earlier ESKD than juvenile - Situs inversus management — universal in NPHP2/INVS; cardiac evaluation; no specific therapy but surgical planning for transplant - Hepatic fibrosis — present in NPHP2; liver surveillance, portal hypertension monitoring; combined kidney-liver transplant if severe - Nutritional support — NG/GT feeding; protein and caloric supplementation - Renin-angiotensin blockade — CKD management; BP control
34	 Nephronophthisis — Juvenile (NPHP1)	Most common NPHP; NPHP1 deletion causing chronic tubulointerstitial nephritis with polydipsia, polyuria, anemia, and ESKD ~age 13	US: Most common genetic cause of ESKD in first 3 decades; ~1/50,000; NPHP1 deletion in ~25% of cases World: ~1/50,000 globally; NPHP1 deletion most frequent cause; ESKD median age ~13 years; ciliopathy spectrum Sources: Saunier et al. Nat Genet 1997; Hildebrandt et al. Kidney Int 1997; Otto et al. JASN 2002	Renal ultrasound (small normal-sized or slightly small echogenic kidneys); NPHP1 deletion by MLPA; genetic panel; retinal exam (Senior-Løken); kidney biopsy	Standard of Care - Renal replacement therapy — ESKD typically by ~13–15 years (juvenile NPHP1); medullary cystic disease; progressive tubulointerstitial nephritis - Kidney transplant — gold standard; curative; no recurrence in graft; deceased or living donor; outcomes excellent - CKD management — ACEi/ARBs for hypertension and proteinuria; anemia management (EPO); mineral bone disease management; nutrition support; growth hormone for CKD-related growth failure

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Ophthalmological surveillance — Senior-Løken syndrome overlap (10–15% of NPHP1); annual exam for retinitis pigmentosa - CNS surveillance — cerebellar involvement in some NPH1 variant syndromes; MRI if neurological symptoms - Avoidance of nephrotoxins — NSAIDs, contrast Emerging ☑ No disease-modifying therapy available for NPHP1 ☑ mTOR pathway — NPHP proteins modulate mTOR/Hedgehog; mTOR inhibitors investigated; no clinical benefit established ☑ CEP290/CC2D2A gene therapy concepts — preclinical; ciliary protein delivery challenging
35	 Nephronophthisis — Adolescent (NPHP3)	NPHP3 mutations causing NPHP with ESKD in adolescence (~age 19); may have extrarenal features (liver fibrosis, retinal dystrophy)	US: Rare; ESKD in late adolescence; less common than NPHP1; NPHP3 mutations World: Rare globally; NPHP3 mutations cause adolescent-onset NPHP; liver and heart involvement reported Sources: Olbrich et al. Nat Genet 2003; Tory et al. JASN 2009; NPHP registry data	Renal ultrasound; NPHP3 genetic testing; liver function; retinal exam; kidney biopsy if uncertain	 Standard of Care - Same approach as juvenile NPHP1 — CKD management, ACEi/ARBs, RRT, transplant - ESKD typically in adolescence to young adulthood (NPHP3 milder than NPHP1) - Extra-renal surveillance — hepatic fibrosis (NPHP3-associated); situs inversus; cardiac malformations in some; retinal degeneration in Senior-Løken overlap - Kidney transplant — curative; no recurrence Emerging: No disease-modifying therapies available
36	 Bardet-Biedl Syndrome — Renal (BBS)	Ciliopathy with primary cilia dysfunction causing obesity, rod-cone dystrophy, polydactyly, cognitive impairment, hypogonadism, and renal anomalies	US: ~1/100,000–160,000 in US; renal anomalies in ~53%; renal failure in ~5–10% World: ~1/100,000–160,000 globally; higher in Arab and Bedouin populations; renal cysts and CAKUT in majority Sources: Forsythe et al. Nat Rev Nephrol 2013; BBS International Registry; Beales et al. Nat Genet 1999	Clinical criteria (primary/secondary features); BBS genetic panel (20+ genes); renal ultrasound; retinal assessment; MRI brain; BMI monitoring	 Standard of Care - CKD/renal anomaly management — structural renal anomalies (cysts, horseshoe, dysplasia) common; ACEi/ARBs for proteinuria/HTN; CKD monitoring per KDIGO; avoid nephrotoxins - Obesity management — severe early-onset obesity is universal in BBS; dietary intervention; behavioral modification; multidisciplinary obesity clinic - Hypertension — antihypertensives per AAP guidelines; obesity-driven - Type 2 diabetes management — early-onset T2D in many; metformin; GLP-1 agonists - Retinal dystrophy — no proven therapy; low vision aids; genetic counseling; avoid vitamin A supplements (counterproductive in some BBS genotypes) - Hypogonadism — testosterone or estrogen replacement post-puberty; endocrinology comanagement - Developmental support — intellectual disability variable; speech/language therapy; occupational therapy - Kidney transplant — for ESKD; good outcomes; obesity management critical peri-transplant FDA-Approved (2020)

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					▢ Setmelanotide (IMCIVREE, Rhythm Pharmaceuticals) — MC4R agonist; FDA approved June 2020 for obesity due to BBS (and other MC4R pathway deficiencies); weight loss of 10–15%; also improves hyperphagia; first pharmacotherapy specifically for BBS obesity; pediatric approval ≥6 years; SC daily injection Emerging ▢ Ciliary biology — BBS proteins form BBSome; downstream therapeutics targeting cAMP, Hedgehog, mTOR; research stage ▢ Gene therapy — large number of BBS genes; gene-specific approaches; preclinical
37	 Joubert Syndrome — Renal (JBTS)	Ciliopathy with cerebellar vermis hypoplasia (molar tooth sign), hypotonia, oculomotor apraxia, and variable nephronophthisis or CAKUT	US: ~1/80,000–100,000 in US; nephropathy in ~35%; ESKD in ~20% by age 20 World: ~1/80,000–100,000 globally; >35 causative genes; molar tooth sign on MRI pathognomonic; renal involvement variable Sources: Romani et al. JASN 2013; Travaglini et al. Ped Nephrol 2013; JBTS International Consortium	Brain MRI (molar tooth sign pathognomonic); renal ultrasound; genetic panel (CEP290, TMEM67, CC2D2A, etc.); retinal exam; developmental assessment	 Standard of Care - Nephronophthisis component — CKD management; ACEi/ARBs; dialysis and transplant for ESKD; kidney transplant does not affect neurological manifestations - Respiratory management — neonatal/infant apnea/hyperpnea ("manta ray" breathing); supplemental oxygen; caffeine for apnea; CPAP; most respiratory symptoms improve with age - Developmental and neurological support — cerebellar vermis aplasia/hypoplasia ("molar tooth sign"); gross motor delay; physiotherapy; occupational therapy; speech therapy; intellectual support - Ophthalmological — oculomotor apraxia, retinal dystrophy; visual rehabilitation; low vision aids - Liver surveillance — hepatic fibrosis in JBTS7/RPGRIP1L and others; portal hypertension monitoring - Genetic testing — >40 genes identified (CEP290, TMEM67/MKS3, AHI1, NPHP1, RPGRIP1L, etc.); genotype informs prognosis Emerging ▢ No disease-modifying therapies available ▢ Ciliopathy pathway research — primary cilia biology; Hedgehog/mTOR modulation; preclinical
38	 Meckel-Gruber Syndrome	Lethal ciliopathy (MKS1, TMEM216, CEP290, etc.) causing renal cystic dysplasia, occipital encephalocele, polydactyly; incompatible with life	US: ~1/13,000–140,000 (wide variation by population); lethal; prenatal diagnosis critical World: 1/13,000 in some Finnish populations; 1/140,000 in others; MKS1, TMEM67, CEP290 common mutations; universally lethal Sources: Mecke & Passarge Neuropediatrics 1971; Barisic et al. Eur J Hum Genet 2015; MKS Consortium	Prenatal ultrasound (renal cysts + encephalocele + polydactyly); fetal MRI; genetic panel for MKS genes; autopsy	 Standard of Care - Lethal condition — Meckel-Gruber syndrome is uniformly lethal (encephalocele + bilateral renal cystic dysplasia + polydactyly); stillbirth or death in early neonatal period - Prenatal diagnosis — ultrasound diagnosis possible from 11–14 weeks gestation; MRI for confirmation; genetic testing (MKS1, TMEM216, CEP290, etc.) - Perinatal palliative care — counseling for comfort-focused care; neonatology consultation; no curative intervention available

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Bereavement support — family counseling; autopsy for genotype-phenotype correlation - Genetic counseling — autosomal recessive; 25% recurrence risk; prenatal diagnosis in subsequent pregnancies; preimplantation genetic testing available
39	 Senior-Løken Syndrome	Ciliopathy combining nephronophthisis with early-onset retinal dystrophy (Leber congenital amaurosis or retinitis pigmentosa); NPHP1/4/5 mutations	US: Rare; ~1/1,000,000 in US; retinal dystrophy may precede renal disease by years World: ~1/1,000,000 globally; NPHP genes (NPHP1, NPHP4, NPHP5, CEP290); retinal degeneration onset before renal failure Sources: Senior et al. Am J Ophthalmol 1961; Otto et al. Nat Genet 2005; Tory et al. JASN 2009	Retinal exam + ERG (retinal dystrophy); renal ultrasound (NPHP pattern); genetic panel (NPHP1, NPHP4, NPHP5/IQCB1, CEP290); kidney biopsy	Standard of Care - Nephronophthisis management — same as NPHP1 for renal component; ACEi/ARBs; CKD monitoring; dialysis; kidney transplant (curative for renal disease; renal outcomes excellent) - Retinal dystrophy — Leber congenital amaurosis or retinitis pigmentosa spectrum; no curative therapy; low vision rehabilitation; dark-adapted glasses; vitamin A supplementation (caution: variable benefit/harm by genotype; discuss with ophthalmologist) - Ophthalmological monitoring — early electroretinogram (ERG) for functional assessment; OCT for structural monitoring; genetic ophthalmology specialist - Genetic counseling — multiple causative genes (NPHP5/IQCB1, NPHP6/CEP290, NPHP1, etc.); autosomal recessive; variable retinal/renal severity by gene Emerging  CEP290 antisense oligonucleotide (sepofarsen, ProQR) — for CEP290-associated retinal dystrophy (LCA10); Phase 2/3 ILLUMINATE trial; intravitreal injection; improves visual acuity in some patients; ongoing trials  Gene therapy for LCA (non-CEP290) — voretigene neparvovec (Luxturna, Spark) FDA approved 2017 for RPE65 mutations (not Senior-Løken); model for future CEP290 delivery
40	 Oro-Facial-Digital Syndrome Type I — Renal	X-linked dominant ciliopathy (OFD1) with oral, facial, and digit anomalies; females affected (lethal in males); renal cysts in ~15–20%	US: ~1/250,000 in US; renal cysts minor feature; X-linked; typically affects females World: ~1/250,000 globally; X-linked; caused by OFD1 mutations; renal cysts in ~15–20%; CNS features prominent Sources: Ferrante et al. Nat Genet 2006; Thauvin-Robinet et al. Orphanet 2011; OFDS1 Registry	OFD1 genetic testing; renal ultrasound; clinical oral/digital/facial assessment; brain MRI (Joubert-like features in some)	Standard of Care - Renal cyst management — similar to ADPKD management when cysts present; ACEi/ARBs for hypertension/proteinuria; nephrology monitoring; dialysis and kidney transplant for ESKD - Oral abnormalities — cleft lip/palate repair; dental management; frequent dental evaluation; jaw anomalies - Cerebral involvement — partial agenesis of corpus callosum; Dandy-Walker malformations; neurodevelopmental monitoring; seizure management; anti-epileptics as needed - Digital anomalies — polydactyly, syndactyly; orthopedic surgical correction - Speech therapy — for oral and neurological components

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Genetic counseling — X-linked dominant; lethal in males; almost exclusively females affected; OFD1 gene mutation; prenatal testing available Emerging - No disease-modifying therapies; supportive multidisciplinary management
41	 Tuberous Sclerosis Complex — Renal (AML / cysts)	TSC1/TSC2 mutations causing renal angiomyolipomas and cysts; risk of hemorrhage from large AMLs and progression to CKD	US: ~1/6,000 in US; renal involvement in >80%; AMLs and cysts nearly universal; RCC risk increased World: ~1/6,000 globally; TSC1/TSC2 mutations; renal AMLs in >80% by adulthood; cystic disease common; RCC in ~2–3% Sources: Northrup & Krueger Ann Neurol 2013; TSC Consensus Conference 2012; Curatolo et al. Lancet 2008	Renal MRI (AMLs — fat signal); TSC1/TSC2 genetic testing or clinical criteria; MRI surveillance every 1–3 years; urinalysis; eGFR	Standard of Care - Everolimus (Afinitor) — mTOR inhibitor; FDA approved for growing TSC-associated AMLs ≥3 cm (EXIST-2 trial); reduces AML volume significantly; also approved for SEGA and renal cell tumors in TSC - Sirolimus — alternative mTOR inhibitor; off-label use in pediatric TSC-AML - Selective embolization (transarterial embolization) — for acute hemorrhage from AML (Wunderlich syndrome) or growing lesions not controlled on mTOR inhibitors - Nephron-sparing surgery — when embolization fails or for diagnostic uncertainty - RAAS blockade — for proteinuria and hypertension - Annual renal MRI surveillance — per ITSC guidelines from age 1 (recommended every 1–3 years) Emerging/Investigational - Nab-sirolimus (Fyarro) — FDA approved 2021 for malignant PEComa; off-label interest in TSC-AML - Longer-term mTOR inhibitor use — emerging data on prevention of renal functional decline - Autologous stem cell approaches — very early for severe multi-organ TSC; not renal-specific - Combination strategies (mTOR + CDK inhibitors) — preclinical
42	 Von Hippel-Lindau — Renal Cysts (pediatric onset)	VHL mutations causing renal cysts and early-onset clear cell RCC; pediatric surveillance important for at-risk family members	US: ~1/36,000 in US; renal cysts from childhood; RCC in 24–45% by 5th decade World: ~1/36,000 globally; VHL mutations; renal cysts childhood onset; RCC rare but reported in pediatric age; surveillance essential Sources: Lonser et al. Lancet 2003; Maher & Kaelin JASN 1997; VHL Alliance guidelines 2020	VHL genetic testing; annual renal MRI from age 8–10 in VHL families; abdominal ultrasound; ophthalmology; CNS imaging (hemangioblastomas)	Standard of Care - VHL renal surveillance — annual renal imaging (MRI preferred; ultrasound for children to minimize radiation); screen for clear cell RCC (common in VHL); kidney cysts usually asymptomatic in childhood - Nephron-sparing surgery — for renal cell carcinoma when lesion ≥3 cm; partial nephrectomy; active surveillance for lesions <3 cm (watchful waiting); minimize nephron loss given bilateral/multifocal RCC risk - Systemic VHL surveillance — annual MRI brain/spine (hemangioblastomas); ophthalmology (retinal hemangioblastomas); biochemical screening for pheochromocytoma (plasma metanephrines); pancreatic endocrine tumor screening

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Multidisciplinary VHL center care — neurosurgery, ophthalmology, endocrinology, nephrology, urology, genetic counseling [FDA-Approved (2021) — MAJOR ADVANCE] - Belzutifan (WELIREG, Merck) — HIF-2α inhibitor; FDA approved August 2021 for VHL-associated RCC, CNS hemangioblastomas, and pancreatic NETs (adults); LITESPARK-004 Phase 2 trial: 49% ORR for VHL-RCC; pediatric data emerging; blocks HIF-2α driven tumor growth; oral once-daily tablet; can eliminate need for surgery in some cases; anemia (common SE); monitor Hgb [Emerging] - Belzutifan pediatric studies — ongoing; pharmacokinetics in adolescents; dose finding - PT2977 (successor HIF-2α inhibitors) — next-generation compounds - Anti-VEGF approaches — bevacizumab for retinal hemangioblastomas; intravitreal injection
43	 Medullary Cystic Kidney Disease (UMOD / MUC1)	UMOD or MUC1 mutations causing chronic tubulointerstitial nephritis and gout in adolescents/young adults; now classified as ADTKD	US: Rare; UMOD and MUC1 mutations; hyperuricemia/gout often precedes renal failure by decades World: Rare globally; UMOD most common cause; MUC1 expansion mutations; autosomal dominant; gout in young adults characteristic Sources: Hart et al. Nat Genet 2002; Kirby et al. Nat Genet 2013; Devuyst & Pirson Ped Nephrol 2008	UMOD/MUC1 genetic testing; renal ultrasound (may appear normal or small); uric acid; kidney biopsy (tubular atrophy, interstitial fibrosis); family history	[Standard of Care] - CKD management — ACEi/ARBs for hypertension/proteinuria; KDIGO CKD stages; monitoring eGFR, proteinuria, BP; typically adult onset (teens–40s) but includes pediatric cases - Gout management — hyperuricemia universal in UMOD mutations; allopurinol (xanthine oxidase inhibitor) first-line; febuxostat alternative; dietary purine restriction; adequate hydration - Avoid nephrotoxins — NSAIDs, contrast, aminoglycosides; NSAIDs particularly harmful given tubulointerstitial damage - Kidney transplant — for ESKD; no disease recurrence in graft; excellent outcomes; gout resolves post-transplant - Genetic counseling — autosomal dominant; UMOD (uromodulin) or MUC1 mutations; genotyping important since MUC1 mutation missed by standard sequencing (VNTR repeat); specialized MUC1 sequencing recommended [Emerging] 4-PBA (4-phenylbutyrate) — chemical chaperone; promotes UMOD protein folding and secretion; Phase 2 trial ongoing for ADTKD-UMOD; reduces intracellular UMOD accumulation; potentially disease-modifying - BRD4 inhibitors — targets MUC1-STAT pathway in ADTKD-MUC1; preclinical
44	 Autosomal Dominant	Umbrella term for hereditary tubulointerstitial nephridites	US: Collectively rare; HNF1B most common in US; ADTKD covers	Genetic panel (MUC1, UMOD, HNF1B, REN); renal	[Standard of Care]

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
	Tubulointerstitial Kidney Disease (ADTKD — MUC1, HNF1B, REN)	caused by MUC1, UMOD, HNF1B, REN, or SEC61A1 mutations; progressive CKD	multiple genes with dominant inheritance World: Rare globally; HNF1B most prevalent cause; MUC1 and UMOD also recognized; slowly progressive to ESKD Sources: Bingham & Ellard Ped Nephrol 2012; ADTKD Consortium data; Eckardt et al. Nat Rev Nephrol 2013	ultrasound; uric acid; liver function (HNF1B); pancreas assessment (HNF1B-related MODY5)	- CKD management — KDIGO stages; ACEi/ARBs for hypertension; monitoring schedule; nutrition; avoid nephrotoxins - Gout treatment — allopurinol for UMOD variant; specific to UMOD subtype; see MCDK/UMOD entry - Kidney transplant — curative; no recurrence; excellent outcomes; all ADTKD subtypes - Genetic counseling — dominant inheritance; multiple genes (UMOD, MUC1, REN, HNF1B) - REN (renin) mutations — severe neonatal anemia, hypo-RAS-related; sodium supplementation, blood transfusion, erythropoietin in infancy/early childhood - HNF1B — complex phenotype (renal cysts, diabetes MODY5, genital tract malformations); multidisciplinary management - KDIGO 2019 ADTKD guidelines — framework for management Emerging 4-PBA — for ADTKD-UMOD (see MCDK/UMOD); Phase 2 ongoing - BRD4/MUC1 pathway — ADTKD-MUC1; investigational
45	 HNF1B Nephropathy / 17q12 Microdeletion Syndrome	HNF1B mutations/17q12 deletion causing renal cysts and diabetes (MODY5), with variable renal dysplasia, CAKUT, hyperuricemia, and cholestasis	US: ~1/10,000–20,000 in US; 17q12 deletion in ~50%; highly variable phenotype (renal + pancreatic/hepatic) World: ~1/10,000–20,000 globally; HNF1B mutations and 17q12 microdeletion; MODY5 diabetes common co-feature Sources: Bingham & Ellard Ped Nephrol 2012; Bellanne-Chantelot et al. JASN 2004; 17q12 Deletion Consortium	HNF1B genetic testing or chromosomal microarray; renal MRI/ultrasound; OGTT (MODY5 diabetes); liver/pancreas imaging; uric acid; urinalysis	Standard of Care - Renal cyst/dysplasia management — CKD monitoring; ACEi/ARBs for hypertension and proteinuria; may present as neonatal/fetal echogenic kidneys; ESKD in subset (variable) - Diabetes (MODY5) management — insulin-dependent diabetes; NOT responsive to sulfonylureas; insulin therapy; endocrinology co-management; onset typically adolescence/young adulthood - Genital tract abnormalities — Müllerian agenesis/bicornuate uterus in females; cryptorchidism in males; reproductive counseling; gynecology/urology referral - Liver/biliary abnormalities — cholestasis in some; hepatology monitoring - Electrolyte monitoring — hypomagnesemia common (tubular Mg wasting); oral magnesium supplementation - Pancreatic exocrine insufficiency — supplemental pancreatic enzymes if present - Kidney transplant — for ESKD; no recurrence; excellent outcomes - Genetic counseling — 50% de novo; MODY5 inheritance; family screening; array CGH for 17q12 deletion (whole gene deletion common)

CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT (CAKUT)

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
46	 Renal Agenesis — Unilateral	Absence of one kidney due to failure of ureteric bud induction; compensatory hypertrophy of contralateral kidney occurs	US: ~1/1,000–2,000 births in US; contralateral kidney anomalies in ~30%; higher in males World: ~1/1,000–2,000 births globally; most common in males; associated with CAKUT syndromes and chromosomal anomalies Sources: Westland et al. Ped Nephrol 2011; Sanna-Cherchi et al. JASN 2013; CAKUT Consortium	Prenatal or postnatal ultrasound; DMSA scan to confirm absence vs. ectopia; VCUG if VUR suspected; blood pressure; urinalysis	Standard of Care - Monitoring — blood pressure annually; urinalysis + protein/creatinine ratio annually from early childhood; eGFR monitoring; avoid nephrotoxins; renal ultrasound every 3–5 years for contralateral compensatory hypertrophy - BP management — ACEi/ARBs at first sign of hypertension or microalbuminuria; compensatory hyperfiltration promotes glomerulosclerosis - Contralateral kidney protection — avoid NSAIDs, aminoglycosides, iodinated contrast; aggressive hydration when ill; careful use of nephrotoxic medications - Diet — avoid excessive protein and sodium intake; DASH principles - CAKUT evaluation — unilateral agenesis frequently associated with other CAKUT anomalies (VUR, contralateral UPJ obstruction); voiding cystourethrogram if UTI; renal ultrasound protocol - Genetic counseling — HNF1B, PAX2, RET mutations in subset; chromosomal microarray if other anomalies present - Long-term counseling — ~40% lifetime risk of hypertension; ~10–15% CKD risk; contact sports counseling (protective padding; kidney guard for high-contact sports)
47	 Renal Agenesis — Bilateral (Potter Sequence)	Complete absence of both kidneys causing oligohydramnios, pulmonary hypoplasia, Potter facies, and limb deformities; lethal without intervention	US: ~1/4,000–5,000 births in US; lethal perinatally from pulmonary hypoplasia; amnioinfusion experimental World: ~1/4,000–5,000 births globally; potters sequence inevitable; outcomes improving with fetal interventions (amnioinfusion) experimentally Sources: Granger et al. Ped Nephrol 2012; Thomas & Frias Ped Rev 2009; NEJM 2023 (amnioinfusion trial)	Prenatal ultrasound (absent kidneys, severe oligohydramnios); MRI fetal; autopsy confirmation postnatal	Standard of Care - Potter sequence — bilateral renal agenesis is uniformly fatal without intervention; pulmonary hypoplasia from oligohydramnios; no kidney function - Experimental renal agenesis — infant dialysis programs at highly specialized centers (Children's Hospital of Philadelphia, etc.); peritoneal dialysis from birth possible - Palliative care — standard approach; neonatology and palliative medicine counseling for families; compassionate care protocol - Prenatal counseling — renal agenesis diagnosed by fetal MRI; detailed counseling on prognosis; options discussion - Amniotic fluid instillation — experimental prenatal therapy to improve lung development; investigational trials underway (CHOP/Cincinnati) - If survival achieved (dialysis protocol): kidney transplant goal; requires sufficient pulmonary development; PD access placed surgically; rare long-term survivors reported Emerging  Prenatal amniotic fluid supplementation (FAST trial) — serial amnioinfusions to restore lung development; Phase

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					2 investigational trials; potential to improve pulmonary hypoplasia and enable neonatal survival to dialysis ▶ Xenotransplantation — pig-to-human kidney; neonatal application highly investigational
48	 Renal Hypoplasia	Small kidney(s) with reduced nephron number but normal architecture; unilateral or bilateral; may progress to CKD	US: ~1/500 births (any degree) in US; bilateral hypoplasia major cause of childhood CKD in US (CKiD data) World: ~1/500 births globally; bilateral hypoplasia in ~0.5–1/10,000; leading cause of CKD in childhood worldwide Sources: Sanna-Cherchi et al. JASN 2012; CKiD Study; Westland et al. CJASN 2014	Prenatal/postnatal ultrasound; kidney size measurement; GFR assessment; DMSA scan (differential function); blood pressure; proteinuria	Standard of Care - CKD monitoring — staged per KDIGO; renal function surveillance from diagnosis; BP monitoring; urinalysis; proteinuria - Antihypertensive therapy — ACEi/ARBs preferred; hyperfiltration injury prevention in solitary or hypoplastic kidney - UTI prevention/treatment — increased susceptibility; low threshold for antibiotic treatment; VCUG if UTI to assess for VUR - Growth monitoring — CKD-associated growth failure; growth hormone therapy if growth failure established - Nutrition — adequate protein and caloric intake; salt restriction if hypertensive - Kidney transplant — for ESKD; bilateral severe hypoplasia may lead to childhood ESKD; excellent post-transplant outcomes
49	 Renal Dysplasia	Abnormal renal parenchymal development with disorganized architecture, primitive ducts, and metaplastic cartilage; may be cystic or non-cystic	US: Most common renal structural anomaly; bilateral dysplasia major cause of pediatric CKD in NAPRTCS World: Most common structural renal anomaly globally; component of many CAKUT phenotypes; bilateral disease leading cause of ESKD Sources: NAPRTCS 2011; Harambat et al. Ped Nephrol 2012; CKiD Study design CJASN 2008	Renal ultrasound (echogenic, disorganized architecture); DMSA (poor function); VCUG (associated VUR); genetic evaluation for syndromic causes	Standard of Care - Serial ultrasound — baseline at diagnosis then every 3–12 months depending on severity; assess for involution (MCDK) vs. progression (cystic dysplasia) - CKD management — KDIGO stages; ACEi/ARBs; bilaterally dysplastic kidneys lead to ESKD; unilateral tolerated if contralateral normal - Hypertension management — frequent comorbidity; standard antihypertensives - UTI management — voiding VCUG to exclude VUR; recurrent UTIs; antibiotic prophylaxis if VUR present - Proteinuria monitoring — annual urine protein/creatinine ratio - Dialysis and transplant — for severe bilateral dysplasia; peritoneal dialysis; transplant when size/weight permit - Genetic evaluation — CAKUT panel genes (RET, PAX2, HNF1B, BMP4, etc.) for bilateral dysplasia; chromosomal microarray for syndromic cases
50	Multicystic Dysplastic Kidney (MCDK)	Complete replacement of one kidney with non-communicating cysts and no functional parenchyma; contralateral kidney compensates	US: ~1/4,000 births in US; unilateral most common; involution by age 5 in most cases World: ~1/4,000 births globally; predominantly unilateral; bilateral lethal; European registry data consistent with US rates	Prenatal/postnatal ultrasound; DMSA (confirms non-function); VCUG (15–20% have contralateral VUR); blood pressure	Standard of Care Observation — most MCDK involute completely by 2–5 years; serial ultrasound every 6–12 months until involution confirmed; no intervention required for asymptomatic involuting MCDK

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			Sources: Kuwertz-Broeking et al. Arch Dis Child 2004; Rabelo et al. Ped Nephrol 2015; MCDK Registry		• Contralateral kidney surveillance — compensatory hypertrophy; monitor for VUR (VCUG in first year if UTI or additional anomalies); hypertension; proteinuria • BP monitoring — annually from age 5; hypertension develops in ~15–20% • Nephrectomy — reserved for: persistently enlarging MCDK after age 2, hypertension, suspected Wilms in a cystic mass, symptoms (pain, UTI) • Renal tumor risk — very small increased risk of RCC or Wilms in MCDK; reassurance; nephrectomy if concerning features • Long-term follow-up — even after involution: annual BP, urinalysis, creatinine through adolescence
51	 Renal Involvement in VACTERL / CHARGE Syndrome	Structural renal anomalies (agenesis, horseshoe, VUR, hydronephrosis) occurring as part of VACTERL or CHARGE association	US: VACTERL ~1/10,000–40,000 US; CHARGE ~1/8,500–10,000 US; renal anomalies in ~50% VACTERL World: VACTERL ~1/10,000–40,000 globally; CHARGE ~1/8,500–10,000; European registries consistent with US data Sources: Solomon et al. Birth Defects Res 2018; Issekutz et al. Am J Med Genet 2005; CHARGE Consortium	Clinical diagnosis; renal ultrasound; VCUG; chromosomal microarray (CHARGE: CHD7 gene); echocardiogram; spine imaging	Standard of Care • Renal anomaly management — 50–80% of VACTERL have renal anomalies (horseshoe, agenesis, dysplasia, VUR); manage per specific anomaly type; CKD monitoring • VUR management — VCUG screening; antibiotic prophylaxis; see VUR entry • Multidisciplinary care — VACTERL is multi-system; cardiology (cardiac anomalies), GI surgery (esophageal atresia/TE fistula, anorectal malformations), orthopedics (vertebral/limb), nephrology coordination • Nephrology surveillance — annual BP, urinalysis, creatinine; early referral if CKD develops • CHARGE syndrome (CHD7) — renal anomalies in 20–40%; same CKD monitoring approach; hearing loss management; ophthalmology; genetics
52	Duplex Kidney / Ureteral Duplication	Incomplete or complete ureteral duplication with two separate collecting systems draining one kidney; upper pole prone to obstruction/ectopia	US: ~1/500 population in US; most common renal anomaly; females more affected; upper pole ureterocele ~20% World: ~1/500 globally; most common congenital urinary tract anomaly; duplicated systems with ureterocele higher in females Sources: Privett et al. Br J Urol 1976; Siomou et al. Ped Nephrol 2011; AAP UTD guidelines 2011	Prenatal ultrasound; postnatal ultrasound; VCUG (lower pole VUR); MAG3 renogram (differential function); MRI urography	Standard of Care • Observation — majority of duplex/ureteral duplication are asymptomatic and managed conservatively • UTI management — upper pole ureterocele often obstructs and causes UTIs; antibiotic prophylaxis for febrile UTI; culture-directed antibiotics for breakthrough UTIs • Ureterocele management — depends on location and function: small orthotopic ureterocele → observation; obstructing ureterocele → endoscopic puncture (first-line); ectopic ureterocele with non-functioning upper pole → upper pole heminephrectomy or complete nephroureterectomy • VUR management — lower pole often refluxes in duplex systems; antibiotic prophylaxis; endoscopic injection (Deflux); ureteral reimplantation for breakthrough infections • Imaging — renal ultrasound + VCUG + MAG3 renal scan to assess function and drainage

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
53	Antenatal Hydronephrosis	Dilatation of the renal pelvis identified on prenatal ultrasound; most common prenatal finding; etiologies range from transient to obstructive	US: ~1–5% of pregnancies have antenatal hydronephrosis; majority resolves; SFU grading standard World: ~1–5% of all pregnancies globally; most common prenatal anomaly detected on ultrasound; majority (50–70%) resolves spontaneously Sources: Nguyen et al. Ped Nephrol 2010; Lee et al. Urology 2006; SFU Classification System	Serial prenatal ultrasound; postnatal ultrasound at 48–72 hours and 4–6 weeks; MAG3 renogram; VCUG; APD measurement (SFU grade)	Standard of Care - Society for Fetal Urology (SFU) grading — Grades 1–4; Grade 1–2 transient physiological in majority (>70% resolves spontaneously) - Postnatal evaluation — renal ultrasound at 48–72 hours and again at 1 month; severe prenatal ANH warrants earlier postnatal scan - VCUG — for bilateral ANH, severe grades, or febrile UTI to rule out VUR/PUV - MAG3 diuretic renography — assess differential renal function and obstruction (UPJ/UVJ) - Antibiotic prophylaxis — amoxicillin in neonates with moderate-severe ANH pending workup; trimethoprim-sulfamethoxazole for older infants; prophylaxis duration per SFU/AAP guidance - Surgical intervention — UPJ obstruction with <40% differential function or progressive dilatation → pyeloplasty; see UPJ/UVJ entries - Observation — mild ANH (APD <10 mm at term) → outpatient follow-up; most resolve
54	Ureteropelvic Junction Obstruction (UPJO)	Obstruction at junction of renal pelvis and ureter causing hydronephrosis; intrinsic (aperistaltic segment) or extrinsic (crossing vessel)	US: ~1/1,500 births in US; most common cause of significant antenatal hydronephrosis; males > females World: ~1/1,500 births globally; bilateral in 10–40%; most common cause of upper tract obstruction requiring intervention Sources: Hussain et al. Ped Nephrol 2014; Sutherland et al. J Urol 2008; EAU Pediatric Guidelines 2021	Renal ultrasound; MAG3 renogram (obstruction pattern, split function); VCUG (exclude VUR); MRI urography for complex cases	Standard of Care - Observation vs. surgery decision — based on: differential renal function (DRF <40% = surgical threshold in most centers), drainage on MAG3 (T½ >20 min), progressive hydronephrosis, recurrent UTIs, symptoms (pain) - Pyeloplasty — gold standard surgical repair; laparoscopic or robotic-assisted dismembered Anderson-Hynes pyeloplasty; >95% success rate; preserves renal function; outpatient or 23-hour stay - Antibiotic prophylaxis — for significant hydronephrosis pending surgery or during observation - Serial MAG3 renography + ultrasound — every 3–6 months during observation; post-pyeloplasty scan at 3–6 months - Endopyelotomy — alternative to open/laparoscopic repair in selected adolescents; lower success rate; less preferred in pediatrics - Congenital UPJ — most common cause; extrinsic compression (crossing vessel) in ~20%
55	Ureterovesical Junction Obstruction	Obstruction at junction of ureter and bladder causing hydroureteronephrosis; includes primary megaureter and obstructed ectopic ureter	US: ~1/5,000 births in US; primary megaureter ~20% of antenatal hydronephrosis referrals World: ~1/5,000 births globally; spontaneous resolution in majority; surgical correction when obstruction is significant Sources: Liu et al. Ped Nephrol 2014; Baskin et al. J Urol 1994;	Renal ultrasound (dilated ureter and pelvis); MAG3 renogram; VCUG (exclude VUR); MRI urography	Standard of Care - Observation — many UVJ obstructions (megaureter) resolve spontaneously, especially primary non-refluxing type; serial ultrasound - Antibiotic prophylaxis — during observation period; febrile UTI management - Surgical intervention — ureteral reimplantation with tapering (for obstructed megaureter); indications:

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			EAU Pediatric Urology Guidelines 2021		declining renal function (<40% DRF), recurrent febrile UTIs despite prophylaxis, progressive dilatation, symptoms - Timing — most centers delay surgery until 12–18 months if no deterioration; neonatal surgery if severe obstruction and infected - MAG3 renography — differential function and drainage assessment; key for surgical decision - Temporary ureteral stent or cutaneous ureterostomy — for urgent decompression in infected obstruction
56	 Vesicoureteral Reflux (VUR) — Primary	Retrograde urine flow from bladder to ureter/kidney due to incompetent ureterovesical junction; predisposes to pyelonephritis and renal scarring	US: ~1% general US population; 30–40% of children post-febrile UTI; familial in ~30% World: ~1% globally; highest in white European-descent children; lower in Black and Asian children; resolves spontaneously in most Sources: Smellie et al. BMJ 1975; Tekgül et al. Eur Urol 2015; RIVUR Trial 2014	VCUG (gold standard); DMSA (renal scarring); MAG3 renogram; grading I–V; urodynamics if bladder dysfunction suspected	Standard of Care - Low-grade VUR (I–II) — observation alone; antibiotic prophylaxis not routinely recommended by 2011 AAP guidelines; parental education on UTI recognition; prompt treatment of febrile UTIs - Higher-grade VUR (III–V) / recurrent febrile UTIs — antibiotic prophylaxis (trimethoprim-sulfamethoxazole or nitrofurantoin); RIVUR trial (2014): prophylaxis reduces UTI recurrence but not renal scarring in uncomplicated VUR; benefit greatest in bladder/bowel dysfunction - Bladder/bowel dysfunction (BBD) — treat aggressively; BBD worsens VUR outcomes; timed voiding, fiber, anticholinergics for OAB component - Endoscopic injection (Deflux/dextranomer-hyaluronic acid) — minimally invasive; 70–80% resolution for Grade II–III; 50–60% for Grade IV; office/day surgery procedure; avoids open surgery - Open ureteral reimplantation (Cohen or Politano-Leadbetter) — >95% success rate; reserved for failed Deflux, Grade V, anatomical abnormalities; gold standard - Robotic-assisted ureteral reimplantation — increasingly used; comparable outcomes to open; shorter stay - Spontaneous resolution — Grade I 80%, Grade II 75%, Grade III 50%, Grade IV 25%, Grade V 15% per year Emerging  Circumcision — reduces UTI risk in uncircumcised males with VUR; reduces breakthrough UTIs  DMSA scan — 4–6 months post-febrile UTI to detect renal scarring; guides management intensity
57	Vesicoureteral Reflux — Secondary	VUR caused by elevated intravesical pressure (neurogenic bladder, PUV, dysfunctional voiding) rather than intrinsic ureterovesical junction abnormality	US: Prevalence depends on underlying bladder pathology; higher grade VUR than primary World: Prevalence determined by underlying pathology globally; neurogenic bladder and PUV major drivers; surgical correction often required	VCUG; urodynamics; identify and treat underlying cause; DMSA for scarring; renal function	Standard of Care Treat underlying cause — neurogenic bladder (clean intermittent catheterization, anticholinergics, Botox), bladder outlet obstruction (PUV ablation), posterior urethral valves; VUR often resolves when primary cause corrected

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			Sources: EAU Pediatric Guidelines 2021; AAP 2011 UTI guidelines; VCUG referral criteria		- CIC for neurogenic bladder — clean intermittent catheterization maintains low bladder pressures; prevents high-pressure VUR - Same antibiotic prophylaxis/surgical considerations as primary VUR if persistent - Bladder pressure management — anticholinergics; intravesical onabotulinumtoxinA for neurogenic DO - Ureteral reimplantation — only after bladder dysfunction maximally treated and only if VUR persists
58	Posterior Urethral Valves (PUV)	Obstructing folds in posterior male urethra causing bilateral hydronephrosis, bladder dysfunction, and CKD; most severe obstructive uropathy in males	US: ~1/5,000–8,000 male births in US; ~15–20% of pediatric ESKD; most severe LUTO in males World: ~1/5,000–8,000 male births globally; European registry data similar; ESKD in 15–20% despite early ablation Sources: Nassr et al. Ped Nephrol 2017; NAPRTCS 2011; ERA-EDTA Registry; PUV Long-term Study	Prenatal ultrasound (bilateral hydronephrosis, thick-walled bladder, keyhole sign); VCUG (confirms valves); cystoscopy; MAG3; creatinine nadir predicts outcome	Standard of Care - Valve ablation — endoscopic cystoscopic ablation (resection or vaporization) of posterior urethral valves; performed urgently/semi-urgently after initial stabilization; primary definitive treatment - Initial stabilization — urethral catheter decompression; treat electrolyte abnormalities and AKI; IV fluids; antibiotics if UTI/urosepsis - Renal function monitoring — PUV is leading cause of congenital obstructive uropathy ESKD; 30–50% reach ESKD by adolescence/young adulthood; serial eGFR, proteinuria, BP - ACEi/ARBs — for hypertension and proteinuria; nephroprotective - VUR management — present in 50% of PUV; antibiotic prophylaxis; usually resolves after valve ablation - Bladder dysfunction management — myogenic bladder damage persists after valve ablation; timed voiding; CIC if elevated PVR; anticholinergics cautiously (may worsen retention) - Nocturnal polyuria — renal concentrating defect common in PUV; desmopressin rarely helpful; fluid management education - Kidney transplant — for ESKD; bladder must be evaluated and optimized pre-transplant; bladder augmentation may be needed; excellent outcomes possible
59	Prune Belly Syndrome (Eagle-Barrett)	Triad of absent abdominal musculature, cryptorchidism, and urinary tract anomalies (megaureter, megacystis, renal dysplasia) exclusively in males	US: ~1/35,000–50,000 male births in US; CKD in ~30%; spectrum from lethal to mild World: ~1/35,000–50,000 male births globally; near-universal male; triad of abdominal wall, cryptorchidism, urinary anomalies Sources: Woodhouse et al. Br J Urol 1982; Hassett et al. J Urol 2012; PBS Consortium data	Prenatal ultrasound; postnatal evaluation: renal ultrasound, VCUG, MAG3; abdominal wall assessment; orchidopexy evaluation	Standard of Care - Urological reconstruction — complex, individualized; urethroplasty for urethral stenosis; orchidopexy for cryptorchidism (bilateral, intra-abdominal); ureteral reconstruction (for severely dilated ureters with infection) - Antibiotic prophylaxis — for recurrent UTIs; culture-directed treatment of febrile UTIs - Clean intermittent catheterization — if bladder dysfunction with elevated PVR - Abdominal wall reconstruction — abdominoplasty for cosmesis and core stability; improves pulmonary mechanics; typically deferred until 6–12 months

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					- Pulmonary management — absent abdominal musculature causes respiratory compromise; chest physiotherapy; treat pulmonary hypoplasia if present - CKD monitoring — renal dysplasia common; ACEi/ARBs; serial renal function; dialysis/transplant for ESKD - Cryptorchidism — orchidopexy early (within first year) to preserve fertility potential and reduce malignancy risk
60	Horseshoe Kidney	Fusion of lower poles of both kidneys across midline; associated with UPJO and VUR; usually incidental finding	US: ~1/400–600 US population; most common fusion anomaly; UPJO in 15–30%; Turner/trisomy 18 association World: ~1/400–600 globally; consistent prevalence worldwide; associated with multiple congenital syndromes Sources: Glodny et al. J Urol 2009; Weizer et al. J Urol 2003; EAU Pediatric Guidelines	Renal ultrasound (bridging isthmus); MAG3 renogram (exclude obstruction); VCUG; CT/MRI for complex anatomy	Standard of Care - Observation — most horseshoe kidneys asymptomatic; incidental finding; no treatment needed - UTI management — increased susceptibility due to altered drainage; urine culture; antibiotic treatment; VCUG to evaluate for VUR - Urolithiasis management — increased stone risk (urinary stasis, altered urodynamics); hydration; stone analysis; metabolic stone evaluation; ESWL (reduced efficacy in horseshoe kidneys); ureteroscopy; PCNL adapted technique - Hypertension management — ACEi/ARBs if hypertension develops; routine BP monitoring - UPJ obstruction — present in 30%; pyeloplasty if indicated (robot-assisted or laparoscopic) - Wilms tumor surveillance — horseshoe kidney has slight increased risk; ultrasound if suspicious lesion - Renal cell carcinoma — increased risk long-term; imaging surveillance in adults
61	Ectopic Kidney	Kidney located outside normal renal fossa (pelvic, iliac, thoracic, crossed fused ectopia); associated with VUR and UPJO	US: ~1/900–1,000 in US; pelvic ectopia most common; crossed fused ectopia ~1/7,500 World: ~1/900–1,000 globally; often incidental; malrotation typical; complications include obstruction and infection Sources: Bauer Urol Clin NA 1975; Glodny et al. J Urol 2009; Walsh Campbell's Urology	Renal ultrasound; MAG3 renogram; VCUG; CT/MRI for complex anatomy	Standard of Care - Observation — most ectopic kidneys function normally; no intervention needed - Hypertension monitoring — annual BP; ectopic kidneys may have anomalous vasculature contributing to renovascular hypertension - UTI management — some ectopic kidneys have drainage impairment; prompt antibiotic treatment; VCUG for VUR - Urolithiasis — similar to horseshoe kidney; hydration; stone management - Pelvic ectopic kidney — important surgical landmark; counsel families; surgical planning for any abdominal/pelvic procedures
62	Urethral Stricture — Pediatric	Narrowing of urethra causing voiding dysfunction, UTIs, and upper tract dilatation; can be congenital, traumatic, or iatrogenic in children	US: Rare in children without prior surgery; iatrogenic post-hypospadias repair most common cause World: Rare globally without prior instrumentation; congenital anterior	Uroflowmetry; VCUG; urethrogram; cystoscopy; urinary tract ultrasound	Standard of Care - Urethral dilation — first-line for short strictures; bougie or catheter-based; temporary relief; high recurrence - Urethroscopy + direct vision internal urethrotomy (DVIU) — for short (≤ 2 cm) strictures; cold-knife or laser incision; 50–60% success for first treatment; decreasing success with repeated procedures

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			urethral valves uncommon; surgical trauma predominant cause Sources: Snodgrass & Prieto J Urol 2009; EAU Pediatric Guidelines 2021; Duel et al. J Urol 1998		- Urethroplasty — gold standard for recurrent/long strictures; anastomotic repair or substitution (buccal mucosal graft); 85–95% success rate; preferred over repeated dilation/DVIU - Urine culture — treat infection before instrumentation - Trauma evaluation — bulbar urethral stricture in boys often post-straddle injury; PFUI (pelvic fracture urethral injury) in pelvic trauma - Posterior urethral valves — separate etiology; managed with valve ablation
63	Bladder Exstrophy — Renal Complications	Bladder exstrophy causes continuous urinary tract anomalies with frequent renal complications (VUR, hydronephrosis, CKD from recurrent infections/obstruction)	US: ~1/30,000–50,000 US births; renal complications in majority without optimal management; VUR universal World: ~1/30,000–50,000 births globally; complex reconstruction required; renal outcomes depend on early surgical management Sources: Mitchell & Bägli J Urol 1996; Gearhart & Mathews J Urol 2000; Exstrophy-Epispadias Complex Consortium	Renal ultrasound; MAG3; VCUG; eGFR; urinalysis; urodynamics post-reconstruction	 Standard of Care - Primary surgical closure — staged reconstruction: primary bladder closure within 72 hours of birth (with or without osteotomy); epispadias repair at 6–12 months; bladder neck reconstruction after bladder capacity achieved (typically 4–5 years) - Modern staged approach vs. complete primary repair (CPR) — complete primary repair at birth increasingly used in specialized centers; comparable outcomes - Continence surgery — bladder neck reconstruction (Mitchell, Young-Dees-Leadbetter); urinary continence achieved in 50–80% by age 5–6 - Bladder augmentation — if bladder capacity insufficient post-reconstruction; ileocystoplasty - CIC — often required post-bladder neck repair for complete emptying - Renal monitoring — VUR present in most; antibiotic prophylaxis; upper tract surveillance - Orthopedic — pelvic osteotomy facilitates closure; physiotherapy; normal ambulation expected
64	Cloacal Anomalies — Renal Involvement	Complex cloaca in females causes renal anomalies (fusion, ectopia, agenesis, hydronephrosis) in ~50%; renal function critical for outcomes	US: ~1/50,000 US females; renal anomalies in ~50%; complex surgical reconstruction needed World: ~1/50,000 females globally; spectrum from simple to complex; renal function crucial determinant of surgical approach and timing Sources: Levitt & Peña Semin Ped Surg 2010; Rich et al. J Pediatr Surg 1988; ARM-Net Registry	Renal ultrasound; MRI pelvis and abdomen; VCUG; MAG3; cystoscopy and genitoscopy; genetic evaluation	 Standard of Care - Multidisciplinary surgical reconstruction — complex staged repair; initial colostomy and mucus fistula at birth; mobilization of common channel; separation of urinary, genital, and GI tracts; typically 12–18 months of age - Urological reconstruction — vaginal reconstruction; ureteral reimplantation if VUR; bladder reconstruction; Malone appendicovesicostomy (MACE) for neurogenic bladder - Renal anomalies — 30–50% have associated CAKUT; renal ultrasound and CKD monitoring - Clean intermittent catheterization — neurogenic bladder common due to sacral deficiency; CIC essential - GI/colorectal management — bowel management program; constipation prevention critical; Malone antegrade colonic enema (MACE) procedure for bowel continence

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					- Gynecological follow-up — vaginal stenosis; Müllerian anomalies; fertility counseling in adolescence - Psychosocial support — lifelong multidisciplinary team
NEUROGENIC BLADDER / FUNCTIONAL UROLOGICAL					
65	 Neurogenic Bladder — Spina Bifida / Myelomeningocele	Bladder dysfunction from spinal cord defect causing detrusor-sphincter dyssynergia, high bladder pressures, VUR, and upper tract damage	US: ~3–4/10,000 births with NTD in US (folate fortification reduced NTD rates); neurogenic bladder in ~90% of MMC World: Globally 1–10/10,000 births; highest in South Asia; NTD rates declining with folate supplementation programs Sources: Khoshnood et al. BMJ 2015; Mitchell et al. Birth Defects Res 2004; MOMS Trial NEJM 2011	Urodynamics (high-pressure bladder, detrusor overactivity, DSD); renal ultrasound; VCUG; MAG3; urinalysis; creatinine; annual assessment	Standard of Care - Clean intermittent catheterization (CIC) — cornerstone; every 3–4 hours; prevents overdistension, reduces infection, protects upper tracts; started from infancy to avoid bladder damage - Anticholinergic medications — oxybutynin (oral or intravesical), tolterodine, solifenacin; reduces detrusor overactivity and high bladder pressures; intravesical oxybutynin better tolerated (fewer systemic effects) - Renal + bladder ultrasound surveillance — every 6–12 months; identify hydronephrosis, VUR, poor bladder compliance early - Urodynamics — annually from birth or at diagnosis; leak point pressure, bladder compliance, capacity; guides CIC and medication adjustments - VUR management — antibiotic prophylaxis; reimplantation if needed after bladder optimization - Upper tract protection — goal bladder storage pressure <40 cmH₂O; compliance monitoring <i>FDA-Approved Pediatric:</i> - OnabotulinumtoxinA (BOTOX, Allergan/AbbVie) — FDA approved 2021 for pediatric neurogenic detrusor overactivity (NDO) in children ≥5 years; 100–200 units intravesical injection into detrusor; urodynamics-guided; improves compliance, reduces incontinence, protects upper tracts; duration 6–12 months; retreatment as needed; LANDMARK approval for pediatric urology - Bladder augmentation (enterocystoplasty) — ileocystoplasty or sigmoid cystoplasty; for refractory neurogenic bladder with inadequate capacity despite CIC + medications + Botox; Mitrofanoff appendicovesicostomy for continent catheterizable channel; bladder neck closure if needed - Mitrofanoff principle — appendicovesicostomy; umbilical or abdominal stoma for CIC; facilitates independent catheterization in adolescents - Malone antegrade colonic enema (MACE) — for neurogenic bowel component - Spinal cord monitoring — surveillance for tethered cord; neurosurgery comanagement; untethering when indicated Emerging

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					▣ Sacral neuromodulation — InterStim; less established in myelomeningocele vs. non-neurogenic OAB; investigational in this population ▣ Beta-3 adrenergic agonist (mirabegron) — approved for adults/OAB; pediatric neurogenic bladder trials ongoing; alternative to anticholinergics ▣ Stem cell therapy for spinal cord injury — very early stage
66	 Neurogenic Bladder — Sacral Agenesis	Sacral agenesis causes variable neurogenic bladder; bowel and bladder dysfunction; associated with maternal diabetes	US: Rare; ~1/25,000 births in US; maternal diabetes risk factor; partial sacral agenesis more common than complete World: Rare globally; ~1/25,000; partial agenesis more frequent; renal/bladder phenotype variable by level of sacral involvement Sources: Boemers et al. J Urol 1994; Wilmshurst et al. Dev Med Child Neurol 2012; EAU Ped Guidelines	Spinal MRI; urodynamics; renal ultrasound; VCUG; MAG3 renogram; electromyography (EUS)	Standard of Care - Same CIC + anticholinergics framework as myelomeningocele neurogenic bladder - Urodynamics — critical given variable bladder phenotype; sacral agenesis spectrum (Type I–IV per Renshaw); Type I (partial unilateral sacral) often mild; Type IV (total sacral agenesis) severe - OnabotulinumtoxinA — same indications as spina bifida neurogenic bladder - Anorectal malformations — frequent comorbidity; bowel management program; constipation management critical (distended rectum worsens bladder dysfunction) - Screening for associated anomalies — maternal diabetes association; cardiac, vertebral, GI anomalies - Bladder augmentation — if conservative measures fail; same surgical principles - MRI sacral/spinal — defines anatomy; surgical planning
67	Dysfunctional Elimination Syndrome	Combination of bladder and bowel dysfunction causing urinary urgency, frequency, incontinence, constipation, and VUR/UTIs in neurologically normal children	US: Common; ~30% of children referred for UTI workup in US; peak age 5–10 years; females predominate World: Common globally; difficult to estimate true prevalence; behavioral and anatomic factors play varying roles across populations Sources: Vande Walle et al. Eur J Ped 2012; Austin et al. J Urol 2016; ICCS Standardization Report	History and voiding diary; uroflowmetry (staccato or plateau pattern); post-void residual; renal/bladder ultrasound; VCUG; urodynamics if needed	Standard of Care - Timed/scheduled voiding — every 2–3 hours; prevents bladder overdistension; cornerstone of management - Bowel management — treat constipation aggressively (dietary fiber, fluids, polyethylene glycol/MiraLax); constipation universally worsens bladder dysfunction (pelvic floor competition) - Pelvic floor biofeedback — for dyssynergia/DSD; teaches relaxation of external sphincter during voiding; highly effective; typically 4–8 sessions - Anticholinergics — oxybutynin for OAB component if behavioral therapy insufficient; start low dose - VUR management — DES worsens VUR outcomes; treat DES first before considering surgery for VUR - Urotherapy — specialist nurse education; bladder diary; voiding diaries; behavioral modification - Urodynamics — for complex/refractory cases; rule out occult neurogenic bladder - Screen for associated stressors — constipation, anxiety, school bathroom avoidance; psychosocial support
68	Overactive Bladder — Pediatric	Urinary urgency with or without incontinence, with	US: Common; ~5–10% of school-age US children; spontaneous	History and voiding diary; uroflowmetry; post-void	Standard of Care

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		frequency and nocturia, in neurologically normal children; detrusor overactivity on urodynamics	resolution typical; peak age 5–7 years World: ~5–10% of school-age children globally; ICCS criteria used internationally; spontaneous resolution in majority Sources: Norgaard et al. J Urol 1998; Neveus et al. ICCS Standardization 2010; Hoebeke et al. Eur Urol 2010	residual; renal ultrasound; urinalysis; urodynamics if refractory	- Behavioral therapy first-line — timed voiding; bladder training; double voiding; fluid management (avoid caffeine, citrus); constipation treatment; urotherapy education - Anticholinergic medications — oxybutynin extended-release (first-line pharmacotherapy); tolterodine; solifenacin (approved for OAB in children ≥5 yr in some countries); darifenacin; side effects: dry mouth, constipation, cognition (oxybutynin IR has most CNS effects; prefer ER formulations) - Mirabegron (beta-3 agonist) — adult approved; pediatric OAB trials (NICE, EU approval pending for pediatric use); fewer anticholinergic side effects; off-label use increasing; favorable tolerability - Urodynamics — for refractory OAB to exclude neurogenic etiology - Constipation treatment — essential; MiraLax; fiber; disimpaction if needed Emerging - OnabotulinumtoxinA (BOTOX) — FDA approved for pediatric NDO ≥5 yr (2021); off-label in non-neurogenic pediatric OAB; evidence growing; typically reserved for refractory cases - Sacral neuromodulation (InterStim Micro) — FDA approved for OAB in adults; increasing pediatric data in refractory non-neurogenic OAB ≥16 yr; office-level evaluation with external stimulation trial
69	 Nocturnal Enuresis — Refractory	Involuntary voiding during sleep in children ≥5 years; primary (never dry) or secondary; refractory defined as failure of standard treatments	US: Common; ~15% of 5-year-olds in US; ~1–2% of adolescents; spontaneous resolution 15%/year World: ~5–15% of 5-year-olds globally; widely prevalent; ICCS defines primary/secondary forms; spontaneous resolution consistent worldwide Sources: Neveus et al. Acta Paediatr 2010; Forsythe & Redmond Dev Med Child Neurol 1974; ICCS Guidelines 2020	Voiding diary; renal/bladder ultrasound; urinalysis; urine osmolality; consider sleep study (OSA); assess bladder capacity; constipation evaluation	Standard of Care - Alarm therapy — bedwetting alarm (enuresis alarm); first-line for motivated families; 70–80% cure rate; conditioning response; typically 8–12 weeks; most effective when combined with behavioral support - Desmopressin (DDAVP) — synthetic vasopressin analogue; reduces nighttime urine production; oral tablet (Nocurna for adults; standard DDAVP for children) or intranasal spray (less preferred due to hyponatremia risk); 60–70% response; use for short-term (camp, sleepovers) or as maintenance; hyponatremia risk — restrict fluid intake after dosing; monotherapy or combined with alarm - Combined alarm + desmopressin — superior to either alone for refractory enuresis; recommended for non-responders - Behavioral interventions — bladder training; timed voiding; constipation treatment; star charts - Treat constipation — constipation in 30–50% of enuretic children; must treat concurrent constipation - Anticholinergics — oxybutynin added to desmopressin for OAB component; helps ~30% of refractory cases

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Exclude organic causes — diabetes insipidus, diabetes mellitus, sleep apnea, spinal cord abnormality, urinary infection; urinalysis, glucose, and post-void residual Emerging  Mirabegron (beta-3 agonist) — combined with desmopressin for refractory nocturnal polyuria + OAB; pediatric trials; off-label
70	Hinman Syndrome (Non-neurogenic Neurogenic Bladder)	Acquired voiding dysfunction with features of neurogenic bladder but no neurological lesion; severe detrusor-sphincter dyssynergia, VUR, and renal damage	US: Rare; ~1% of children with severe voiding dysfunction referred to US tertiary centers; males predominate World: Rare globally; behavioral and psychological overlay on dysfunctional voiding; recognized in all healthcare settings Sources: Allen JAMA 1977; Hinman & Bauman J Urol 1973; Khen-Dunlop et al. J Urol 2006	Urodynamics (high-pressure dyssynergia); renal ultrasound; VCUG; MRI spine (exclude lesion); psychiatric/psychological assessment	 Standard of Care - CIC (clean intermittent catheterization) — essential for elevated post-void residual; prevents hydronephrosis; cornerstone - Anticholinergics — for OAB/detrusor overactivity component; oxybutynin; solifenacin - Biofeedback pelvic floor rehabilitation — central component; treats learned sphincter dyssynergia; typically 6–12 sessions - Psychosocial evaluation and therapy — Hinman-Allen syndrome has psychological overlay in many cases; child psychiatry/psychology referral; address family dynamics and anxiety - Constipation treatment — aggressive; MiraLax; bowel program - Urodynamics — essential for diagnosis confirmation and monitoring; high PVR, trabeculated bladder, absent neurological lesion - Renal monitoring — CKD possible from long-standing high-pressure bladder; ACEi/ARBs; renal function surveillance - Bladder augmentation — last resort for severe bladder damage despite maximum conservative therapy

TUBULOPATHIES

71	 Cystinosis — Nephropathic (CTNS)	Lysosomal cystine transport defect causing intracellular cystine accumulation; renal Fanconi syndrome in infancy progressing to ESKD by ~10 years untreated	US: ~1/100,000–200,000 in US; CTNS mutations; most common metabolic cause of renal Fanconi in children World: ~1/100,000–200,000 globally; higher in Brittany, France (1/26,000); Northern European founder mutation common Sources: Gahl et al. NEJM 2002; Town et al. Nat Genet 1998; Cystinosis Registry European Network	Leukocyte cystine measurement (elevated); CTNS genetic testing; slit-lamp (corneal crystals); kidney biopsy (cystine crystals); urine amino acids/phosphate/glucose (Fanconi)	 Standard of Care - Cysteamine (Cystagon — immediate release; Procysbi — delayed-release, twice-daily dosing) — cystine-depleting agent; dramatically slows progression to ESKD; initiated at diagnosis regardless of renal function; Procysbi FDA approved 2013 (improved tolerability, 12-hourly vs. 6-hourly) - Electrolyte and phosphate supplementation — for Fanconi syndrome (phosphate, potassium bicarbonate, sodium bicarbonate, vitamin D, carnitine) - Rickets management — phosphate + calcitriol - Renal transplant — at ESKD; does NOT cure extrarenal disease (cystine continues to accumulate in transplanted organs)
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#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Cysteamine eye drops — for corneal crystals; separate formulation required (Cystadrops — EU approved; compounded in US) - Multidisciplinary follow-up — ophthalmology (corneal crystals), endocrine (thyroid, growth, diabetes), neurology (swallowing, cognition), muscle Emerging/Investigational ☑ Gene therapy (lentiviral CTNS) — Phase 1/2 trials (UCLA; NCT03897361); autologous HSCT with gene-corrected cells; most advanced pipeline for cystinosis ☑ Rapamycin (sirolimus) — mTOR pathway dysregulated in cystinosis; preclinical/early clinical data ☑ Cysteamine prodrugs — improved CNS penetration for neurological cystinosis ☑ Stem cell therapy — investigational for muscle/swallowing manifestations
72	 Lowe Syndrome / Oculocerebrorenal Syndrome (OCRL)	X-linked OCRL1 mutations causing congenital cataracts, intellectual disability, and renal Fanconi syndrome; exclusively affects males	US: ~1/500,000 in US; X-linked; intellectual disability and cataracts distinguish from other Fanconi causes World: ~1/500,000 globally; X-linked; OCRL mutations; Fanconi syndrome onset in infancy; neurological involvement uniform Sources: Loi et al. Cell 1992; Bökenkamp & Ludwig Ped Nephrol 2011; Lowe Syndrome Association	OCRL genetic testing; urine amino acids/phosphate (Fanconi pattern); ophthalmology (cataracts, glaucoma); developmental assessment; leukocyte OCRL enzyme activity	 Standard of Care - Fanconi syndrome management — proximal tubular dysfunction; oral supplementation of: potassium (KCl), bicarbonate/citrate (Bicitra, Polycitra-K), phosphate, calcium, vitamin D; replace all wasted solutes - Citrate supplementation — prevents kidney stones (calcium oxalate/phosphate); Polycitra or potassium citrate - ACE inhibitors / ARBs — for proteinuria; partial effect; CKD progression management - Kidney transplant — for ESKD; renal disease alone is transplantable; extra-renal complications (intellectual disability, cataracts) persist; donor evaluation and informed consent around global prognosis - Ophthalmological — congenital cataracts present at birth; early surgical extraction (within weeks of birth); contact lens/glasses for aphakia; glaucoma (50%): topical beta-blockers, prostaglandins, trabeculectomy; lifelong ophthalmology - Neurological — intellectual disability universal; hypotonia; seizure management (anti-epileptics); developmental therapy; behavioral management (self-injurious behavior in ~50%) - Physical/occupational therapy — hypotonia management; motor development support - Nutrition — NG/GT feeding if swallowing difficulties; adequate caloric/protein intake
73	 Dent Disease Type 1 (CLCN5)	X-linked CLCN5 mutations causing low-molecular weight proteinuria, hypercalciuria,	US: Rare; ~1/30,000 males in US estimated; females carriers with mild proteinuria; ESKD in 30–80% of males	Urine LMW proteins (beta-2 microglobulin, retinol-binding protein); urinalysis (proteinuria, hematuria); urine calcium; renal	 Standard of Care High fluid intake — dilutes calcium; reduces stone formation; 2–3 L/m²/day; key preventive measure

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		nephrocalcinosis, and progressive CKD in males	World: Rare globally; CLCN5 mutations; males primarily affected; European registries report ESKD by 30–50 yrs in 30–80% Sources: Thakker RV Nephron 2000; Wrong et al. QJM 1994; Dent Disease International Registry	ultrasound (nephrocalcinosis); CLCN5 genetic testing	- Thiazide diuretics — hydrochlorothiazide; reduces urinary calcium excretion; first-line anti-hypercalciuric; monitor electrolytes (hypokalemia risk); combined with citrate supplementation - Potassium citrate — prevents calcium stone formation; alkalizes urine; supplement for metabolic acidosis component - Vitamin D and calcium — avoid excessive supplementation (worsens hypercalciuria); standard replacement only if deficient - Dietary sodium restriction — high sodium increases calcium excretion; reduce dietary NaCl - Phosphate supplementation — for hypophosphatemia when present; phosphate wasting variable in Dent 1 vs. 2 - ACE inhibitors — for proteinuria reduction; some benefit but limited evidence - Monitor urinary calcium, creatinine, phosphate, protein annually — guide supplementation - Kidney transplant — for ESKD (30–80% reach ESKD by adulthood); does not recur in graft Emerging - mTOR pathway — CLCN5 (CIC-5) regulates receptor-mediated endocytosis; mTOR inhibitors investigated in animal models; no clinical data
74	 Dent Disease Type 2 (OCRL)	OCRL mutations causing Dent disease phenotype (LMWP, hypercalciuria) with milder or absent Lowe syndrome neurological and ocular features	US: Rare; ~15% of all Dent disease from OCRL mutations; allelic with Lowe, milder phenotype World: Rare globally; OCRL mutations; milder than classic Lowe syndrome; distinguished by absence of neurological features Sources: Hoopes et al. Am J Hum Genet 2005; Bökenkamp & Ludwig Ped Nephrol 2011	Urine LMW proteins; urine calcium; OCRL genetic testing; ophthalmology (subtle lens changes); developmental assessment	Standard of Care - Same management as Dent Disease Type 1 — high fluid intake, thiazide diuretics, potassium citrate, dietary NaCl restriction - Distinguish from Lowe Syndrome — OCRL mutations cause both Dent 2 (mild) and Lowe syndrome (severe); Dent 2 has kidney involvement WITHOUT significant ocular or neurological features; genotype-phenotype correlation imperfect - Fanconi syndrome management — if proximal tubular dysfunction more prominent than Dent 1; bicarbonate, phosphate, potassium supplementation - Ophthalmological exam — even though Dent 2 lacks overt cataracts, subclinical lenticular changes possible; annual slit-lamp exam recommended - Kidney transplant — for ESKD; favorable outcomes
75	 Bartter Syndrome Type 1 (SLC12A1)	Antenatal/neonatal Bartter syndrome due to NKCC2 mutations; severe polyhydramnios, prematurity, hypokalaemic alkalosis, hypercalciuria	US: Rare; ~1/1,000,000 in US; most severe Bartter; antenatal polyhydramnios; nephrocalcinosis common World: ~1/1,000,000 globally; SLC12A1 mutations; antenatal Bartter; dehydration, failure to thrive in neonates	Urine/serum electrolytes (hypokalemia, metabolic alkalosis, elevated chloride excretion); renal ultrasound (nephrocalcinosis); genetic testing (SLC12A1)	Standard of Care - Indomethacin (or other COX inhibitor) — prostaglandin inhibition; reduces polyuria and electrolyte wasting; most effective single agent; start at ~0.5–2 mg/kg/day; monitor renal function and GI toxicity; limit long-term use - Potassium supplementation — oral KCl supplementation; may require high doses (2–4 mEq/kg/day)

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			Sources: Simon et al. Nat Genet 1996; Nozu et al. J Hum Genet 2017; Bartter Syndrome Consortium		- Aldosterone antagonists (spironolactone, eplerenone) — reduces potassium wasting; potassium-sparing; combined with KCl - Amiloride — potassium-sparing via ENaC block; alternative or adjunct - Sodium and fluid supplementation — especially in neonatal/antenatal Bartter; NG tube feeding; NaCl supplements - Magnesium supplementation — for hypomagnesemia if present (more common in Type 3) - ACEi/ARBs — reduces hyperaldosteronism-driven hypokalemia; adjunct therapy - Regular monitoring — serum electrolytes (K, Cl, Na, Mg), bicarb, renin, aldosterone; renal function; growth - Neonatal intensive care — most present with severe antenatal polyhydramnios + prematurity; NICU admission; IV electrolyte replacement before oral tolerated - Prostaglandin use timing — indomethacin initiated after stabilization; ibuprofen alternative if renal concerns - Outcome — medullary nephrocalcinosis common in all antenatal Bartter types; renal function typically preserved Emerging - Chloride channel replacement — SLC12A1 (NKCC2) gene; no gene therapy available - Vasopressin analogues — partial antidiuretic effect in some subtypes; investigational
76	 Bartter Syndrome Type 2 (KCNJ1)	ROMK channel mutations; antenatal Bartter syndrome with neonatal hyperkalemia (paradoxical) transitioning to hypokalemia after neonatal period	US: Very rare; antenatal onset; transient hyperkalemia in neonatal period; KCNJ1 mutations World: Very rare globally; KCNJ1 mutations; antenatal onset similar to Type 1; hyperkalemia distinguishes neonatal phase Sources: Simon et al. Cell 1996; Nozu et al. J Hum Genet 2017; Bartter Syndrome Consortium	Electrolytes (neonatal hyperkalemia → hypokalemia); antenatal/postnatal ultrasound; genetic testing (KCNJ1)	Standard of Care - Indomethacin (or other COX inhibitor) — prostaglandin inhibition; reduces polyuria and electrolyte wasting; most effective single agent; start at ~0.5–2 mg/kg/day; monitor renal function and GI toxicity; limit long-term use - Potassium supplementation — oral KCl supplementation; may require high doses (2–4 mEq/kg/day) - Aldosterone antagonists (spironolactone, eplerenone) — reduces potassium wasting; potassium-sparing; combined with KCl - Amiloride — potassium-sparing via ENaC block; alternative or adjunct - Sodium and fluid supplementation — especially in neonatal/antenatal Bartter; NG tube feeding; NaCl supplements - Magnesium supplementation — for hypomagnesemia if present (more common in Type 3) - ACEi/ARBs — reduces hyperaldosteronism-driven hypokalemia; adjunct therapy - Regular monitoring — serum electrolytes (K, Cl, Na, Mg), bicarb, renin, aldosterone; renal function; growth

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Neonatal hyperkalemia early — KCNJ1/ROMK mutations paradoxically cause neonatal hyperkalemia before evolving to hypokalemia in infancy; important to recognize and not treat the transient hyperkalemia aggressively - Otherwise identical management to Type 1 antenatal Bartter - Nephrocalcinosis monitoring — medullary; renal ultrasound annually
77	 Bartter Syndrome Type 3 (CLCNKB)	ClC-Kb chloride channel mutations; most clinically variable Bartter type; ranges from classic Bartter to Gitelman-like phenotype	US: ~1/500,000 in US; most common Bartter type; variable onset infancy to adolescence World: ~1/500,000 globally; CLCNKB mutations; most frequent Bartter variant; variable severity and age of onset Sources: Simon et al. Nat Genet 1997; Nozu et al. J Hum Genet 2017; Bartter Syndrome Consortium	Serum/urine electrolytes (hypokalemia, metabolic alkalosis, hypocalciuria variable); genetic testing (CLCNKB); renal ultrasound	Standard of Care - Indomethacin (or other COX inhibitor) — prostaglandin inhibition; reduces polyuria and electrolyte wasting; most effective single agent; start at ~0.5–2 mg/kg/day; monitor renal function and GI toxicity; limit long-term use - Potassium supplementation — oral KCl supplementation; may require high doses (2–4 mEq/kg/day) - Aldosterone antagonists (spironolactone, eplerenone) — reduces potassium wasting; potassium-sparing; combined with KCl - Amiloride — potassium-sparing via ENaC block; alternative or adjunct - Sodium and fluid supplementation — especially in neonatal/antenatal Bartter; NG tube feeding; NaCl supplements - Magnesium supplementation — for hypomagnesemia if present (more common in Type 3) - ACEi/ARBs — reduces hyperaldosteronism-driven hypokalemia; adjunct therapy - Regular monitoring — serum electrolytes (K, Cl, Na, Mg), bicarb, renin, aldosterone; renal function; growth - Classic presentation — hypokalemia, hypochloremia, metabolic alkalosis; typically diagnosed in infancy/early childhood; milder than antenatal forms - Hypomagnesemia more prominent — MgCl₂ or Mg oxide supplementation important - Indomethacin — mainstay; may need lifelong use; monitor renal function - Overlap with Gitelman syndrome — CLCNKB mutations can mimic Gitelman phenotype; genetic testing essential
78	 Bartter Syndrome Type 4 (BSND / CLCNKA+B)	Barttin or digenic CLCKA+B mutations causing severe antenatal Bartter syndrome with sensorineural hearing loss	US: Very rare; antenatal presentation; deafness distinguishes from other Bartter types; BSND mutations World: Very rare globally; BSND mutations (barttin subunit); deafness co-feature; most severe electrolyte imbalances	Electrolytes; audiogram (SNHL); genetic testing (BSND or both CLCNKA and CLCNKB); antenatal ultrasound	Standard of Care - Indomethacin (or other COX inhibitor) — prostaglandin inhibition; reduces polyuria and electrolyte wasting; most effective single agent; start at ~0.5–2 mg/kg/day; monitor renal function and GI toxicity; limit long-term use - Potassium supplementation — oral KCl supplementation; may require high doses (2–4 mEq/kg/day)

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			Sources: Birkenhager et al. Nat Genet 2001; Nozu et al. J Hum Genet 2017; Bartter Consortium		- Aldosterone antagonists (spironolactone, eplerenone) — reduces potassium wasting; potassium-sparing; combined with KCl - Amiloride — potassium-sparing via ENaC block; alternative or adjunct - Sodium and fluid supplementation — especially in neonatal/antenatal Bartter; NG tube feeding; NaCl supplements - Magnesium supplementation — for hypomagnesemia if present (more common in Type 3) - ACEi/ARBs — reduces hyperaldosteronism-driven hypokalemia; adjunct therapy - Regular monitoring — serum electrolytes (K, Cl, Na, Mg), bicarb, renin, aldosterone; renal function; growth - Sensorineural hearing loss — bilateral, severe/profound; early hearing aids; cochlear implant evaluation; deaf education planning; ENT audiology comanagement - Most severe antenatal Bartter form — combines renal (Bartter) + cochlear (BSND encodes barttin expressed in stria vascularis) manifestations - Early neonatal management — IV electrolytes; intensive supplementation; NG tube feeding - Long-term electrolyte management — chronic; significant supplementation required; compliance monitoring
79	 Bartter Syndrome Type 5 / Antenatal (CASR)	Activating CASR mutations causing neonatal severe hyperparathyroidism-like or Bartter-like syndrome with hypercalciuria and nephrocalcinosis	US: Very rare; CASR gain-of-function mutations; AD hypocalcemia distinguishable from Bartter phenotype World: Very rare globally; CASR activating mutations; heterozygous mutations cause autosomal dominant hypocalcemia type 1 Sources: Watanabe et al. Nat Genet 2002; Nozu et al. J Hum Genet 2017	Calcium, PTH, electrolytes; CASR genetic testing; renal ultrasound (nephrocalcinosis); urinary calcium	Standard of Care - Indomethacin (or other COX inhibitor) — prostaglandin inhibition; reduces polyuria and electrolyte wasting; most effective single agent; start at ~0.5–2 mg/kg/day; monitor renal function and GI toxicity; limit long-term use - Potassium supplementation — oral KCl supplementation; may require high doses (2–4 mEq/kg/day) - Aldosterone antagonists (spironolactone, eplerenone) — reduces potassium wasting; potassium-sparing; combined with KCl - Amiloride — potassium-sparing via ENaC block; alternative or adjunct - Sodium and fluid supplementation — especially in neonatal/antenatal Bartter; NG tube feeding; NaCl supplements - Magnesium supplementation — for hypomagnesemia if present (more common in Type 3) - ACEi/ARBs — reduces hyperaldosteronism-driven hypokalemia; adjunct therapy - Regular monitoring — serum electrolytes (K, Cl, Na, Mg), bicarb, renin, aldosterone; renal function; growth - Hypocalcemia management — CRITICAL: activating CaSR mutations cause hypocalcemia PLUS Bartter-like salt-wasting; calcium and vitamin D supplementation

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					carefully titrated; aggressive calcium supplementation risks nephrocalcinosis - Calcimimetics (cinacalcet) — CONTRAINDICATED in activating CASR mutations (would worsen hypocalcemia); important drug-disease interaction awareness - PTH monitoring — suppressed PTH is expected; do not treat with PTH analogs without specialist guidance - Calcitriol + calcium — judicious supplementation; avoid over-supplementation Emerging - CaSR antagonists (calcilytics) — investigational for activating CASR Bartter/hypocalcemia; NPS2143 and related compounds; Phase 2 research stage
80	 Gitelman Syndrome (SLC12A3)	Autosomal recessive SLC12A3 mutations causing thiazide-sensitive NaCl cotransporter defect; hypokalemia, hypomagnesemia, metabolic alkalosis, hypocalciuria	US: ~1/40,000 in US; most common inherited tubular disorder; adolescent presentation with muscle cramps, fatigue World: ~1/40,000 globally; SLC12A3 mutations; widespread in Southern European, Chinese, Taiwanese populations Sources: Simon et al. Nat Genet 1996; Gitelman Alliance data; Blanchard et al. KI 2015	Serum electrolytes (hypokalemia, hypomagnesemia, metabolic alkalosis); urinary calcium (low); renin/aldosterone; genetic testing (SLC12A3)	 Standard of Care - Magnesium supplementation — ESSENTIAL; hypomagnesemia universal and symptomatic (muscle cramps, fatigue, tetany); oral MgCl₂ or Mg oxide; divided doses to improve absorption; target serum Mg >0.6 mmol/L; IV Mg for severe symptomatic hypomagnesemia - Potassium supplementation — oral KCl; dietary potassium-rich foods; target K >3.0 mmol/L; chronic supplementation often required - Low dietary sodium — paradoxically reduces urinary potassium wasting via reduced distal delivery - Aldosterone antagonists (spironolactone, eplerenone) — potassium-sparing; reduces K wasting; effective adjunct; monitor for hyperkalemia - Amiloride — potassium-sparing; highly effective in Gitelman (blocks ENaC downstream); some experts prefer over spironolactone - Indomethacin — less effective than in Bartter; limited role; used occasionally for residual symptoms - RAAS blockade — ACEi/ARBs occasionally used for additional potassium sparing; limited evidence - Monitoring — serum K, Mg, creatinine, BP every 3–6 months; urine electrolytes annually <i>Important: Avoid medications worsening hypomagnesemia/hypokalemia — aminoglycosides, amphotericin B, cisplatin, prolonged diuretics</i> Emerging - Eplerenone — selective mineralocorticoid receptor antagonist; fewer anti-androgen side effects than spironolactone; increasingly preferred in adolescent males - Magnesium transport research — SLC12A3 (NCCT) loss-of-function; SLC41A3 Mg transporter involvement studied
81	Fanconi Syndrome — Idiopathic	Generalized proximal tubular dysfunction causing	US: Rare as isolated idiopathic entity in US; most cases have	Urine amino acids, glucose, phosphate, uric acid, LMW	 Standard of Care

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		glycosuria, aminoaciduria, phosphaturia, bicarbonaturia, and LMWP without identified cause	underlying cause (cystinosis, Dent, Lowe, drugs) World: Rare as primary globally; most cases secondary; heavy metals, chemotherapy, tenofovir major acquired causes Sources: Klootwijk et al. JASN 2015; Foreman Ped Nephrol 2019; Fanconi Syndrome review KI 2020	proteins, bicarbonate; serum electrolytes; exclude secondary causes; genetic evaluation	- Replace all wasted solutes — essential; guided by urine losses: - Oral bicarbonate/citrate (Polycitra, Bicitra) — metabolic acidosis; 1–3 mEq/kg/day - Oral phosphate — hypophosphatemia; 25–75 mg/kg/day in divided doses - Oral potassium — hypokalemia; KCl or Polycitra-K - Oral/IV sodium — hyponatremia - Vitamin D (calcitriol or alfacalcidol) — for impaired 1-hydroxylation and phosphate deficiency-related bone disease; rickets prevention/treatment - Calcium supplementation — with vitamin D - Hydration — polyuria management; adequate fluid intake - Find underlying cause — idiopathic is a diagnosis of exclusion; evaluate for cystinosis (the most common cause in children — CTNS gene testing; slit-lamp exam for crystals), Wilson disease, galactosemia, hereditary fructose intolerance, Lowe syndrome, drug toxicity (tenofovir, expired tetracycline, cisplatin) - Treat underlying etiology when identified — cystinosis → cysteamine (see #70); Wilson → penicillamine/zinc; HFI → fructose elimination
82	 Fanconi-Bickel Syndrome (SLC2A2)	SLC2A2 (GLUT2) mutations causing proximal tubular Fanconi syndrome plus glycogen accumulation in liver/kidney with hepatomegaly and growth retardation	US: Very rare; ~200 cases reported worldwide; AR; SLC2A2 (GLUT2) mutations; infancy onset World: Extremely rare globally; ~200 total cases; GLUT2 transport defect; hepatic and renal manifestations; diet management Sources: Santer et al. Nat Genet 1997; Oe et al. JIMD Rep 2011; Fanconi-Bickel Registry	Urine amino acids, glucose, phosphate (Fanconi pattern); blood glucose (postprandial hyperglycemia, fasting hypoglycemia); liver ultrasound; SLC2A2 genetic testing	Standard of Care - Galactose-restricted diet — galactose is GLUT2 substrate; restriction reduces accumulation; not fully effective but reduces hepatic glycogen accumulation - Uncooked cornstarch — prevents hypoglycemia between meals; slow-release glucose via GGCS pathway; given every 4–6 hours; same as GSD type I management - Fanconi syndrome replacement — oral bicarbonate, phosphate, potassium, vitamin D, calcium supplementation as for idiopathic Fanconi - Liver management — hepatomegaly with glycogen/galactose accumulation; monitor liver function; avoid hepatotoxins - Growth support — poor growth common; nutrition optimization; growth hormone if severe CKD develops - Monitor renal function — CKD progression variable; some patients develop ESKD; ACEi/ARBs for proteinuria
83	 Tyrosinemia Type I — Renal Fanconi (FAH)	FAH mutations causing hepatic and renal tubular dysfunction; succinylacetone accumulation causes Fanconi syndrome and hepatic failure	US: ~1/100,000–120,000 in US; most common in Quebec (~1/16,000); FAH mutations; liver failure prominent World: ~1/100,000–120,000 globally; highest in French-Canadian (Quebec) and Scandinavian populations	Urine succinylacetone (pathognomonic); urine amino acids (Fanconi pattern); plasma tyrosine; liver function; AFP; FAH genetic testing; newborn screening in many countries	Standard of Care - NTBC (nitisinone / Orfadin) — transforms disease; inhibits 4-hydroxyphenylpyruvate dioxygenase; virtually eliminates toxic metabolite accumulation; initiated immediately at diagnosis or NBS positive result; dramatically reduces need for liver transplant - Low tyrosine / low phenylalanine diet — essential adjunct to NTBC; prevents tyrosinemia-related neurological crises

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			Sources: Lindstedt et al. Lancet 1992; NTBC treatment trial; HT1 International Registry		- Renal Fanconi management — phosphate, bicarbonate, electrolyte supplementation while Fanconi active (usually resolves with NTBC) - Newborn screening — mandatory; early treatment prevents severe disease - Liver transplant — for NTBC non-responders, established HCC, or late diagnosis with cirrhosis; curative for hepatic and renal disease Emerging/Investigational - Gene therapy (mRNA or viral vector for FAH gene) — Phase 1/2 investigational programs; potentially curative without liver transplant - Base editing / prime editing — preclinical for FAH correction; highly precise - NTBC dose optimization — ongoing studies (lower doses may reduce ophthalmologic adverse effects)
84	 Hereditary Fructose Intolerance — Renal Fanconi (ALDOB)	ALDOB mutations causing fructose-1-phosphate accumulation; liver failure, hypoglycemia, and proximal renal tubular Fanconi syndrome after fructose exposure	US: ~1/20,000–30,000 in Europe (US less well characterized); Fanconi reversible with fructose restriction World: ~1/20,000–30,000 in European populations; ALDOB mutations; reversible with dietary fructose avoidance Sources: Bouteldja & Timson J Inherit Metab Dis 2010; Baker & Winegrad Lancet 1970	Urine amino acids, glucose, phosphate (after fructose); hypoglycemia after fructose; liver enzymes; ALDOB genetic testing; enzyme assay (liver biopsy)	Standard of Care - Strict dietary fructose/sucrose/sorbitol elimination — definitive and curative for renal Fanconi; complete elimination resolves or prevents renal tubular damage; sorbitol found in many medications and "sugar-free" products (hidden sources) - Dietitian support — careful food labeling; fructose-free diet education; hidden fructose sources (HFCS, fruit juices, certain vegetables) - Fanconi syndrome supplementation — if already established tubular damage: bicarbonate, phosphate, potassium, vitamin D, calcium (same as idiopathic Fanconi); improves with dietary elimination - Emergency management — hypoglycemia in acute fructose exposure; IV dextrose; glucagon - Medical alert bracelet — for IV fructose-containing solutions (avoid sucrose, sorbitol in IV medications) - Liver function monitoring — hepatomegaly with repeated exposures; LFTs; coagulopathy
85	 Renal Tubular Acidosis Type 1 — Distal (ATP6V1B1 / ATP6V0A4)	Defective H ⁺ excretion in distal tubule causing hyperchloremic normal anion gap metabolic acidosis, hypocitraturia, nephrocalcinosis, and rickets	US: ~1/100,000 in US; AR forms with deafness (ATP6V1B1); distal RTA causing nephrocalcinosis if untreated World: ~1/100,000 globally; AR forms more prevalent in consanguineous populations; sensorineural deafness subtype common in Middle East Sources: Karet et al. Nat Genet 1999; Alper JASN 2002; RTA Consensus Review Ped Nephrol 2021	Blood gas (hyperchloremic metabolic acidosis); urine pH (inappropriately alkaline >5.5 during acidosis); urinary citrate; renal ultrasound (nephrocalcinosis); genetic testing	Standard of Care - Oral alkali therapy — first-line and highly effective; potassium citrate (Polycitra-K, Urocit-K) preferred: treats acidosis AND replaces potassium AND prevents stones (citrate chelates calcium); dose 1–3 mEq/kg/day - Sodium bicarbonate — alternative if potassium overload concern; divided doses - Potassium supplementation — hypokalemia universal; potassium citrate provides both K⁺ and alkali; aggressive replacement in severe hypokalemia - Calcium supplementation + vitamin D — for metabolic bone disease (rickets); calcitriol for established rickets

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Stone prevention — citrate supplementation dramatically reduces nephrolithiasis and nephrocalcinosis; lifelong therapy - Audiological evaluation — ATP6V1B1 mutations associated with sensorineural hearing loss; audiogram at diagnosis and annually; hearing aids if needed - Monitoring — serum electrolytes (K, HCO₃, Cl), urine pH, urine calcium/creatinine ratio every 3–6 months; renal ultrasound annually for nephrocalcinosis <i>Key goal: maintain serum HCO₃ >22 mEq/L; urine pH should lower appropriately after ammonium chloride loading</i>
86	 Renal Tubular Acidosis Type 2 — Proximal (SLC4A4)	Defective bicarbonate reabsorption in proximal tubule; usually part of Fanconi syndrome; SLC4A4 mutations cause isolated proximal RTA with ocular anomalies	US: Rare in isolation in US; more common as part of Fanconi syndrome; SLC4A4 mutations rare World: Rare globally in isolation; SLC4A4 mutations cause isolated proximal RTA with ocular features (glaucoma, cataracts) Sources: Igarashi et al. Nat Genet 1999; Ota et al. JASN 2013; RTA Consensus Review Ped Nephrol 2021	Blood gas (metabolic acidosis); urine pH (appropriately acidic at nadir); fractional excretion of bicarbonate elevated; genetic testing; ophthalmology for SLC4A4	 Standard of Care - Large-dose alkali therapy — MUCH larger doses needed than dRTA; 5–15 mEq/kg/day; proximal tubule wastes bicarbonate → need high doses to exceed renal threshold; potassium citrate or sodium bicarbonate - Potassium supplementation — bicarb supplementation increases urinary K loss (alkalinization drives K excretion); supplement KCl or potassium citrate simultaneously - Thiazide diuretic — paradoxically reduces bicarbonate wasting by inducing volume contraction (increases proximal reabsorption); use cautiously with adequate K replacement - Phosphate supplementation — if hypophosphatemia present (especially in syndromic proximal RTA/Fanconi) - Treat underlying cause — isolated SLC4A4 mutations (rare); more commonly RTA2 part of Fanconi syndrome; diagnose and treat etiology - Vitamin D (calcitriol) — for bone disease/rickets component - Note: Urine pH paradoxically low in severe metabolic acidosis (below threshold); rises when alkali dose exceeds reabsorptive capacity
87	 Renal Tubular Acidosis Type 4 — Hyperkalemic	Aldosterone deficiency or resistance causing hyperchloremic metabolic acidosis with hyperkalemia; may be due to hyporeninism, adrenal disorders, or tubular resistance	US: Most common RTA type overall in US; common in pediatric CKD; rarely primary in children World: Most prevalent RTA globally in CKD context; primary form rare in children; obstructive uropathy and adrenal causes predominant Sources: Battle & Haque Kidney Int 2012; Campese & Karubian AJKD 1990; RTA review Ped Nephrol 2021	Blood electrolytes (hyperkalemia + metabolic acidosis); plasma renin, aldosterone; urine pH; screen for obstruction; adrenal function tests	 Standard of Care - Treat underlying cause — most common: congenital adrenal hyperplasia, obstructive uropathy (PUV, VUR), CKD, calcineurin inhibitor toxicity, hypoaldosteronism, pseudohypoaldosteronism - Fludrocortisone — for mineralocorticoid deficiency forms (hypoaldosteronism, CAH-related); 0.05–0.2 mg/day; increases renal sodium retention and potassium excretion; effective for low-aldosterone types - Dietary potassium restriction — reduce dietary K to 1–2 mEq/kg/day - Patiromer (Veltassa) or sodium zirconium cyclosilicate (Lokelma) — potassium binders; adult approved; pediatric use growing for refractory hyperkalemia; oral; well-tolerated

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Sodium bicarbonate — corrects metabolic acidosis; also lowers serum K transiently via redistribution - Loop diuretics (furosemide) — increases urinary potassium excretion; volume-replete patients - ACEi/ARB — AVOID or reduce dose; worsens hyperkalemia and low aldosterone state - Treat obstructive uropathy — relief of obstruction often resolves RTA4
88	 Primary Hyperoxaluria Type 1 (AGXT)	AGXT mutations causing hepatic overproduction of oxalate; oxalate deposition in kidneys and systemic organs causing nephrocalcinosis, nephrolithiasis, and ESKD	US: ~1–3/1,000,000 in US; ~1–2% of pediatric dialysis patients; infantile oxalosis rapidly fatal World: ~1–3/1,000,000 globally; higher in consanguineous populations; European OHO registry; most severe primary hyperoxaluria Sources: Hoppe et al. Nephrol Dial Transplant 2011; Cochat & Rumsby NEJM 2013; OHO Registry data	Urine oxalate (markedly elevated); plasma oxalate; AGXT genetic testing; renal ultrasound (nephrocalcinosis); kidney biopsy (oxalate crystals); liver enzyme assay	Standard of Care - Lumasiran (Oxlumo, Alnylam) — RNAi; inhibits HAO1 (glycolate oxidase); FDA approved Nov 2020 for PH1 in all ages including infants; reduces urinary oxalate by ~65%; subcutaneous monthly (loading) then quarterly; ILLUMINATE-A (adults/children) and ILLUMINATE-B (infants/children <6 yr) Phase 3 trials - High fluid intake — ≥ 3 L/m²/day; dilutes urine oxalate - Oral citrate (potassium citrate) — inhibits calcium oxalate crystallization - Vitamin B6 (pyridoxine) — for pyridoxine-responsive patients (~30%; AGXT p.Gly170Arg mutation); dose 5–20 mg/kg/day; test all patients at diagnosis - Combined liver-kidney transplant — definitive cure (corrects hepatic enzyme defect + replaces damaged kidney); increasingly replaced by lumasiran but still needed in advanced disease - Intensive hemodialysis — for established oxalosis to remove systemic oxalate deposits before transplant FDA-Approved (October 2023)  Nedosiran (Rivfloza, Novo Nordisk) — RNAi; inhibits LDHA (lactate dehydrogenase A); FDA approved Oct 2023 for PH1 (adults and children ≥ 9 yr) AND PH2; once-monthly subcutaneous injection; alternative RNAi mechanism to lumasiran; PHYOX2 Phase 3 trial Emerging/Investigational  Stirapentol — LDH inhibitor; early stage  Oxalobacter formigenes (probiotic) — intestinal oxalate degradation; Phase 2/3; not yet approved  Gene therapy (AAV-mediated AGXT correction) — preclinical/early clinical  Next-gen RNAi (LCPI-003) — investigational Standard of Care - High fluid intake — ≥ 3 L/m²/day - Oral citrate — inhibits calcium oxalate and calcium oxalate monohydrate (whewellite) crystallization - Dietary oxalate restriction — low-oxalate diet
89	 Primary Hyperoxaluria Type 2 (GRHPR)	GRHPR mutations causing glyoxylate reductase deficiency; less severe than PH1; oxalate and L-glycerate excretion elevated	US: Very rare; ~10% of PH cases; milder than PH1; ESKD less common World: Very rare globally; GRHPR mutations; European OHO Registry reports ~10% of PH cases; milder renal progression	Urine oxalate and L-glycerate; GRHPR genetic testing; renal ultrasound; plasma oxalate	Standard of Care - High fluid intake — ≥ 3 L/m²/day - Oral citrate — inhibits calcium oxalate and calcium oxalate monohydrate (whewellite) crystallization - Dietary oxalate restriction — low-oxalate diet

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			Sources: Cramer et al. J Biol Chem 1999; Hoppe et al. Nephrol Dial Transplant 2011		- Isolated kidney transplant — for ESKD; oxaluria improves post-transplant but glyoxylate/glycerate overproduction continues; recurrence risk lower than PH1 - Monitor renal function — frequent eGFR and 24-hour urinary oxalate assessments FDA-Approved (October 2023) - Nedosiran (Rivfloza, Novo Nordisk) — RNAi targeting LDHA; FDA approved Oct 2023 specifically for PH2 as well as PH1 (children ≥9 yr and adults); reduces urinary oxalate; once-monthly subcutaneous injection Emerging/Investigational - Lumasiran — mechanism targets HAO1 (upstream of GRHPR defect); potential role being explored but primary indication is PH1 - CRISPR/gene therapy for GRHPR — preclinical - Oxalobacter formigenes — investigational for all PH types
90	 Primary Hyperoxaluria Type 3 (HOGA1)	HOGA1 mutations causing mild hyperoxaluria; rarely leads to ESKD; mostly associated with recurrent nephrolithiasis	US: Rarest PH; HOGA1 mutations; nephrolithiasis rather than ESKD; mildest course World: Rarest PH type globally; HOGA1 mutations; milder phenotype; stone disease predominant; ESKD exceptional Sources: Belostotsky et al. Am J Hum Genet 2010; Hoppe et al. Nephrol Dial Transplant 2011	Urine oxalate; HOGA1 genetic testing; renal ultrasound; stone composition analysis	Standard of Care - High fluid intake — ≥3 L/m²/day; dilutes urine oxalate - Oral citrate (potassium citrate) — inhibits calcium oxalate stone formation - Dietary oxalate restriction — low-oxalate diet - Nephron-sparing urologic stone management — ureteroscopy, ESWL, PCNL as appropriate; avoid open surgery when possible - Monitor renal function — PH3 generally milder; ESKD less common than PH1/PH2 but possible - Vitamin B6 — not effective (HOGA1 pathway not B6-responsive) Emerging/Investigational - RNAi approaches targeting HOGA1 pathway — preclinical; no clinical trials as of 2025 - Nedosiran — mechanism via LDHA; theoretical relevance to PH3 pathway; not yet studied in PH3 - Increased recognition of PH3 phenotype — natural history studies ongoing to clarify ESKD risk
91	 Renal Glucosuria (SLC5A2)	Benign tubular glucose transport defect causing persistent glucosuria with normal blood glucose; SLC5A2 mutations (same as SGLT2)	US: ~0.2–0.6% of US population; benign in heterozygotes; homozygous SLC5A2 causes significant glycosuria World: ~0.2–0.6% globally; SLC5A2 mutations; related to SGLT2 transporter; benign; linked to naturally low blood glucose Sources: Santer & Calado JASN 2010; Kleta et al. Diabetes 2004	Urinalysis (persistent glucose with normal blood glucose); multiple urine samples; SLC5A2 genetic testing; exclude diabetes mellitus	Standard of Care - No specific treatment required — isolated benign renal glucosuria is a benign condition; no metabolic consequences; blood glucose is NORMAL (not diabetes) - Education and reassurance — emphasize normal blood glucose; not diabetes; benign prognosis; glucosuria detected incidentally on urinalysis - Differentiate from diabetes mellitus — blood glucose, HbA1c, fasting glucose normal; urine glucose elevated due to low renal threshold for glucose (SLC5A2 mutation)

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- SGLT2 inhibitors — SLC5A2 (SGLT2) is the pharmacological target of SGLT2 inhibitor drugs; benign renal glucosuria is essentially the natural SGLT2 inhibited phenotype; no therapy needed - Annual monitoring — urinalysis; confirm no proteinuria develops; monitor weight/growth; recheck blood glucose if clinical concern for diabetes - Genetic counseling — autosomal dominant or recessive; SLC5A2 mutations; spectrum of severity
92	 Nephrogenic Diabetes Insipidus — X-linked (AVPR2)	AVPR2 mutations causing resistance to ADH/vasopressin in collecting duct; massive polyuria, polydipsia, and hypernatremia in male infants	US: ~1/250,000 US males; AVPR2 X-linked; neonatal hypernatremia if unrecognized; high morbidity World: ~1/250,000 males globally; most cases in North America and Europe from founder mutations; AVPR2 gene Sources: Bichet et al. Nat Rev Nephrol 2009; Robben et al. JASN 2006; NDI Foundation Registry	Urine osmolality (low despite high serum osmolality); water deprivation test + desmopressin (no response); AVPR2 genetic testing; MRI (absent posterior pituitary bright spot)	Standard of Care - High fluid intake — unrestricted; free water access critical; prevents dehydration/hypernatremia; mainstay especially in infants - Amiloride — potassium-sparing diuretic; blocks ENaC; reduces polyuria by ~50%; preferred over hydrochlorothiazide alone; mechanism: increases intracellular cAMP via Na⁺ reduction - Hydrochlorothiazide (HCTZ) — paradoxical antidiuresis via volume contraction; often combined with amiloride; reduces urine output 30–50% - Indomethacin (or other COX inhibitor) — reduces prostaglandin-mediated antagonism of AVP; adds 30–50% additional reduction when combined with diuretics; monitor renal function and GI toxicity; limit long-term use - Low-salt, low-protein diet — reduces obligate solute load; reduces urine output - Newborn screening / early diagnosis — prevents hypernatremia-related intellectual disability; early hydration critical <i>Nomenclature Update (2022/2025): International expert consensus (Nat Rev Nephrol 2025; Levchenko et al.) recommends the term "arginine vasopressin resistance" (AVP-R) instead of "nephrogenic DI" to better reflect pathophysiology and reduce stigma</i> Emerging/Investigational - Pharmacological chaperones (AVPR2 antagonists as chaperones) — for class II misfolding mutations (90% of AVPR2 defects); tolvaptan-based chaperones show partial rescue in vitro; clinical development challenging due to competitive binding; relcovaptan (SR49059) was early candidate (halted due to CYP450 interactions); new chaperone candidates in development - Sildenafil (PDE5 inhibitor) — case reports of reduced urine output in children via cGMP-AQP2 pathway; adult studies show no consistent benefit; results conflicting - AVPR2 agonists — bypass misfolded receptor; activate downstream cAMP regardless of mutation class; preclinical data promising

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					▶ Gene therapy (AAV-AVPR2) — preclinical; X-linked inheritance means gene delivery to collecting duct principal cells; early stage ▶ Metformin / statins — activate AQP2 independently of AVP; preclinical data; not established clinically
93	 Nephrogenic Diabetes Insipidus — Autosomal (AQP2)	AQP2 mutations affecting aquaporin-2 water channel; autosomal recessive (most) or dominant; milder than X-linked AVPR2 NDI	US: Rare; AQP2 mutations in ~10% of hereditary NDI; autosomal recessive or dominant forms World: Rare globally; AQP2 mutations; both AR and AD inheritance; males and females equally affected Sources: Deen et al. Science 1994; Bichet et al. Nat Rev Nephrol 2009	Urine osmolality; water deprivation test + desmopressin (no response); AQP2 genetic testing; serum electrolytes	 Standard of Care • High fluid intake — unrestricted; free water access critical; prevents hypernatremic dehydration • Hydrochlorothiazide + amiloride combination — reduces urine output via volume contraction; same mechanism as X-linked NDI • Low-salt, low-protein diet — reduces solute load; reduces urine output • Indomethacin — adjunctive; reduces polyuria; monitor renal function; limit duration <i>Nomenclature Update: International expert consensus recommends "arginine vasopressin resistance" (AVP-R) over "nephrogenic DI" (Levtchenko et al., Nat Rev Nephrol 2025)</i> Emerging/Investigational ▶ Direct AQP2 channel activators — bypass both receptor and cAMP pathway; several small molecules identified in high-throughput screening; preclinical ▶ Gene therapy (AAV-AQP2) — autosomal recessive (AR) form has two defective copies; gene delivery more complex; early preclinical ▶ AQP2 trafficking enhancers — PGE2-EP4 agonists; secretin analogues; fluvastatin-secretin combination shown to increase AQP2 trafficking in mouse models; investigational ▶ mRNA therapy for AQP2 — hepatic or renal-targeted delivery; preclinical ▶ CRISPR/base editing — potentially curative for specific AQP2 mutations; very early
94	 Pseudohypoaldosteronism Type 1 — Renal (NR3C2)	Autosomal dominant NR3C2 mutations causing renal mineralocorticoid resistance; salt-wasting nephropathy with elevated renin/aldosterone	US: Rare; renal type milder than systemic; neonatal salt-wasting presentation World: Rare globally; NR3C2 mutations; mineralocorticoid receptor insensitivity; systemic (SCNN1B/G) form more severe Sources: Geller et al. NEJM 1998; Chang et al. Nat Genet 1996; Pseudohypoaldosteronism review JASN	Serum electrolytes (hyponatremia, hyperkalemia); markedly elevated renin and aldosterone; urinary sodium (salt wasting); NR3C2 genetic testing	 Standard of Care • High-dose sodium supplementation — massive salt wasting; NaCl 10–20 mEq/kg/day; NG supplementation in infants; often challenging; gradual reduction as patients mature and some improvement occurs with age (renal form improves; systemic form does not) • Renal form (NR3C2 heterozygous) — autosomal dominant; milder; improves with age; aldosterone supplementation sometimes used; NR3C2 is renal mineralocorticoid receptor • High-dose fludrocortisone — INEFFECTIVE in renal PHA1 (receptor is non-functional); avoid; only useful in adrenal insufficiency (secondary PHA)

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Monitor electrolytes — hyperkalemia, hyponatremia, metabolic acidosis; frequent monitoring in infancy; normalize by 1–2 years of age in renal form - Indomethacin — reduces urinary sodium wasting in some reports; adjunct - Differentiate from systemic PHA1 (SCNN1B/G mutations) — systemic form affects lung and sweat glands in addition to kidney; does not improve with age; see entry #93 systemic
95	 Pseudohypoaldosteronism Type 2 / Gordon Syndrome (WNK1/4)	WNK1/4 or KLHL3/CUL3 mutations causing chloride shunting in distal nephron; hypertension, hyperkalemia, metabolic acidosis with low renin/aldosterone	US: Rare; familial hyperkalemic hypertension; WNK1/WNK4 mutations; childhood to adulthood presentations World: Rare globally; WNK kinase pathway mutations; European kindreds well-described; pseudohypoaldosteronism type 2 Sources: Wilson et al. Science 2001; Boyden et al. Science 2012; Gordon Syndrome Registry	Serum electrolytes (hyperkalemia, metabolic acidosis); blood pressure elevated; low renin and aldosterone; WNK1/WNK4/KLHL3/CUL3 genetic testing	Standard of Care - Thiazide diuretics — HIGHLY EFFECTIVE and essentially curative; hydrochlorothiazide 1–3 mg/kg/day; thiazides inhibit NCC (the sodium-chloride cotransporter that is constitutively overactivated by WNK1/4 mutations); normalizes hyperkalemia, metabolic acidosis, and hypertension rapidly - Dietary potassium restriction — reduces hyperkalemia while establishing thiazide therapy - Dietary sodium restriction — adjunct; reduces chloride load - Monitoring — electrolytes normalize rapidly on thiazides; titrate dose to maintain normal K, HCO₃, and BP - Note: Fludrocortisone is INEFFECTIVE (PHA2 is caused by excessive NCC activity, not aldosterone deficiency) - Genetic counseling — WNK1, WNK4, KLHL3, CUL3 mutations; autosomal dominant; family screening
96	 Liddle Syndrome (SCNN1B/G)	Gain-of-function mutations in ENaC beta/gamma subunits causing constitutive sodium reabsorption; severe hypertension, hypokalemia, low renin/aldosterone in adolescents	US: Rare; important monogenic hypertension cause in US; autosomal dominant; SCNN1B/SCNN1G World: Rare globally; autosomal dominant; ENaC gain-of-function; family histories of severe early hypertension and hypokalemia Sources: Liddle et al. Trans Assoc Am Physicians 1963; Shimkets et al. Cell 1994; Botero-Velez et al. NEJM 1994	Blood pressure (severe hypertension); electrolytes (hypokalemia); low renin and aldosterone; SCNN1B/G genetic testing; genetic family screening	Standard of Care - Amiloride — HIGHLY EFFECTIVE; directly blocks ENaC (epithelial sodium channel, the upregulated channel in Liddle); normalizes hypertension, hypokalemia, and metabolic alkalosis; 5–20 mg/day; first-line drug - Triamterene — alternative potassium-sparing diuretic; blocks ENaC directly; similar mechanism to amiloride; less commonly used - Low dietary sodium — reduces sodium retention and BP - Spironolactone/eplerenone — INEFFECTIVE in Liddle (spironolactone is an aldosterone antagonist; Liddle has low aldosterone; ENaC is overactive independent of aldosterone; spironolactone does not block ENaC directly); common diagnostic pitfall - Monitoring — BP, serum K, HCO₃ normalize with amiloride; follow lifelong - Genetic counseling — SCNN1B or SCNN1G mutations; autosomal dominant; gain-of-function ENaC

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
97	 Apparent Mineralocorticoid Excess (HSD11B2)	HSD11B2 mutations causing cortisol to activate mineralocorticoid receptors; severe hypertension, hypokalemia, low renin/aldosterone; cortisol acts as mineralocorticoid	US: Rare; autosomal recessive; severe childhood hypertension with low renin/aldosterone; HSD11B2 mutations World: Rare globally; HSD11B2 mutations; acquired form from licorice ingestion more common than genetic; Mediterranean distribution Sources: White et al. Lancet 1997; Mune et al. Nat Genet 1995; AME Syndrome Registry	Blood pressure; electrolytes (hypokalemia); low renin/aldosterone; urine cortisol:cortisone ratio (elevated); HSD11B2 genetic testing; exclude licorice use	Standard of Care - Spironolactone / eplerenone — competitive mineralocorticoid receptor antagonism; blocks cortisol-driven mineralocorticoid receptor activation; reduces hypertension and hypokalemia; first-line pharmacotherapy - Low dietary sodium + high dietary potassium — adjunct BP control - Dexamethasone — suppresses ACTH-driven cortisol production; reduces cortisol available for receptor binding; used in some protocols - Amiloride — alternative if spironolactone not tolerated; direct ENaC block - Carbenoxolone — inhibits 11β-HSD2 (the deficient enzyme); experimental and not recommended due to toxicity - Kidney transplant — for ESKD; 11β-HSD2 expressed in kidney; transplant corrects the metabolic defect; hypertension resolves post-transplant; favorable outcomes - Monitoring — BP, K, HCO₃, plasma renin, aldosterone (both low), cortisol/cortisone ratio in urine (elevated ratio diagnostic); serial renal function - Genetic counseling — HSD11B2 mutations; autosomal recessive; homozygous/compound heterozygous

TUBULOINTERSTITIAL DISORDERS

98	Acute Tubulointerstitial Nephritis (ATIN) — Drug-induced	Immune-mediated tubular and interstitial inflammation triggered by medications (NSAIDs, antibiotics, PPIs, anticonvulsants, checkpoint inhibitors); AKI	US: ~10–15% of biopsied AKI in US; antibiotics and NSAIDs most common; checkpoint inhibitors emerging World: ~10–15% of biopsied AKI globally; drug-induced predominant in high-income countries; infection-related in resource-limited settings Sources: Praga & González JASN 2010; Moledina & Perazella JASN 2017; ERA-EDTA ATIN data	Kidney biopsy (interstitial infiltration with lymphocytes/eosinophils); eosinophiluria (variable); urinalysis (white cell casts); drug exposure history; eGFR trend	Standard of Care - Identify and discontinue causative drug — CRITICAL; NSAIDs, beta-lactam antibiotics, PPIs, sulfonamides, vancomycin, checkpoint inhibitors (anti-PD1/CTLA4 — increasingly recognized); drug withdrawal is first and most important step; renal recovery often begins within days–weeks - Corticosteroids — prednisone 1 mg/kg/day (max 60 mg) for 4–8 weeks then taper; most evidence from observational studies; AKI Collaborative recommends steroids for moderate-severe ATIN (creatinine >2$\times$ baseline) if no improvement 5–7 days after drug withdrawal; some evidence for checkpoint inhibitor-ATIN - Supportive care — IV hydration if dehydrated; electrolyte correction; avoid further nephrotoxins - Renal biopsy — for diagnosis confirmation, severity assessment, and treatment decisions especially if unclear etiology or no recovery - Monitor renal recovery — eGFR, urinalysis, urine eosinophils (variable sensitivity); follow-up 3 months post-recovery Emerging
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#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					 Checkpoint inhibitor nephritis — anti-PD1/CTLA4 → ATIN increasingly common; steroid + mycophenolate for refractory cases; rituximab case reports; guidance from IKIUS 2023 consensus
99	 Acute Tubulointerstitial Nephritis — Immune / Sarcoid-associated	Immune-mediated ATIN in absence of drug trigger; may be associated with sarcoidosis, SLE, Sjögren, IgG4-related disease, or idiopathic	US: Rare in children in US; sarcoidosis-associated ATIN more common in adolescents; IgG4-RD rare in pediatrics World: Rare globally in children; sarcoidosis prevalence varies geographically; IgG4-related disease predominantly adult Sources: Drent et al. Lancet 2013; Jayne & Rasmussen Semin Nephrol 2014	Kidney biopsy; chest imaging (sarcoidosis); serum ACE; ANA, anti-Ro/La; serum IgG4; calcium (hypercalcemia in sarcoidosis); calcium in urine	Standard of Care - Identify and treat underlying condition — systemic sarcoidosis, anti-TBM antibody disease, IgG4-related kidney disease, SLE, Sjögren syndrome; disease-specific therapy - Corticosteroids — prednisone 0.5–1 mg/kg/day (max 60 mg); 3–6 months tapering course; cornerstone for sarcoid-ATIN, anti-TBM, IgG4-RKD - IgG4-related kidney disease — rituximab increasingly used for steroid-sparing or refractory IgG4-RKD; excellent response (>90% remission); IgG4 levels monitor response - Hydroxychloroquine — sarcoidosis adjunct; steroid-sparing; monitor ophthalmological toxicity - ACEi/ARBs — for proteinuria component and CKD management - Renal biopsy — diagnostic and guides treatment intensity - Monitor IgG4 levels, ACE level, calcium (sarcoid); CBC; serial renal function
100	 TINU Syndrome (Tubulointerstitial Nephritis and Uveitis)	Autoimmune condition causing acute TIN + bilateral uveitis; predominantly affects adolescent females; modified CRP-peptide antibodies pathogenic	US: Rare; ~1–5% of TIN in US; median age ~15 years; females predominate; uveitis may precede nephritis World: Rare globally; ~1–5% of TIN series worldwide; HLA association; relapsing uveitis common; TINU Consortium data Sources: Mandeville et al. Am J Ophthalmol 2001; Mackensen et al. JASN 2007; TINU Consortium 2020	Kidney biopsy (interstitial nephritis); ophthalmology (bilateral anterior uveitis); anti-modified CRP antibodies; urinalysis; eGFR	Standard of Care - Corticosteroids (systemic) — prednisone 1 mg/kg/day; required for most TINU (especially bilateral panuveitis); 3–6 month course with taper; renal ATIN responds more readily than ocular component - Topical ophthalmological treatment — corticosteroid eye drops (prednisolone acetate); mydriatics (tropicamide, cyclopentolate) for posterior synechiae prevention; initial treatment for anterior uveitis - Ophthalmology co-management — CRITICAL; uveitis may persist or relapse independently of renal disease; can precede or follow renal disease; regular follow-up slit-lamp exams - Systemic immunosuppression for refractory uveitis — methotrexate, mycophenolate mofetil, azathioprine; used for steroid-sparing in chronic recurrent uveitis - Uveitis-specific biologics — adalimumab (HUMIRA) FDA approved 2016 for non-infectious intermediate/posterior uveitis in adults; increasingly used in pediatric TINU; anti-TNF therapy for refractory cases - Renal monitoring — urinalysis, eGFR, beta-2 microglobulin (tubular marker) to monitor renal recovery
101	Chronic Tubulointerstitial Nephritis	Progressive interstitial fibrosis and tubular atrophy from multiple etiologies	US: Common end-stage pathway in US; aristolochic acid and heavy	Kidney biopsy (tubular atrophy, interstitial fibrosis); eGFR trend; urinalysis; renal	Standard of Care

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		(chronic obstruction, chronic infections, analgesics, Chinese herbs, heavy metals, genetic); leads to CKD	metal nephropathy uncommon in US children World: Common end-stage pathway globally; aristolochic acid nephropathy major issue in Asia (Chinese herbs); heavy metals in industrial settings Sources: De Broe Kidney Int 2012; Pozdzik et al. Nephrol Dial Transplant 2013; ERA-EDTA ATIN data	ultrasound (small kidneys); identify and remove cause; heavy metal screening	- Identify and treat underlying etiology — most important step; causes: analgesic nephropathy (NSAIDs), lithium toxicity, calcineurin inhibitor toxicity, chronic reflux nephropathy, obstructive uropathy, sickle cell, radiation nephritis, idiopathic - Remove offending agent — NSAID cessation, lithium reduction/substitution (valproate/quetiapine); CNI dose reduction; often prevents further progression - CKD management per KDIGO — ACEi/ARBs for proteinuria; BP control; anemia management (EPO); CKD-MBD management; dietary modifications - Electrolyte monitoring — tubulointerstitial disease causes tubular dysfunction; RTA may develop; monitor K, HCO₃, phosphate, sodium; supplement as needed - Dialysis and transplant — for ESKD; no disease recurrence unless etiology not corrected (e.g., recurrent obstruction, active autoimmune) - Specific: Aristolochic acid nephropathy — irreversible; no treatment; avoid herbal supplements containing aristolochic acid

NEPHROLITHIASIS / NEPHROCALCINOSIS

102	 Hypercalciuria — Idiopathic	Excessive urinary calcium excretion without identifiable systemic cause; most common metabolic cause of nephrolithiasis in children	US: ~2–4% of US children; accounts for ~40–50% of calcium oxalate stones; familial in ~50% World: ~2–4% globally in children with urolithiasis; genetic predisposition variable; dietary calcium intake modifies expression Sources: Moe & Pearle Nat Rev Nephrol 2011; Penido & Ukarapol Ped Nephrol 2012; ROKS Consortium	24-hour or spot urine calcium:creatinine ratio; serum calcium, PTH, vitamin D; renal ultrasound; stone composition; genetic evaluation if refractory	Standard of Care - High fluid intake — dilutes urinary calcium; 2–3 L/m²/day; plain water preferred; first and most important measure - Dietary sodium restriction — high sodium intake drives urinary calcium excretion via NHE3 competition; <2 g/day sodium; reduces calcium excretion significantly - Dietary calcium — NORMAL intake recommended; calcium restriction paradoxically INCREASES calcium oxalate stone formation (reduces intestinal oxalate binding); do not restrict dairy - Thiazide diuretics (hydrochlorothiazide 0.5–1 mg/kg/day) — reduces urinary calcium reabsorption in distal tubule; first-line pharmacotherapy for symptomatic hypercalciuria; highly effective; combined with potassium citrate to prevent hypokalemia - Potassium citrate — prevents stones by chelating calcium; alkalinizes urine; reduces calcium oxalate and calcium phosphate crystallization; Urocit-K, Polycitra-K - Dietary oxalate restriction — for calcium oxalate stones; reduce spinach, nuts, chocolate, beets - Protein moderation — high animal protein increases urinary calcium and uric acid; moderate restriction Emerging  Genetic hypercalciuria — CLDN14, CASR variants in some familial cases; precision management based on genotype
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#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
103	 Cystinuria (SLC3A1 / SLC7A9)	Defective renal and intestinal cystine transporter causing cystine stones; recurrent large stones, staghorn calculi, and progressive CKD	US: ~1/7,000 in US; ~5–7% of pediatric kidney stones; autosomal recessive; SLC3A1 (type A) and SLC7A9 (type B) World: ~1/7,000 globally; higher in Libyan Jewish and Spanish populations; urological complications common Sources: Dello Strologo et al. JASN 2002; Calonge et al. Nat Genet 1994; Cystinuria Consortium	Urine cystine screening (nitroprusside test); 24-hour urine cystine quantification; stone analysis (hexagonal crystals); genetic testing	Standard of Care - High fluid intake — CRITICAL; urine output >3 L/m²/day; dramatically reduces cystine supersaturation; cornerstone of management; distribute throughout day and night (nocturia/night drinking important) - Urinary alkalinization — potassium citrate; target urine pH 7.0–7.5; cystine solubility dramatically increases above pH 6.5; essential adjunct - Dietary sodium restriction — reduces cystinuria; sodium promotes renal cystine excretion - Dietary methionine/protein restriction — reduces cystine production; moderate animal protein restriction - D-penicillamine — thiol agent; forms mixed disulfide with cystine (more soluble); 2nd line due to toxicity (nephrotic syndrome, lupus-like, bone marrow); vitamin B6 co-supplementation; use for refractory cases - Tiopronin (Thiola) — thiol agent; better tolerated than D-penicillamine; forms cysteine-tiopronin disulfide; first-choice thiol agent; doses 10–50 mg/kg/day - Captopril — thiol group; reduces cystinuria; modest effect; used adjunctively - Extracorporeal shock wave lithotripsy (ESWL) — less effective for cystine stones (hard); ureteroscopy preferred; PCNL for large staghorn calculi - Genetic counseling — SLC3A1 (type A) or SLC7A9 (type B) mutations; autosomal recessive
104	Enteric Hyperoxaluria	Secondary hyperoxaluria from increased intestinal oxalate absorption in fat malabsorption (short bowel syndrome, IBD, bariatric surgery)	US: Common in pediatric short bowel syndrome in US; increasingly recognized in pediatric IBD World: Common globally in short bowel syndrome and malabsorptive conditions; fat malabsorption key mechanism Sources: Asplin & Coe Kidney Int 1999; Kumar et al. Am J Kidney Dis 2011	24-hour urine oxalate (elevated but less than PH1/2); serum oxalate; stool fat; intestinal workup; stone analysis	Standard of Care - Treat underlying GI disease — Crohn disease (medical therapy, bowel resection), short bowel syndrome, bariatric surgery (Roux-en-Y, sleeve gastrectomy); reducing fat malabsorption reduces oxalate absorption - Dietary oxalate restriction — reduce spinach, nuts, beets, chocolate, wheat bran; low-oxalate diet - Dietary fat restriction — excess fat saponifies calcium, leaving oxalate free for absorption; low-fat diet critical - Oral calcium carbonate with meals — binds dietary oxalate in gut; prevents absorption; give WITH food not separately; 500–2000 mg with each meal - High fluid intake — dilutes urinary oxalate; target urine output >2 L/day - Potassium citrate — urinary alkalinization; reduces calcium oxalate crystallization - Cholestyramine — bile acid sequestrant; reduces oxalate absorption in fat malabsorption states; adjunct - Probiotics (Oxalobacter formigenes) — gut bacteria that degrade oxalate; investigational in clinical trials; not commercially available

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					Pyridoxine (B6) — reduces endogenous oxalate synthesis; modest effect in enteric hyperoxaluria (larger effect in PH1)
105	 Hyperuricosuria — Pediatric	Elevated uric acid excretion causing uric acid or mixed uric acid-calcium stones; associated with high animal protein intake, enzyme defects, or uricosuric drugs	US: ~5–10% of pediatric kidney stones in US; uric acid stones uncommon in children; Lesch-Nyhan association World: ~5–10% of pediatric stones globally; uric acid stones more common in warm climates and with high protein intake Sources: Lesch & Nyhan Am J Med 1964; Moe et al. Clin J Am Soc Nephrol 2011; EAU Urolithiasis Guidelines	24-hour urine uric acid; serum uric acid; urine pH (acidic favors uric acid stones); HPRT enzyme activity; stone analysis; dietary history	Standard of Care - High fluid intake — dilutes uric acid in urine; reduces supersaturation; 2–3 L/m²/day - Dietary purine restriction — reduce meat, organ meats, shellfish, legumes; moderate animal protein - Urinary alkalinization — potassium citrate; target urine pH 6.0–6.5; uric acid stones dissolve in alkaline urine; caution not to over-alkalinize (increases calcium phosphate risk) - Allopurinol — xanthine oxidase inhibitor; reduces uric acid production; for persistent symptomatic hyperuricosuria unresponsive to dietary measures; 10 mg/kg/day; monitor CBC and liver function - Febuxostat — alternative XO inhibitor; adult data; limited pediatric data; used if allopurinol intolerant - Diagnose secondary causes — Lesch-Nyhan syndrome, glycogen storage disease, tumor lysis syndrome, high purine diet, hemolytic anemia; treat etiology
106	 Xanthinuria / Xanthine Stones (XDH)	Xanthine oxidase deficiency (XDH mutations) causing hypouricemia and xanthinuria; xanthine stones (radiolucent); rarely causes AKI	US: Very rare; ~150 cases worldwide; autosomal recessive; XDH deficiency; xanthine stones World: Very rare globally; ~150 total reported cases; XDH mutations; type I isolated, type II combined with sulfite oxidase deficiency Sources: Ichida et al. J Clin Invest 1997; Simmonds et al. J Inherit Metab Dis 1995	Serum uric acid (very low); urine xanthine and hypoxanthine; stone analysis (xanthine stones); XDH genetic testing; SO activity (type II)	Standard of Care - High fluid intake — dilutes xanthine in urine; prevents xanthine crystallization; 2–3 L/m²/day; cornerstone - Low purine diet — reduces xanthine production; limit meats, organ meats, shellfish; vegetarian or low-purine diet - Avoid allopurinol — allopurinol CONTRAINDICATED in xanthinuria Type I (XDH deficiency); increases xanthine accumulation; worsens xanthinuria; important drug-disease interaction - Urinary alkalinization — potassium citrate; increases xanthine solubility; target urine pH 7.0+ - Monitor renal function — xanthine stones can cause obstructive nephropathy; renal ultrasound annually - Genetic counseling — XDH (xanthine dehydrogenase) mutations; autosomal recessive; usually benign if dietary measures followed - Type II xanthinuria — combined xanthine + sulfite oxidase deficiency; more severe; neurological involvement; sulfite restriction; no specific therapy
107	 2,8-Dihydroxyadenine Urolithiasis (APRT deficiency)	Adenine phosphoribosyltransferase deficiency causing 2,8-DHA crystalluria; stone formation and AKI from crystal deposits in kidney	US: Rare globally; ~1/50,000–100,000; highly prevalent in Iceland (~1/1,000); often misdiagnosed as uric acid stones World: ~1/50,000–100,000 globally; Icelandic founder mutation well-	Urine microscopy (reddish-brown 2,8-DHA crystals); urine amino acids; APRT enzyme activity (erythrocytes); APRT genetic testing; stone analysis	Standard of Care Allopurinol — ESSENTIAL and highly effective; inhibits xanthine oxidase → reduces production of 2,8-DHA by blocking adenine conversion; normalizes urinary 2,8-DHA; must be started as soon as diagnosis made and continued lifelong; dose 10–20 mg/kg/day; prevents stone

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			studied; 2,8-DHA crystals on urine microscopy pathognomonic Sources: Bollée et al. Nephrol Dial Transplant 2012; Edvardsson et al. KI 1993; APRT Deficiency Registry		recurrence and CKD progression; delayed diagnosis leads to irreversible CKD - High fluid intake — dilutes 2,8-DHA; 2–3 L/m²/day; adjunct to allopurinol - Dietary restriction of adenine — reduce organ meats, red meats; low-purine diet adjunct - Febuxostat — alternative to allopurinol if intolerant; reduces DHA production; use with monitoring - Urinary alkalinization — limited benefit (2,8-DHA has low solubility across all pH ranges); potassium citrate as adjunct - Dialysis and transplant — for ESKD from DHA nephropathy; allopurinol continues post-transplant to prevent graft DHA crystalline nephropathy - Genetic counseling — APRT mutations; autosomal recessive; most common in Japan and Iceland
108	Struvite Urolithiasis (infection stones)	Magnesium ammonium phosphate stones caused by urease-producing organisms (Proteus, Klebsiella) alkalinizing urine; associated with recurrent UTIs and structural anomalies	US: ~5–10% of pediatric stones in US; recurrent UTI and neurogenic bladder predispose; staghorn calculi World: ~5–10% globally; Proteus mirabilis and urease-producing organisms predominant; higher in developing world with untreated UTI Sources: Griffith & Osborne Kidney Int 1987; EAU Urolithiasis Guidelines 2022	Stone analysis (struvite composition); urine culture; renal imaging (CT preferred for stone mapping); urodynamics if neurogenic bladder suspected	Standard of Care - Complete stone removal — CRITICAL; struvite stones harbor bacteria within the stone matrix; antibiotic therapy alone cannot sterilize stone; percutaneous nephrolithotomy (PCNL) preferred for large/staghorn struvite stones; ureteroscopy for ureteral stones; ESWL less effective for large struvite - Antibiotic therapy — culture-directed; treat before and after stone manipulation; long-term suppressive antibiotics post-procedure to prevent recurrence if predisposing conditions persist - Identify and treat predisposing conditions — neurogenic bladder (CIC + anticholinergics); foreign body (catheter, stent); anatomical obstruction; VUR; underlying UTI source - Urease inhibitor (acetohydroxamic acid, AHA) — inhibits bacterial urease; reduces struvite stone growth; significant toxicity (hemolytic anemia, DVT); rarely used in pediatrics; adults with recurrent staghorn calculi - Bladder catheter management — change catheters regularly; maintain sterile technique - Irrigation/dissolution — urinary alkalization; Renacidin irrigation through nephrostomy tube (PCN drain) for residual stone dissolution - Prevention — adequate hydration; urine culture surveillance; treat bacteriuria promptly
109	Nephrocalcinosis — Prematurity / Furosemide-associated	Calcium deposits in renal parenchyma in premature infants receiving furosemide; hypercalciuria from immature tubular function and diuretic use	US: ~10–40% of premature US infants receiving furosemide; often resolves; risk with repeated dosing World: ~10–40% of premature infants globally receiving	Renal ultrasound (echogenic medullary pyramids); urine calcium:creatinine ratio; serum calcium, phosphate; review furosemide exposure	Standard of Care Minimize furosemide exposure — furosemide is primary cause of nephrocalcinosis in premature infants; use thiazide diuretics instead when chronic diuretic needed for CLD; furosemide → hypercalciuria → medullary nephrocalcinosis

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			furosemide; consistent rates across Western neonatal ICUs Sources: Jacinto et al. Pediatrics 2000; Hufnagle et al. Pediatrics 1982; Neonatal nephrocalcinosis review		- Switch to chlorothiazide/hydrochlorothiazide — for chronic diuretic requirement; significantly reduces hypercalciuria; prevents nephrocalcinosis progression - Spirololactone — combined with thiazide; potassium-sparing; used in BPD management - Adequate calcium and phosphate nutrition — prevent metabolic bone disease of prematurity - High fluid intake — dilutes urinary calcium; adequate hydration; 150–180 mL/kg/day in NICU - Vitamin D — standard supplementation; avoid over-supplementation (hypercalcemia/hypercalciuria) - Serial renal ultrasound — monitor nephrocalcinosis resolution (most resolves by 6–12 months) - Monitor renal function — occasional CKD if extensive nephrocalcinosis; annual creatinine surveillance
110	 Distal RTA-associated Nephrocalcinosis	Nephrocalcinosis and nephrolithiasis as complication of untreated or undertreated distal (Type 1) RTA; hypocitraturia + alkaline urine promote calcium phosphate deposition	US: Common complication when dRTA is unrecognized in US; nephrocalcinosis distinguishes Type 1 from others World: Common globally when dRTA is untreated; medullary nephrocalcinosis on imaging key distinguishing feature Sources: Alper JASN 2002; Karet et al. Nat Genet 1999; RTA and Nephrocalcinosis review Ped Nephrol	Renal ultrasound; urine citrate (low); urine calcium; blood gas; urine pH; identify underlying dRTA cause	Standard of Care - Same as Distal RTA Type 1 treatment — potassium citrate is cornerstone; treats acidosis AND prevents nephrocalcinosis progression AND prevents new stone formation - Potassium citrate (Polycitra-K, Urocit-K) — 1–3 mEq/kg/day; long-term; prevents calcium crystallization; most important intervention - Early and sustained alkali therapy — critical; nephrocalcinosis can stabilize or partially regress with aggressive alkali therapy; starts at diagnosis - Monitor nephrocalcinosis — renal ultrasound every 6–12 months; document stability or regression - Calcium and vitamin D — for bone disease; balanced supplementation; avoid excessive calcium (exacerbates calcium deposition if acidosis not corrected) - Monitor renal function — nephrocalcinosis → potential CKD; eGFR annually
COMPLEMENT-MEDIATED / THROMBOTIC MICROANGIOPATHY					
111	 Atypical HUS (aHUS) — CFH/CFI/C3/CFB/MCP mutations	Complement dysregulation causing TMA with microangiopathic hemolytic anemia, thrombocytopenia, and AKI; highly recurrent without treatment	US: ~2/million/year in US; CFH mutations most common (~20–30%); ESKD in ~50% without eculizumab World: ~2/million/year globally; European aHUS Registry well-characterized; CFH most common mutation; eculizumab transformed outcomes Sources: Fremeaux-Bacchi et al. JASN 2013; Loirat & Fremeaux-Bacchi Orphanet J Rare Dis 2011; aHUS Registry	CBC (microangiopathic hemolytic anemia, thrombocytopenia); renal function; complement panel (C3, C4, CH50, factor H/I); genetic panel; anti-CFH antibodies; ADAMTS13 (exclude TTP)	Standard of Care - Eculizumab (Soliris, Alexion) — anti-C5 complement inhibitor; FDA approved 2011 for aHUS; preferred for ACUTE episode (most evidence; faster onset); biweekly IV infusions; requires mandatory meningococcal vaccination ≥2 weeks before initiation; lifelong in most patients with high-risk mutations (CFH, CFI, C3, CFB, THBD) - Ravulizumab (Ultomiris, Alexion/AstraZeneca) — next-generation anti-C5; FDA approved for aHUS (adults and pediatric ≥1 month) 2019; extended dosing: every 4 weeks (<20 kg) or every 8 weeks (>20 kg); 3–4× longer

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					half-life than eculizumab; complete TMA response in 78% of children in Phase 3 trials; increasingly preferred for MAINTENANCE over eculizumab; 2-year Phase 3 data (2024) confirm sustained hematologic and renal improvement - Plasma infusion — bridge therapy before complement inhibitor available; fresh frozen plasma contains normal complement factors; less effective than C5 inhibitors - Kidney transplant — for ESKD; requires peri-transplant complement inhibition to prevent recurrence (risk varies by mutation: CFH >50% recurrence without eculizumab, CFI >30%, MCP <5%) - Meningococcal + encapsulated bacteria prophylaxis — mandatory vaccinations and/or prophylactic penicillin/amoxicillin before and during complement inhibition Note: Genetic testing for CFH, CFI, CFB, C3, MCP/CD46, THBD, CFHR1–5, anti-CFH antibodies mandatory before or at initiation of treatment; guides duration and discontinuation decisions Emerging/Investigational - Iptacopan (Fabhalta, Novartis) — Factor B inhibitor (proximal alternative pathway); FDA approved for C3G 2025; investigational for aHUS; Phase 2/3 trials - Danicopan (Factor D inhibitor) — oral; Phase 2/3 in aHUS/C3G - Subcutaneous C5 inhibitors — improved convenience; in development - Crovalimab (Veopoz, Roche) — novel anti-C5 antibody; FDA approved for PNH 2023; investigational for aHUS; enables SC self-injection - Cemdisiran — siRNA targeting complement C3 in liver; investigational for aHUS - Discontinuation protocols — evidence accumulating for selected low-risk patients (MCP/CD46 mutations); individualized approach per KIDRAP and SHARE guidelines
112	 Atypical HUS — Anti-CFH Antibody Mediated	Acquired aHUS from autoantibodies against complement factor H; occurs predominantly in children aged 5–12; associated with CFHR1/3 deletion	US: ~5–10% of aHUS in US; most common in children 5–12 years; CFHR1/3 deletion in ~90% World: 5–10% of aHUS globally in children; European cohort data; CFHR1/3 homozygous deletion most frequent finding Sources: Dragon-Durey et al. JASN 2005; Józsi et al. Blood 2008; aHUS Registry Europe	Anti-CFH IgG antibody titer; CFHR1/3 deletion by MLPA; C3 (low); complement panel; renal function; ADAMTS13	Standard of Care - Plasma exchange (plasmapheresis) — removes anti-CFH antibodies AND supplies normal CFH; preferred initial therapy; typically 5–10 sessions - Rituximab — depletes anti-CFH antibody-producing B cells; key adjunct; prevents recurrence; usually given early in course; repeat doses for relapse - Eculizumab / Ravulizumab — complement inhibition during acute phase; bridges to definitive immunosuppression effect; increasingly used with immunosuppression rather than alone

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Corticosteroids — immunosuppressive; reduces anti-CFH Ab titers; combined with rituximab - Mycophenolate mofetil — maintenance immunosuppression post-rituximab; long-term B cell suppression - Monitor anti-CFH antibody titers — guide treatment duration and identify relapse - CFHR1-3 deletion screening — anti-CFH Ab most common in patients with homozygous CFHR1-3 deletion; helps identify genetic risk
113	Shiga toxin HUS (STEC-HUS)	Most common HUS in children; Shiga toxin from E. coli O157:H7 or O104:H4 triggers endothelial TMA; diarrhea-associated HUS	US: ~2–3/100,000 children/year in US; peak age 1–5 years; summer predominance; ESKD in 5–10% World: ~2–3/100,000 children/year in high-income countries; higher in Argentina (5–7/100,000) — world's highest rate; E. coli O157:H7 Sources: Tarr et al. NEJM 2005; Garg et al. NEJM 2003; STEC-HUS International Registry	CBC (hemolytic anemia, thrombocytopenia); stool culture/PCR for STEC; Shiga toxin PCR/EIA in stool; serum Shiga toxin-producing E. coli antibodies; renal function; LDH; complement (normal)	Standard of Care - Supportive care — cornerstone; careful fluid management; avoid hypovolemia and hypervolemia; antihypertensive therapy; dialysis (peritoneal or hemo) for renal failure; red cell transfusion for severe anemia (caution: may exacerbate hypertension) - NO antibiotics — risk of increased Shiga toxin release from lysed STEC bacteria; antibiotics contraindicated in active STEC-HUS diarrheal illness; meta-analyses do not support benefit - NO antimotility agents (loperamide) — increases toxin absorption time; contraindicated - Early pediatric nephrology involvement — critical given rapid progression to renal failure <i>Eculizumab Controversy:</i> - Eculizumab use in STEC-HUS remains controversial; was used extensively after 2011 E. coli O104:H4 Germany outbreak; subsequent systematic reviews and RCT data show no consistent benefit over supportive care in typical STEC-HUS; 2024 guidelines do not recommend routine use; may have benefit in select cases with severe neurological involvement or refractory disease; individualized, expert-center decision only Emerging/Investigational - Anti-Stx2 monoclonal antibody (Urtoxazumab) — targets Shiga toxin 2; Phase 2 trials; reduces Stx2-mediated injury; investigational - CBLB502 (entolimod, TLR5 agonist) — reduces complement-mediated TMA; preclinical/early clinical - Dialysis optimization — continuous renal replacement therapy (CRRT) in severe pediatric STEC-HUS; evolving protocols - FimH antagonists — prevent STEC intestinal adherence; investigational prevention approach
114	 DGKE-associated HUS	DGKE mutations causing infantile-onset recurrent TMA/HUS; not complement-mediated; diacylglycerol	US: Very rare; DGKE mutations described in ~20 families globally; presents in first year of life	Genetic testing (DGKE); normal complement; recurrent TMA episodes	Standard of Care Emerging/Investigational: Eculizumab — less effective (non-complement mechanism); investigational use ongoing

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		kinase epsilon deficiency activates platelets	World: Very rare globally; <50 cases in literature; DGKE mutations; persistent/relapsing TMA episodes from infancy Sources: Lemaire et al. Nat Genet 2013; Bruneau et al. JASN 2018	from infancy; renal function; ADAMTS13 normal	
115	 TTP — Congenital / Upshaw-Schulman Syndrome	ADAMTS13 deficiency from mutations causing congenital TTP; presents in neonatal period or childhood with TMA triggered by illness/pregnancy	US: Very rare; ~1/million in US; AR; episodes triggered by infection, fever, or pregnancy World: ~1/million globally; ADAMTS13 deficiency; European TTP Registry; episodes from childhood with complement triggers Sources: Levy et al. Nature 2001; Upshaw NEJM 1978; International TTP Registry	ADAMTS13 activity (<10%); ADAMTS13 inhibitor absent (distinguishes from acquired); ADAMTS13 genetic testing; CBC (TMA); LDH	Standard of Care - Fresh frozen plasma (FFP) — ADAMTS13 replacement; historically the primary treatment; on-demand for acute TTP events AND prophylactic infusions; achieves ~19% of normal ADAMTS13 activity; allergic reactions common; relied on donated plasma supply FDA-Approved (November 2023) — CRITICAL NEW THERAPY - Recombinant ADAMTS13 (Adzynma, Takeda) — FDA approved Nov 2023 for congenital TTP (adults + children); first disease-specific therapy for cTTP; 40 IU/kg IV; restores ~100% of normal ADAMTS13 activity vs. ~19% with FFP; Phase 3 crossover trial (NEJM 2024, Scully et al.); ZERO acute TTP events vs. 1 with FFP; adverse events 9% vs. 48% with FFP; no TTP events in Phase 3; no neutralizing antibodies detected; 2-year real-world data (Japan, 2025) confirm efficacy in ESRD patients; ISTH 2025 guidelines: STRONG recommendation for rADAMTS13 over FFP for cTTP prophylaxis in remission (moderate certainty evidence) <i>Standard Approach with Adzynma:</i> - rADAMTS13 (Adzynma) 40 IU/kg IV — prophylactic every 2–3 weeks OR on-demand for acute events; replaces FFP as first-choice where accessible - FFP — alternative if rADAMTS13 unavailable; still effective but inferior Emerging/Investigational - Extended-half-life recombinant ADAMTS13 — to allow less frequent prophylactic dosing; in development - Gene therapy (ADAMTS13) — hepatocyte-targeted; preclinical - Caplacizumab — off-label for acute cTTP episodes with severe symptoms (ISTH 2025: conditional recommendation to use caplacizumab for acute cTTP if available, as part of shared decision-making)
116	TTP — Acquired Pediatric	Autoimmune ADAMTS13 inhibitor causing TTP in children; rare compared to adults; requires urgent treatment	US: Rare in children in US; peak in adolescent females; anti-ADAMTS13 antibodies; systemic involvement World: Rare globally in children; ADAMTS13 antibody-mediated;	ADAMTS13 activity (<10%); ADAMTS13 inhibitor (IgG); CBC (TMA); LDH; renal function; neurological assessment	Standard of Care Therapeutic plasma exchange (TPE) — cornerstone; removes anti-ADAMTS13 autoantibodies AND replaces functional ADAMTS13; daily until platelet count normalized; typically 5–10 sessions; median 5 exchanges required before platelet normalization (UK pediatric registry)

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			European Registry shows pediatric peak in adolescence Sources: Scully et al. Blood 2012; Veyradier et al. Blood 2004; International TTP Registry		- Corticosteroids — high-dose IV methylprednisolone then oral prednisone; immunosuppression reduces autoantibody production; standard adjunct to TPE - Rituximab (anti-CD20) — B-cell depletion; reduces autoantibody production; added at diagnosis in many centers (particularly for severe/refractory disease); ~75% of patients in UK TTP Registry received rituximab <i>FDA-Approved Pediatric Expansion (January 5, 2026) — CRITICAL NEW APPROVAL:</i> - Caplacizumab (Cablivi, Sanofi) — anti-VWF nanobody; inhibits VWF-platelet interaction; FDA EXPANDED approval to pediatric patients ≥12 years (January 5, 2026) for acquired TTP in combination with TPE and immunosuppression; previously approved for adults only (2019); Orphan Drug Designation; pediatric approval supported by retrospective data (30 patients 2–18 yr): 80% remission; UK TTP Registry (16 peds <18 yr, including 4 under age 12): no refractory cases, 1 relapse; youngest treated 33 months (off-label) - Dosing: 10 mg IV loading then 10 mg SC daily; for <40 kg: 5 mg daily; continue for 30 days after last TPE; extend if ADAMTS13 activity remains severely deficient - Key benefit: faster platelet normalization; reduces TPE sessions; reduces hospital stay <i>ADAMTS13 Monitoring:</i> - Serial ADAMTS13 activity levels guide caplacizumab duration; ADAMTS13 recovery may be delayed by caplacizumab (median 35 days in UK registry) — this is normal and expected; extend treatment until ADAMTS13 ≥20% - ISTH 2025 focused update: no change to prior recommendations for iTTP management; caplacizumab endorsed as part of standard TPE+IS+rituximab protocol Emerging/Investigational - Caplacizumab without TPE — Phase 2/3 trials ongoing (CABLIVI TPE-free); some centers adopting for mild-moderate iTTP - Immune checkpoint inhibitor-associated TTP — increasing recognition; rituximab + caplacizumab regimens - Tolerization approaches for anti-ADAMTS13 Ab — investigational
117	 Cobalamin C Deficiency — TMA (MMACHC)	MMACHC mutations causing impaired cobalamin metabolism; presents as TMA in neonates/infants with elevated methylmalonic acid and homocysteine	US: Rare; most common organic acidemia globally; TMA in infancy; MMACHC mutations World: Rare globally; Cobalamin C the most prevalent intracellular cobalamin defect; TMA presentation distinguishes from HUS	Plasma amino acids (elevated homocysteine); urine organic acids (elevated methylmalonic acid); MMACHC genetic testing; complement (normal); ADAMTS13 (normal)	Standard of Care Hydroxocobalamin (OHCbl) — high-dose intramuscular or subcutaneous; first-line and most effective; crosses blood-brain barrier; preferred over cyanocobalamin; doses 1–2 mg/day IM in acute phase, then 1 mg 3× weekly maintenance; dramatically improves TMA and metabolic parameters

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			Sources: Martinelli et al. NEJM 2011; Van Hove et al. JIMD 2002; MMACHC Registry		- Betaine — reduces homocysteine (transmethylation pathway); oral; adjunct to OHCbl; 100–500 mg/kg/day - Folate (folinic acid/leucovorin) — improves folate-dependent methylation; adjunct; folate cycle impaired in CblC - Carnitine — secondary carnitine deficiency common; L-carnitine supplementation; 100 mg/kg/day - Riboflavin (B2) — supports methylmalonyl pathway enzymes - Dietary protein restriction — reduces methylmalonate accumulation; moderate restriction; maintain adequate nutrition - Monitoring — homocysteine levels (goal <50 μmol/L), methylmalonate levels, CBC, renal function, ophthalmological (pigmentary retinopathy universal in CblC) - Ophthalmological management — maculopathy/retinopathy begins in infancy; low vision aids; no proven retinoprotective therapy Emerging - mRNA therapy for MMACHC — Moderna mRNA-3705 Phase 2 (now reclassified for PA/MMA; CblC program separate); AAV-MMACHC gene therapy preclinical 5-HIAA targeting — neurological disease mechanism research
118	 Interferon-pathway TMA (TMEM173 / STING)	Gain-of-function STING mutations (SAVI syndrome) causing type I interferon overactivation with vasculopathy, TMA, and pulmonary fibrosis	US: Very rare; <50 cases reported worldwide including US; pediatric onset; STING pathway World: Very rare globally; <50 reported cases; TMEM173/STING gain-of-function; type I interferonopathy spectrum Sources: Bertoli-Avella et al. NEJM 2018; Liu et al. Science 2014; STING-associated vasculopathy registry	STING/TMEM173 genetic testing; type I interferon signature; renal biopsy (TMA); pulmonary CT; skin biopsy (vasculopathy)	Standard of Care Emerging/Investigational: JAK inhibitors (ruxolitinib, baricitinib — target interferon signaling) — investigational for STING-associated TMA
119	HUS — Secondary (drug / transplant-associated)	TMA caused by immunosuppressants (calcineurin inhibitors, mTOR inhibitors), quinine, gemcitabine, or post-transplant in setting of antibody-mediated rejection	US: Uncommon in US; CNI-TMA in ~3–5% of transplant recipients; dose-related and complement-mediated forms World: 3–5% of transplant recipients globally; consistent across US, European transplant registries Sources: Noris & Remuzzi JASN 2010; Thadhani et al. Kidney Int 1996; ERA-EDTA transplant TMA data	TMA triad; drug exposure history; calcineurin inhibitor levels; renal biopsy; complement panel; anti-donor antibodies (transplant context)	Standard of Care - Identify and remove causative agent — CRITICAL; calcineurin inhibitors (tacrolimus, cyclosporine — TMA in 1–5% transplant recipients), mTOR inhibitors (sirolimus, everolimus), gemcitabine, bevacizumab, quinine, cocaine, clopidogrel; dose reduction or switch immunosuppressant - Plasma exchange — for drug-associated TMA with severe TMA; effective for removing drug and/or pathogenic factors; daily until TMA resolves; LESS effective than for iTTP - Eculizumab / Ravulizumab — complement activation demonstrated in CNI-TMA and transplant-associated

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					TMA; complement inhibition increasingly used for severe transplant-associated TMA; evidence from case series; no RCT; individualized decision • Platelet transfusions — AVOID in active TMA (fuels microthrombus formation); only for life-threatening bleeding • Supportive care — dialysis for AKI; antihypertensives; transfusion for severe anemia • Post-stem cell transplant TMA — eculizumab evidence growing; ravulizumab; high mortality without treatment • Identify complement mutation — some "secondary" HUS patients have underlying complement mutations revealed by drug trigger; genetic testing for CFH/CFI/CFB/C3/MCP recommended in severe or relapsing drug-TMA
HYPERTENSION — PEDIATRIC					
120	 Essential / Primary Hypertension — Pediatric	Hypertension without identifiable secondary cause; increasingly common in adolescents with overweight/obesity; familial component	US: ~3–5% of US children; prevalence rising with obesity; primarily adolescents; rare in young children World: ~3–5% globally; variability by country; higher in US/Australia due to obesity epidemic; rising worldwide Sources: Hansen et al. Hypertension 2007; Flynn & Kaelber AAP 2017 guidelines; NHANES data	Blood pressure ≥95th percentile on ≥3 occasions; ambulatory blood pressure monitoring (ABPM); exclude secondary causes; echocardiogram; lipids; fasting glucose	Standard of Care • Lifestyle modification first-line — DASH diet; sodium restriction; weight loss (if overweight/obese); exercise; 6-month trial for Stage 1 without comorbidities • Pharmacological therapy — initiate immediately for: Stage 2 HTN, symptomatic HTN, Stage 1 with end-organ damage (LVH), comorbid CKD/diabetes/proteinuria • First-line drug classes (evidence-based): - ACEi (lisinopril, fosinopril, enalapril) — preferred in CKD, proteinuria, diabetes - ARB (losartan, candesartan) — preferred if ACEi intolerant; same CKD indications - Long-acting calcium channel blockers (amlodipine) — preferred post-transplant (first 6–12 months); well-tolerated; no monitoring requirements - Thiazide/thiazide-type diuretics (hydrochlorothiazide, chlorthalidone) — for volume-dependent or low-renin hypertension • BP targets (2017 AAP): <90th percentile for age/sex/height (children <13 yr); <130/80 mmHg (adolescents ≥13 yr) • ABPM (ambulatory BP monitoring) — recommended for confirmation, white coat HTN, masked HTN, CKD surveillance; BP load and nocturnal dipping assessment • Beta-blockers — NOT first-line in pediatric primary HTN; used for specific indications (migraine, portal hypertension) • ACEi/ARB contraindicated in adolescent females with child-bearing potential — teratogenicity warning Emerging/Investigational ▣ SGLT2 inhibitors — growing evidence for renal hemodynamic BP-lowering effects; relevant in obese adolescents with metabolic syndrome; off-label use

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					▶ GLP-1 agonists (liraglutide, semaglutide) — weight loss–mediated BP reduction; approved for pediatric obesity; indirect BP benefit ▶ Telehealth + home BP monitoring — improving adherence and treatment titration; clinical effectiveness research ongoing ▶ Finerenone — non-steroidal MRA; BP lowering in CKD/diabetes context; pediatric trials emerging ▶ Renalase modulation — novel pathway; early preclinical for pediatric renal hypertension
121	 Obesity-related Hypertension — Pediatric	Hypertension directly attributable to obesity; insulin resistance, increased sympathetic activity, and renin-angiotensin activation; common metabolic syndrome component	US: ~30–40% of obese US children have hypertension; increasing prevalence with obesity epidemic World: ~30–40% globally in obese children; metabolic syndrome association consistent across populations Sources: Dietz & Robinson NEJM 1998; Reinehr et al. Int J Obes 2006; WHO Global Obesity data	Anthropometric measures (BMI, waist circumference); fasting insulin, glucose (insulin resistance); lipids; ABPM; echocardiogram (LVH)	Standard of Care • Weight loss — primary and most effective intervention; 5–10% reduction lowers BP significantly; lifestyle, behavioral, dietary interventions • DASH diet + sodium restriction — especially effective in obesity-related HTN; reduces diet-induced renin-angiotensin activation • Physical activity — aerobic exercise ≥1 hour/day; reduces BP by 3–5 mmHg • Screen for sleep-disordered breathing — obstructive sleep apnea common in pediatric obesity; OSA treatment reduces BP • Pharmacotherapy per 2017 AAP guidelines — if lifestyle fails after 6 months or if Stage 2/end-organ damage; ACEi/ARB, CCB, or thiazide per renal/metabolic comorbidities Emerging/Investigational ▶ GLP-1 receptor agonists: - Liraglutide (Saxenda, Novo Nordisk) — FDA approved for obesity ≥12 yr (2020); significant weight loss + BP reduction; 3 mg SC daily - Semaglutide SC (Wegovy, Novo Nordisk) — FDA approved for obesity ≥12 yr (2022); superior weight loss to liraglutide; FLOW trial demonstrates renal and BP protection in adults; 2.4 mg weekly - Tirzepatide (Zepbound, Lilly) — GLP-1/GIP dual agonist; SURMOUNT-2 data; FDA approved obesity in adults (2023); adolescent trials underway ▶ SGLT2 inhibitors — renal hemodynamic BP lowering; weight loss benefit; off-label in obese hypertensive adolescents ▶ Bariatric surgery — highly effective for severe obesity (BMI ≥35 with comorbidities); sustained hypertension remission documented
122	 Renovascular Hypertension — Fibromuscular Dysplasia	Arterial dysplasia causing renal artery stenosis and renin-dependent hypertension; most common	US: ~0.5–1% of pediatric hypertension is renovascular; FMD more common than atherosclerosis in US children	Doppler ultrasound renal arteries; MRA; CTA; captopril renogram; renal arteriography (gold standard); plasma renin	Standard of Care • Percutaneous transluminal angioplasty (PTA) — first-line treatment for pediatric FMD-related renovascular hypertension; balloon dilation without stenting preferred in

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		cause of renovascular hypertension in children	World: ~0.5–1% of pediatric HTN globally; FMD and Takayasu arteritis major causes; genetic forms rare Sources: Tullus et al. Lancet 2008; Stanley et al. J Vasc Surg 1995; ASPIRE Registry	activity; peripheral vein renin sampling	children (stents poorly tolerate growth); 70–80% cure or significant improvement in BP; single dilations or serial dilations - ACEi/ARBs — for BP control and renal protection; reduce renin-driven hypertension; monitor renal function closely (risk of acute bilateral RAS-associated AKI with RAAS blockade) - Surgical revascularization — aortorenal bypass (saphenous vein, autologous artery graft) for: failed angioplasty, long-segment disease, ostial lesions, very young children; excellent long-term outcomes in specialized pediatric vascular surgery centers (Cincinnati Children's, CHOP, Great Ormond Street) - Calcium channel blockers — amlodipine for BP control when RAAS blockers not tolerated or pre-angioplasty; well-tolerated; commonly used - Multivessel/bilateral screening — renal artery assessment; mesenteric/cealic vessels; fibromuscular dysplasia can be systemic - Captopril renography or renal vein renins — diagnose lateralizing renovascular hypertension
123	 Renal Artery Stenosis — Takayasu / Vasculitis-associated	Large vessel vasculitis causing renal artery stenosis; Takayasu arteritis most common in young females; middle aortic syndrome in children	US: Rare; Takayasu most common large-vessel vasculitis in US children; RAS in 30–70% of Takayasu World: Rare globally; highest prevalence in Japan, India, Middle East; pediatric Takayasu well-described in Asian registries Sources: Ozen et al. ARD 2010; Eleftheriou et al. Rheumatology 2015; TAKI Consortium	MRA; CTA angiography; FDG-PET (active inflammation); ESR/CRP; ANCA; renal arteriography; captopril renogram; ophthalmology	Standard of Care - Immunosuppression for active Takayasu — corticosteroids (prednisone 1–2 mg/kg/day); first-line for acute inflammatory phase; methotrexate, azathioprine, or mycophenolate as steroid-sparing agents - Biologics for refractory Takayasu — tocilizumab (IL-6 receptor antagonist) FDA approved for adult GCA/Takayasu; increasingly used for pediatric Takayasu; abatacept (CTLA4-Ig); TNF inhibitors (infliximab); rituximab - BP management — ACEi/ARBs; calcium channel blockers; monitor renal function; bilateral RAS risk with RAAS blockade - Surgical/endovascular revascularization — reserved for quiescent disease; active inflammation precludes safe repair; PTA or bypass grafting for significant RAS causing uncontrolled HTN, renal ischemia, or renovascular HTN - Surveillance — MRA or CTA annually; PET-CT for activity assessment; inflammatory markers (CRP, ESR, IL-6) - Antiplatelet therapy — aspirin for ischemic risk reduction in vessel disease Emerging  Secukinumab (IL-17 inhibitor) — investigational for refractory Takayasu  Ustekinumab (IL-12/23) — case reports for refractory large vessel vasculitis

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
124	 Renal Artery Stenosis — Neurofibromatosis-associated	NF1 causes RAS through neurofibromas or dysplastic arteriopathy of renal arteries; important and often underrecognized cause of renovascular hypertension in children with NF1	US: ~1% of NF1 patients develop renovascular HTN; NF1 frequency ~1/3,000 in US World: ~1/3,000 globally for NF1; vascular complications in ~5%; renovascular HTN ~1% of NF1 patients; US and European data Sources: Friedman et al. Am J Med Genet 2002; Oderich et al. J Vasc Surg 2007; NF1 Registry	NF1 clinical diagnosis (café-au-lait spots, Lisch nodules); Doppler ultrasound; MRA; renal arteriography; captopril renogram; peripheral renin studies	Standard of Care - ACEi/ARBs — for BP control; renovascular mechanism; monitor renal function (bilateral NF1-RAS risk with bilateral disease) - Surgical revascularization — NF1-associated renal artery stenosis requires surgery (not PTA/angioplasty); NF1 vessels are friable, dysplastic, prone to restenosis after angioplasty; aortorenal bypass preferred; experienced pediatric vascular surgery center essential - Calcium channel blockers — amlodipine for BP; peri-operative control - Multi-organ NF1 management — NPI, bone disease, plexiform neurofibromas, learning difficulties; multidisciplinary - Surveillance — aortic involvement possible; renal artery imaging; abdominal imaging for pheochromocytoma (NF1 association) Emerging  Selumetinib (Koselugo) — MEK inhibitor FDA approved 2020 for NF1 plexiform neurofibromas ≥2 yr; no evidence for vascular NF1 lesions; research stage
125	 Glucocorticoid-Remediable Aldosteronism (CYP11B1/B2)	Chimeric gene fusion of CYP11B1/CYP11B2 causing aldosterone production under ACTH control; familial hypertension, hypokalemia, suppressed renin, elevated aldosterone	US: Rare; ~1% of primary aldosteronism cases in US; important cause of familial early-onset HTN World: Rare globally; CYP11B1/B2 chimeric gene; autosomal dominant; HTN responsive to dexamethasone suppression Sources: Sutherland et al. Lancet 1966; Stowasser et al. JASN 2003; GRA Registry	Blood pressure; electrolytes (hypokalemia); renin/aldosterone (low renin, high aldosterone); dexamethasone suppression test; CYP11B1/CYP11B2 fusion gene detection (Southern blot or long-range PCR)	Standard of Care - Low-dose dexamethasone — cornerstone; suppresses ACTH → suppresses aldosterone (produced by chimeric CYP11B1/B2 fusion gene driven by ACTH-responsive promoter); 0.125–0.25 mg/day; restores normoaldosteronism; corrects hypokalemia and hypertension; long-term maintenance; minimize dose to avoid glucocorticoid side effects (growth suppression) - Mineralocorticoid receptor antagonists (spironolactone, eplerenone) — add if BP not fully controlled on dexamethasone alone; eplerenone preferred in adolescent males (fewer anti-androgen effects) - Amiloride — blocks ENaC; potassium-sparing; adjunct or alternative to MRA - Potassium supplementation — hypokalemia correction; oral KCl - Genetic testing (CYP11B1/B2 chimeric gene) — Southern blot or long-range PCR; confirmatory in all suspected cases; distinguishes from other primary aldosteronism - Family screening — autosomal dominant; first-degree relatives at 50% risk; screen for hypertension and hypokalemia; genetic testing of relatives
126	 Monogenic Hypertension — Other (Liddle, AME, Gordon)	Genetic causes of low-renin, low-aldosterone hypertension not separately listed; family history of	US: Collectively rare in US; important because treatment differs from essential hypertension World: Collectively rare globally; each condition responds to specific	Electrolytes; renin/aldosterone; genetic panel (SCNN1B/G, HSD11B2, WNK1/4, KLHL3, CUL3); urine steroid profile;	Standard of Care Liddle Syndrome — amiloride/triamterene; low sodium diet (see Liddle entry #95)

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		early-onset hypertension key	treatment targeting pathogenic mechanism Sources: Lifton et al. Cell 2001; Dluhy & Williams Kidney Int 1994; Monogenic HTN review Nat Rev Nephrol 2018	consider in children with treatment-resistant hypertension + low renin	- Apparent Mineralocorticoid Excess — spironolactone/eplerenone; dexamethasone (see AME entry #96) - Gordon Syndrome (PHA2) — thiazide diuretics (see PHA2 entry #94) - GRA — dexamethasone + MRA (see GRA entry #124) - All forms: dietary sodium restriction; genetic confirmation of diagnosis; family screening; tailored pharmacotherapy based on specific molecular mechanism - General principle — mechanism-based treatment essential; wrong class of antihypertensive will be ineffective or harmful; genetic diagnosis guides therapy - Monitor electrolytes, renin, aldosterone — confirms correct diagnosis and treatment response
127	White Coat Hypertension — Pediatric	Elevated clinic blood pressure with normal ambulatory blood pressure monitoring (ABPM); not associated with end-organ damage; may progress to sustained hypertension	US: ~30–40% of children with elevated US clinic BP have white coat HTN on ABPM; important to diagnose World: ~30–40% globally with elevated clinic BP have white coat HTN; ABPM essential for definitive diagnosis Sources: Flynn et al. Hypertension 2014; Lurbe et al. J Hypertens 2016; AAP 2017 Pediatric HTN Guidelines	ABPM (24-hour; daytime mean below 95th percentile); echocardiogram (normal LV mass); renal ultrasound; repeat clinic measurements; home monitoring	Standard of Care - Ambulatory blood pressure monitoring (ABPM) — ESSENTIAL for diagnosis; 24-hour ABPM shows normal BP outside clinical setting; daytime and nighttime loads normal; distinguishes WCH from true hypertension; 2017 AAP guideline strongly recommends ABPM confirmation - No pharmacological treatment — WCH does not require antihypertensive medications; reassurance - Lifestyle counseling — DASH diet, reduced sodium, regular physical activity, healthy weight - Repeat ABPM every 1–2 years — monitor for progression to true hypertension (WCH is a risk factor for future sustained HTN); follow through adolescence - Address anxiety — explain white coat effect; reduce anxiety around measurements; home BP monitoring with validated oscillometric device; compare to ABPM - Reassurance — WCH in children generally benign; stress reduction; normalization of BP expected outside clinic
128	Masked Hypertension — Pediatric	Normal clinic blood pressure but elevated ABPM; associated with end-organ damage and cardiovascular risk; prevalence underestimated	US: ~6–10% of US children with normal clinic BP have masked HTN; higher in CKD, obesity World: ~6–10% globally; consistently higher in CKD (~30–40%) and obese children; associated with organ damage Sources: Lurbe et al. Hypertension 2005; Stabouli et al. Ped Nephrol 2011; Masked HTN meta-analysis 2019	ABPM (daytime mean $\geq$ 95th percentile with normal clinic); home blood pressure monitoring; echocardiogram; renal function	Standard of Care - ABPM — diagnostic; normal or borderline clinic BP but elevated 24-hour or nighttime BP on ABPM; daytime mean $\geq$135/85 or 24-hour $\geq$130/80 in adolescents; subclinical target organ damage (LVH, microalbuminuria) risk similar to sustained HTN - Antihypertensive therapy — indicated if masked HTN confirmed; same drug classes as true HTN: ACEi/ARBs (especially if CKD), CCB, thiazide; lower threshold for treatment vs. WCH due to similar cardiovascular risk - Lifestyle interventions — weight loss, sodium restriction, DASH diet, physical activity; critical first-line measures - Monitor for end-organ damage — echocardiography for LVH annually; urinalysis for microalbuminuria; optic disc evaluation in severe cases

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Repeat ABPM — every 6–12 months to assess treatment response and disease evolution - High-risk groups — CKD, obesity, premature birth, coarctation history, sleep apnea — screen for masked HTN in all
129	Hypertensive Emergency — Pediatric	Severe hypertension with acute end-organ damage (hypertensive encephalopathy, seizures, acute heart failure, retinopathy, AKI); requires immediate IV therapy	US: ~1–2% of US pediatric HTN referrals; secondary HTN most common; AKI and neurological complications common World: ~1–2% globally of pediatric HTN referrals; consistent rates in US and European pediatric intensive care data Sources: Deal et al. J Pediatr 1992; Flynn et al. Hypertension 2007; Pediatric Emergency HTN Collaborative	Blood pressure (severely elevated ≥ 99 th percentile + 12 mmHg); neurological assessment; echocardiogram; renal function; fundoscopy; CT/MRI brain; CBC (TMA)	Standard of Care - Nicardipine IV — dihydropyridine CCB; preferred IV agent for pediatric hypertensive emergency; continuous infusion; titratable; does not cause reflex tachycardia; minimal adverse effects; preferred in renal/CKD-associated emergency - Labetalol IV — alpha/beta blocker; effective for catecholamine excess (pheochromocytoma, cocaine); safe in most pediatric emergencies; bolus or infusion - Hydralazine IV or IM — vasodilator; use in pregnancy-associated HTN; less titratable; alternative in resource-limited settings - Sodium nitroprusside IV — most rapidly titratable; reserve for extreme emergencies (hypertensive encephalopathy, severe aortic coarctation crisis); cyanide toxicity risk with prolonged use; requires intensive monitoring - Esmolol IV — ultra-short-acting beta-blocker; for acute aortic dissection, perioperative HTN, thyroid storm - Furosemide IV — if volume overload component (acute glomerulonephritis emergency) <i>Key principles:</i> TARGET: Reduce MAP by 25% in first 6–8 hours; then gradually to normal over 24–48 hours; avoid precipitous drops (risk of ischemic stroke, vision loss) - Identify and treat underlying cause — do NOT lower BP before securing IV access and ruling out AKI/GN/pheochromocytoma/coarctation - Oral agents — NOT appropriate for true hypertensive emergency; use IV only - Transfer to PICU — all hypertensive emergencies; neurological monitoring essential Emerging/Investigational ☒ Clevidipine IV — ultrashort-acting CCB (clevidipine); used in adult cardiac surgery; pediatric data emerging for intraoperative/postoperative hypertension ☒ Fenoldopam IV — dopamine receptor agonist; renal vasodilation; may offer renal protection vs. other agents; used in some pediatric centers ☒ Diagnosis-specific protocols — coarctation rescue (prostaglandin, balloon dilation), aortic surgery protocols; increasingly protocol-driven in pediatric centers

ELECTROLYTE / MINERAL DISORDERS

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
130	 Hypomagnesemia with Secondary Hypocalcemia (TRPM6)	TRPM6 mutations causing neonatal severe hypomagnesemia and secondary hypocalcemia; seizures in neonates if untreated	US: Very rare; ~50 cases reported worldwide; AR; neonates with refractory hypocalcemia/seizures World: Very rare globally; ~50 total reported cases; TRPM6 mutations; autosomal recessive; responds to high-dose Mg supplementation Sources: Walder et al. Nat Genet 2002; Schlingmann et al. Nat Genet 2002	Serum magnesium (very low); serum calcium (low — secondary); PTH (elevated or normal); TRPM6 genetic testing; urine magnesium (tubular leak)	Standard of Care - High-dose magnesium supplementation — LIFELONG; oral MgCl₂ or Mg gluconate; IV MgSO₄ for acute severe hypomagnesemia (seizures); goal serum Mg >0.65 mmol/L; TRPM6 mutations require supraphysiological Mg doses to overcome impaired absorption; divided doses (3–4x/day) improve absorption - Calcium supplementation + vitamin D — secondary hypocalcemia due to PTH resistance from hypomagnesemia; calcium normalizes as Mg is replenished; calcitriol if persistent hypocalcemia - IV magnesium for acute crisis — seizures, tetany; 0.2 mL/kg of 50% MgSO₄ IV slowly; follow with continuous infusion - Monitor neurological development — hypomagnesemia seizures in infancy can cause neurodevelopmental injury if diagnosis delayed; early treatment critical - Genetic counseling — TRPM6 (transient receptor potential melastatin member 6) mutations; autosomal recessive
131	 Familial Hypomagnesemia with Hypercalciuria and Nephrocalcinosis (CLDN16/19)	CLDN16 or CLDN19 mutations disrupting paracellular magnesium/calcium reabsorption in thick ascending limb; nephrocalcinosis, nephrolithiasis, ocular coloboma (CLDN19)	US: Very rare in US; CLDN16/19 mutations; severe ocular involvement with CLDN19; progressive CKD World: Very rare globally; CLDN16 more prevalent in Mediterranean; CLDN19 associated with severe macular coloboma Sources: Simon et al. Nat Genet 1999; Konrad et al. Nat Genet 2006; FHHNC Registry	Serum magnesium (low); urine magnesium/calcium (elevated); renal ultrasound (nephrocalcinosis); ophthalmology (CLDN19: coloboma); CLDN16/19 genetic testing	Standard of Care - Magnesium supplementation — high-dose oral MgCl₂; improves symptoms but does NOT halt progression to ESKD; incomplete response due to tubular impermeability (CLDN16/CLDN19 mutations affect tight junctions of thick ascending limb) - Thiazide diuretics — modest reduction in urinary calcium; slows nephrocalcinosis progression partially - Potassium citrate — urinary alkalinization; reduces nephrocalcinosis and stone burden - ACEi/ARBs — renoprotective; CKD management - Ophthalmological — CLDN19 mutations cause severe ocular involvement (macular coloboma, nystagmus, myopia); regular ophthalmology; CLDN16 mutations have less ocular disease - Kidney transplant — ~50% reach ESKD by adolescence/young adulthood; transplant corrects renal disease; tubular defect not present in donor kidney; excellent outcomes; Mg and Ca normalize post-transplant - Genetic counseling — CLDN16 or CLDN19 mutations; autosomal recessive; family screening
132	 X-linked Hypophosphatemic Rickets (PHEX)	PHEX mutations causing increased FGF23 production leading to phosphate wasting, hypophosphatemia, and rickets; most common inherited form of rickets	US: ~1/20,000 in US; X-linked dominant; males more severely affected; most common genetic rickets in US World: ~1/20,000 globally; highest in North America and Europe;	Serum phosphate (low); TmP/GFR (reduced); FGF23 (elevated); ALP (elevated); PTH (normal); wrist X-ray (metaphyseal changes); PHEX genetic testing	Standard of Care Burosumab (Crysvita, Ultragenyx) — anti-FGF23 monoclonal antibody; FDA approved for pediatric XLH ≥1 year (2018); SC injection every 2 weeks (pediatric) or every 4 weeks (adult); directly addresses pathophysiology; superior to conventional therapy for rickets healing, linear growth, phosphate normalization

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			PHEX mutations; burosumab (anti-FGF23) approved 2018 Sources: HYP Consortium Nat Genet 1995; Carpenter et al. JASN 2011; Imel et al. Lancet 2019		- Oral phosphate + active vitamin D (calcitriol / alfacalcidol) — conventional therapy; still used where burosumab unavailable; inferior to burosumab but widely used in resource-limited settings; risk of nephrocalcinosis with this approach - Growth hormone — for growth failure unresponsive to phosphate/burosumab; consider in pre-pubertal patients with severe growth impairment - Orthopedic management — corrective osteotomy for severe bowing; dental care (abscesses) - ENT — hearing evaluation; ossicular involvement in some Emerging/Investigational ▢ Setrusumab (anti-sclerostin) — Phase 2 ORBIT trial for OI; potential relevance to bone disease in XLH ▢ Long-term burosumab data in adults — maturing; transition from pediatric protocols ▢ Burosumab in pregnancy — case reports accumulating; safety data emerging ▢ Gene therapy (AAV-PHEX) — preclinical
133	 Autosomal Recessive Hypophosphatemic Rickets (DMP1/ENPP1)	DMP1 or ENPP1 mutations causing FGF23 excess and hypophosphatemic rickets; ENPP1 mutations also cause calcification of soft tissues (GACI)	US: Very rare in US; ENPP1 mutations cause GACI in infancy transitioning to hypophosphatemia later World: Very rare globally; DMP1 and ENPP1 mutations; GACI-to-ARHR2 phenotypic transition established Sources: Lorenz-Depiereux et al. Nat Genet 2006; Nitschke et al. Nat Genet 2012	Serum phosphate; FGF23 (elevated); ALP; DMP1/ENPP1 genetic testing; X-ray; vascular imaging for GACI (ENPP1)	Standard of Care - Burosumab (Crysvita) — anti-FGF23; FDA approved for XLH (PHEX); expanding evidence base for ARHR1 (DMP1) and ARHR2 (ENPP1); use is off-label for ARHR but increasingly adopted given shared FGF23-excess pathophysiology - Oral phosphate + active vitamin D — conventional therapy for ARHR when burosumab not available/approved - ENPP1 deficiency (ARHR2) — may co-present with generalized arterial calcification of infancy (GACI); complex management Emerging/Investigational ▢ Asfotase alfa (Strensiq) — enzyme replacement for hypophosphatasia (ALPL); different condition but relevant to mineralization; approved for hypophosphatasia including pediatric ▢ Bisphosphonates — used for GACI (ENPP1 deficiency) to reduce arterial calcification; emerging data ▢ Inositol hexaphosphate — ENPP1 substrate; may reduce PPI deficiency-related ectopic calcification; early stage ▢ Gene therapy (AAV-DMP1, AAV-ENPP1) — preclinical
134	 Autosomal Dominant Hypophosphatemia (FGF23)	FGF23 gain-of-function mutations or KL mutations causing hypophosphatemia; milder than XLH but similar mechanism; variable penetrance	US: Rare; autosomal dominant FGF23 mutations; variable penetrance; spectrum from mild to rickets World: Rare globally; FGF23 gain-of-function mutations; autosomal	Serum phosphate; FGF23 (elevated); PHEX (normal); FGF23/KL genetic testing; ALP; X-ray if symptomatic	Standard of Care Burosumab (Crysvita) — anti-FGF23; FDA-approved for XLH; off-label use in ADHR (FGF23 gain-of-function) with same rationale; increasingly used in symptomatic ADHR

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			dominant; clinical severity variable within families Sources: Autosomal Dominant Hypophosphatemia ADHR Consortium Nat Genet 2000; White et al. JASN 2001		- Oral phosphate + active vitamin D — for mild/asymptomatic cases or where burosumab not available - Symptomatic management only — some patients have spontaneous remission (iron-deficiency corrects FGF23 cleavage impairment in ADHR); oral iron supplementation important adjunct Emerging/Investigational - Iron supplementation — specific to ADHR; FGF23 mutations resist cleavage by PHEX; iron deficiency worsens disease by further stabilizing mutant FGF23; iron repletion may partially normalize phosphate - Anti-FGF23 strategies (burosumab) — growing clinical experience in ADHR - Gene therapy — preclinical
135	 Tumoral Calcinosis — Hyperphosphatemic (GALNT3/FGF23/KL)	Loss-of-function in FGF23/KL pathway causing reduced FGF23 activity; hyperphosphatemia and ectopic calcification (tumoral calcinosis); periarticular calcium deposits	US: Very rare in US; autosomal recessive; childhood onset; deposits around large joints World: Very rare globally; GALNT3, FGF23, and KL (klotho) mutations; higher in Arab populations; calcification deposits pathognomonic Sources: Topaz et al. Nat Genet 2004; Ichikawa et al. J Clin Invest 2007	Serum phosphate (elevated); FGF23 (inappropriately normal or low); TmP/GFR; genetic testing (GALNT3, FGF23, KL); X-ray; ultrasound of deposits	Standard of Care Emerging/Investigational: Novel therapies under investigation — anti-FGF23 agents paradoxically contraindicated; targeted approaches in development
136	 Idiopathic Infantile Hypercalcemia (CYP24A1 / SLC34A1)	CYP24A1 mutations causing impaired vitamin D catabolism; SLC34A1 mutations causing phosphate transport defect; both result in hypercalcemia in infancy with nephrocalcinosis	US: Rare; ~1/30,000 in US for CYP24A1 deficiency; worsened by vitamin D supplementation World: ~1/30,000 globally for CYP24A1 deficiency; SLC34A1 mutations cause fetal variant; ascertainment improving with genetic testing Sources: Schlingmann et al. NEJM 2011; Dinour et al. NEJM 2010; IIH Registry	Serum calcium (elevated); PTH (suppressed); 25-OH vitamin D; 1,25 vitamin D; CYP24A1/SLC34A1 genetic testing; renal ultrasound (nephrocalcinosis)	Standard of Care Emerging/Investigational: Fluconazole (inhibits CYP24A1 — reduces 1,25 vitamin D production) — off-label emerging use
137	 Familial Hypocalciuric Hypercalcemia (CASR)	Heterozygous inactivating CASR mutations causing mild asymptomatic hypercalcemia with low urine calcium; usually benign; important to distinguish from primary hyperparathyroidism	US: ~1/78,000 in US; CASR loss-of-function mutations; often incidental hypercalcemia; neonatal form severe World: ~1/78,000 globally; FHH types 1–3 (CASR, GNA11, AP2S1); hypercalciuria low; NSHPT (homozygous) serious Sources: Pollak et al. Cell 1993; Carel et al. Eur J Endocrinol 2010; FHH International Consortium	Serum calcium (mildly elevated); PTH (high normal or mildly elevated); urine calcium:creatinine ratio (low — FE _{Ca} <0.01); CASR genetic testing; family history of hypercalcemia	Standard of Care Emerging/Investigational: Cinacalcet for symptomatic FHH — off-label, emerging indication

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
INFECTIONS / INFLAMMATORY					
138	Urinary Tract Infection — Febrile (first / recurrent)	Bacterial infection of upper urinary tract in children; important to differentiate from lower UTI; risk of renal scarring with recurrent febrile UTIs	US: ~1–3% of boys, 3–8% of girls by age 7 in US; recurrent in ~30%; most common serious bacterial infection in young children World: ~1–3% of boys, 3–8% of girls globally; E. coli >80%; higher recurrence rates in girls; AAP and NICE guidelines Sources: Roberts & Downs Ped Rev 2005; Shaikh et al. Arch Dis Child 2016; UTI-LTF Study NEJM 2010	Urine dipstick (leukocyte esterase, nitrite); urine culture (catheter specimen or SPA in young children); renal ultrasound; DMSA for scarring; VCUG after first febrile UTI in young children	Standard of Care Emerging/Investigational: Antibiotic prophylaxis for VUR — role debated; current RIVUR and CUTIE trial guidance informing practice
139	Pyelonephritis — Acute	Bacterial upper tract infection causing fever, loin pain, and renal parenchymal inflammation; risk of renal scarring and CKD	US: Common; ~5% of febrile US children <2 years; E. coli most common; DMSA scintigraphy standard World: Common globally; prevalence varies with circumcision rates and VUR screening practice Sources: Hoberman et al. NEJM 2003; RIVUR Trial NEJM 2014; NICE UTI Guidelines 2007	Urine culture; blood culture if systemically unwell; DMSA scan (2–3 months after episode for scarring); renal/bladder ultrasound; inflammatory markers	Standard of Care - IV antibiotics (inpatient) — for: infants <3 months, ill-appearing, vomiting, urosepsis; ceftriaxone or ampicillin-gentamicin; oral step-down after 48–72 hours of afebrile clinical improvement - Oral antibiotics (outpatient) — for older children without systemic illness; trimethoprim-sulfamethoxazole or cephalosporins; 10–14 days total; culture-guided E. coli most common — gram-negative coverage essential; adjust when culture sensitivities return - Imaging — renal ultrasound at diagnosis or during admission; VCUG at 4–6 weeks post-fever to evaluate for VUR in children <2 years or recurrent UTI; DMSA scan 4–6 months post-APN for renal scar detection - Fever management — antipyretics; assess for urosepsis; blood cultures if febrile and ill-appearing - Antibiotic prophylaxis — for VUR if detected on post-APN workup; evidence-based per 2011 AAP guidelines - AAP 2011 guideline — recommends VCUG only for febrile UTI with abnormal renal ultrasound; revised 2017 to expand back to all febrile UTI <2 yr; culture-confirmed UTI required before workup
140	Xanthogranulomatous Pyelonephritis	Rare chronic renal infection with granulomatous destruction of renal parenchyma; associated with staghorn calculi, obstruction, and Proteus/E. coli infection; mimics renal tumor	US: Rare in US children; ~1.4% of renal infections; <200 pediatric cases reported globally World: Rare globally; <200 total pediatric cases in literature; mimics renal mass; Proteus most common pathogen Sources: Kim et al. J Urol 2001; Loffroy et al. Eur Radiol 2007	Renal imaging (CT: characteristic appearance with enlarged kidney containing 'bear paw' calculi and hypodense areas); urine culture; surgical pathology (foamy macrophages, lipid-laden)	Standard of Care - Nephrectomy — definitive treatment; XGP is a destructive chronic granulomatous infection; kidney rarely salvageable; open or laparoscopic nephrectomy; specimen reveals staghorn calculus, pus, xanthoma cells (lipid-laden macrophages) - Antibiotics — preoperative and postoperative; treat Proteus, E. coli, Pseudomonas (most common organisms); 4–6 weeks of antibiotics peri-operatively; fluoroquinolones or culture-directed therapy - Percutaneous drainage — for abscess or perinephric involvement; temporizing before surgery

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Diagnose staghorn calculus — almost universal trigger; struvite stones; obstruction - Imaging — CT abdomen/pelvis preferred; demonstrates "bear paw sign" (multiple renal abscesses) - Fistula management — if XGP fistulizes to bowel or retroperitoneum; complex surgical repair
141	Renal Abscess — Pediatric	Focal bacterial infection forming purulent collection within renal parenchyma or perinephric space; may follow UTI or hematogenous spread	US: Rare in US children; ~1–2% of serious UTIs; <i>S. aureus</i> hematogenous; gram-negative rods ascending World: Rare globally; structural anomalies major predisposing factor; <i>S. aureus</i> and gram-negatives predominant pathogens Sources: Coelho et al. J Urol 2007; Mangat & Nagler Ped Nephrol 2012	Renal ultrasound (hypoechoic collection); CT (ring-enhancing lesion); urine and blood cultures; urinalysis; inflammatory markers	Standard of Care - IV antibiotics — empirical broad-spectrum (cefazolin or ceftriaxone ± metronidazole for anaerobes); culture-guided adjustment; <i>Staphylococcus aureus</i>, gram-negative enteric organisms; 4–6 weeks total antibiotics - Percutaneous drainage (interventional radiology) — for abscesses ≥3 cm; CT-guided drainage; relieves obstruction and obtains culture specimen; most can be managed without surgery - Surgical drainage — for: failed percutaneous drainage, multiloculated abscess, perinephric extension, necrotizing fasciitis; open or laparoscopic - Blood cultures — bacteremia common; IV antibiotics until defervescence then oral step-down - Identify predisposing factors — VUR, urinary obstruction, instrumentation, skin infection hematogenous seeding (<i>S. aureus</i>), immunosuppression, diabetes - Imaging — ultrasound for initial diagnosis; CT for staging and drainage guidance; repeat imaging at 48–72 hours to assess treatment response
142	Schistosomiasis — Renal	<i>Schistosoma haematobium</i> infection causing obstructive uropathy, bladder fibrosis, hydronephrosis, and renal impairment; endemic in Africa and Middle East	US: Rare in US (no endemic <i>S. haematobium</i>); occurs in immigrants from endemic regions World: ~200 million worldwide affected; <i>S. haematobium</i> causes urogenital schistosomiasis; endemic in Africa, Middle East Sources: Colley et al. Lancet 2014; WHO Schistosomiasis data 2022; Kjetland et al. Lancet Infect Dis 2012	Urine microscopy (terminal hematuria, <i>S. haematobium</i> eggs); serology; renal and bladder ultrasound (thickened bladder wall, hydronephrosis); cystoscopy for bladder disease	Standard of Care - Praziquantel — drug of choice; curative for active <i>Schistosoma haematobium</i> infection; 40 mg/kg/day divided in 2 doses × 1 day; kills adult worms; highly effective; repeat dose at 4–6 weeks - Prevention of secondary infection — education; avoid freshwater exposure in endemic areas - Surgical management — for obstructive uropathy (ureteral fibrosis, bladder contracture, uretero-vesical junction obstruction); ureteral reimplantation; bladder augmentation; ureterolysis - Bladder carcinoma surveillance — <i>S. haematobium</i> carcinogen; squamous cell carcinoma of the bladder risk in chronic infection; cystoscopy in adults with long history; not routine in pediatric acute infection - Kidney involvement — glomerulonephritis from immune complex deposition; <i>S. mansoni</i> more commonly causes glomerulonephritis; ACEi/ARBs for proteinuria; CKD monitoring - World Health Organization (WHO) mass drug administration programs — praziquantel preventive

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					chemotherapy in endemic school-age populations; reduces transmission and disease burden
143	Renal Tuberculosis — Pediatric	Hematogenous spread of <i>M. tuberculosis</i> to kidneys; typically presents years after primary infection with frequency, hematuria, pyuria, and progressive renal destruction	US: Rare in US children due to low TB burden; primarily in immigrant/refugee populations World: Significant in high-TB-burden regions; renal TB in ~4–8% of pulmonary TB; endemic in South Asia, Sub-Saharan Africa Sources: Global TB Report WHO 2022; Abbara & Davidson Nat Rev Urol 2011	Urine TB culture (early morning specimens ×3); TB PCR (urine); renal ultrasound/CT (cavitation, calcification); chest X-ray; Mantoux/IGRA; IVU (calyceal destruction)	Standard of Care
144	BK Virus Nephropathy (non-transplant)	BK polyomavirus reactivation causing tubulointerstitial nephritis in immunocompromised non-transplant patients (primary immunodeficiency, chemotherapy, HIV)	US: Rare outside transplant or immunocompromised settings in US; increasing recognition in chemotherapy patients World: Rare globally outside transplant; primary BKV infection nearly universal; nephropathy requires immunosuppression Sources: Hirsch et al. NEJM 2002; Randhawa & Khaleel Virchows Arch 2004	Urine BK virus PCR (>7 log copies/mL suggests nephropathy); serum BK PCR; kidney biopsy (BK virus inclusions, tubulitis); urine decoy cells (screening)	Standard of Care Emerging/Investigational: Cidofovir (use with extreme caution — nephrotoxic); brincidofovir — investigational
145	COVID-19 / MIS-C-associated Nephropathy — Pediatric	Kidney involvement in COVID-19 (AKI, collapsing FSGS) and multisystem inflammatory syndrome in children (MIS-C) with TMA, AKI, hypovolemia	US: ~5–10% of hospitalized pediatric COVID-19 have AKI in US; MIS-C ~1/3,000–4,000 infected; renal in 60–70% of MIS-C World: ~5–10% globally; MIS-C rates similar across high-income countries; renal involvement in 60–70% of MIS-C cases Sources: Feldstein et al. NEJM 2020; Götzinger et al. Lancet Child Adolesc Health 2020; CDC MIS-C surveillance	Urinalysis; creatinine; AKI staging; SARS-CoV-2 PCR/serology; inflammatory markers (ferritin, CRP, IL-6); MIS-C diagnostic criteria; TMA screen for severe cases; kidney biopsy if proteinuria persists	Standard of Care Emerging/Investigational: Biologics (anakinra, infliximab) for refractory MIS-C — emerging use
ACUTE KIDNEY INJURY					
146	AKI — Sepsis-associated	Acute kidney injury caused by sepsis-induced hemodynamic compromise, endothelial injury, and inflammatory mediators; most common AKI cause in PICU	US: ~10–15% of PICU patients in US; AKI in ~30–50% of septic shock; <i>S. aureus</i> and gram-negatives predominant World: ~30–50% globally in septic shock; consistent rates in high-income country ICUs; mortality increases 2–8× with AKI Sources: Symons et al. Clin J Am Soc Nephrol 2007; Goldstein & Devarajan Ped Nephrol 2011	Serum creatinine (KDIGO staging); urine output; blood and urine cultures; inflammatory markers; urinalysis; renal ultrasound; novel biomarkers (NGAL, KIM-1)	Standard of Care Emerging/Investigational: Furosemide continuous infusion for fluid management — role refined by SPARK/FISST trials

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
147	AKI — Nephrotoxin-induced	AKI caused by direct tubular toxicity from drugs or contrast agents; aminoglycosides, vancomycin, contrast media, NSAIDs, and chemotherapy most common in children	US: ~20–30% of pediatric AKI in US; aminoglycosides ~10% of hospital-acquired AKI; vancomycin toxicity increasing World: ~20–30% globally in pediatric AKI cohorts; drug exposure varies by local antimicrobial patterns Sources: Schneider et al. JASN 2010; Blowey et al. Ped Nephrol 2019; Goldstein nephrotoxin score 2011	Creatinine trend after drug exposure; urinalysis (granular casts in ATN); drug levels (aminoglycosides, vancomycin); novel biomarkers (NGAL, KIM-1, TIMP-2×IGFBP7); renal ultrasound	Standard of Care - Identify and discontinue nephrotoxin — IMMEDIATE; aminoglycosides, vancomycin, NSAIDs, contrast agents, cisplatin, amphotericin B, calcineurin inhibitors, acyclovir, tenofovir; most important intervention - IV fluid resuscitation — maintain renal perfusion; avoid over-hydration; goal: normovolemia; urine output ≥0.5–1 mL/kg/hr - Drug level monitoring — aminoglycoside peak/trough; vancomycin AUC-guided dosing (2019 consensus recommends AUC-guided over trough monitoring to reduce nephrotoxicity); adjust dosing interval for reduced renal function - Alternative antibiotics — aztreonam or cephalosporins instead of aminoglycosides; rifamycins instead of vancomycin when possible; isolate sensitivities - Contrast-associated AKI prevention — IV pre-hydration with NS or sodium bicarbonate; N-acetylcysteine (controversial; reasonable in high-risk patients); low-osmolar/iso-osmolar contrast; minimize contrast volume - Cisplatin nephroprotection — aggressive IV hydration pre- and post-cisplatin; magnesium supplementation (cisplatin causes Mg wasting); sodium thiosulfate (cytoprotective); PREVENT with high diuresis Emerging  Sodium thiosulfate — nephroprotection with cisplatin; Phase 3 data in pediatric oncology (Children's Oncology Group ACCL1031); reduces cisplatin-induced hearing loss AND nephrotoxicity
148	AKI — Ischemic / Hypoxic (cardiac surgery, birth asphyxia)	AKI from renal hypoperfusion during cardiac surgery (on-pump procedures), birth asphyxia, or severe hypovolemia; tubular injury from ischemia-reperfusion	US: ~30–40% of post-cardiac surgery US children; ~60–70% with severe birth asphyxia/HIE; high AKI mortality World: Consistent globally; post-cardiac surgery AKI rates ~40% across international registries; HIE-associated AKI similar Sources: Sethi et al. J Am Heart Assoc 2017; Askenazi et al. Ped Nephrol 2019; TOBY Trial HIE data	Creatinine and urine output post-operatively or post-asphyxia; KDIGO staging; urinalysis; renal Doppler ultrasound; novel biomarkers (urine NGAL)	Standard of Care Emerging/Investigational: Mild therapeutic hypothermia for HIE — not contraindicated; neonatal RenalGuard and delayed cord clamping under study
149	Cortical Necrosis — Neonatal	Irreversible ischemic necrosis of renal cortex in neonates; caused by severe birth asphyxia, septic shock, or DIC; leads to permanent renal insufficiency	US: Rare; ~2% of neonatal AKI in US; bilateral leads to ESKD; devastating outcomes World: Rare globally; bilateral cortical necrosis most often from perinatal asphyxia and renal vein thrombosis	Renal ultrasound (echogenic cortex, loss of corticomedullary differentiation); CT (cortical enhancement loss); DMSA scan (no function); creatinine; coagulation studies	Standard of Care - Supportive care — primarily palliative/supportive; cortical necrosis represents irreversible ischemic destruction of renal cortex; prognosis guarded - Renal replacement therapy — acute dialysis (peritoneal dialysis in neonates) for severe AKI; bridge to recovery if partial cortical injury

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			Sources: Lerner et al. Pediatrics 1992; Askenazi et al. Ped Nephrol 2019		- IV fluids and vasopressors — manage underlying cause (septic shock, severe asphyxia, severe dehydration); restore perfusion pressure - Treat underlying cause — maternal placental abruption, HUS, severe birth asphyxia, septicemia - Anticoagulation — renal vein thrombosis as comorbidity; heparin if thrombosis identified and clinical stability permits - Long-term monitoring — partial cortical necrosis may result in CKD; hypertension; renal function surveillance throughout childhood; ACEi/ARBs if hypertension/proteinuria develops
150	 Renal Vein Thrombosis — Neonatal	Thrombosis of renal vein in neonates; presents with hematuria, renal mass, and thrombocytopenia; risk factors include dehydration, sepsis, maternal diabetes, polycythemia	US: ~2.2/100,000 live births in US; most common neonatal vascular emergency; left kidney more common World: ~2.2/100,000 globally; consistent data from neonatal registries; thrombophilia screen and maternal conditions key Sources: Zigman et al. Arch Dis Child 2000; Schmidt et al. Ped Nephrol 2004	Renal Doppler ultrasound (absent venous flow, enlarged kidney); CT/MRI angiography; blood count (thrombocytopenia); coagulation studies; thrombophilia screen	Standard of Care - Anticoagulation — LMWH (low-molecular-weight heparin: enoxaparin) for non-bilateral RVT without significant hemorrhage; duration 6–12 weeks minimum; CHEST 2012/2018 and ASH 2018 guidelines recommend therapeutic anticoagulation for bilateral RVT or unilateral RVT extending into IVC - Supportive care — hydration; electrolyte correction; blood transfusion for severe hemorrhagic anemia; dialysis if renal failure - Conservative observation — for unilateral RVT without IVC extension in stable neonate; monitor with serial ultrasound - Systemic thrombolytics (tPA) — for life-threatening bilateral RVT with massive renal failure; use cautiously given neonatal bleeding risk; specialist consultation - Serial renal ultrasound with Doppler — monitoring for thrombus resolution; assess renal blood flow recovery - Long-term monitoring — renal atrophy common; CKD risk; hypertension; annual BP, urinalysis, creatinine through childhood
151	 Renal Vein Thrombosis — Non-neonatal Pediatric	Renal vein thrombosis in older children; usually associated with nephrotic syndrome, hypercoagulable states, dehydration, or central venous catheters	US: Uncommon in US; mostly in nephrotic syndrome (especially membranous) and Factor V Leiden thrombophilia World: Uncommon globally; hypercoagulable states major risk; nephrotic syndrome patients at highest risk Sources: Llach et al. Ann Intern Med 1980; Kerlin et al. Blood 2003	Renal Doppler ultrasound; CT/MRI venography; thrombophilia screen (protein C/S, antithrombin, APLA, Factor V Leiden); coagulation studies; urinalysis	Standard of Care Emerging/Investigational: Thrombolysis for massive bilateral RVT with AKI — limited pediatric data
152	 Renal Artery Thrombosis — Neonatal / Catheter-associated	Thrombosis of renal artery from umbilical arterial catheter positioning at aortic bifurcation; also from polycythemia,	US: ~3–5% of UAC placements develop aortic/renal thrombosis in US NICUs; majority asymptomatic	Renal Doppler ultrasound; CT/MRI arteriography; blood pressure monitoring; DMSA scan (differential function);	Standard of Care Remove umbilical artery catheter — immediate; UAC-associated RAT accounts for majority of neonatal cases;

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		hypercoagulable states; causes renovascular hypertension	World: ~3–5% of UAC placements globally; renovascular hypertension may emerge as only manifestation Sources: Boo et al. Ann Trop Paediatr 1999; Greenway et al. Acta Paediatr 2004	thrombophilia screen; UAC position review	catheter tip repositioned or removed; most important intervention - Anticoagulation — LMWH (enoxaparin); therapeutic dosing; duration per extent and clinical response; ASH/CHEST guidelines - Thrombolytics (tPA, urokinase) — for critical ischemia, major bilateral RAT, or IVC/aortic involvement; high bleeding risk; specialist decision - BP management — renovascular hypertension from renin release by ischemic kidney; aggressive antihypertensive needed; captopril with close renal function monitoring; nicardipine IV for hypertensive emergency - Nephrectomy — for persistently non-functioning kidney causing refractory hypertension; consider after 4–6 weeks if no recovery - Supportive AKI management — fluid balance; electrolytes; dialysis if needed - Serial Doppler ultrasound — monitor renal artery patency and renal perfusion recovery
153	Hepatorenal Syndrome — Pediatric	Functional AKI in children with severe hepatic failure; intense renal vasoconstriction without structural kidney injury; reversible if liver recovers or transplant performed	US: Uncommon in US; complicates severe liver failure from biliary atresia, acute liver failure, cirrhosis World: Uncommon globally; pathophysiology involves arterial vasodilation and renal vasoconstriction; liver transplant definitive treatment Sources: Ginès et al. NEJM 2009; Nayar & Bhardwaj Ped Gastroenterol 2019	Clinical diagnosis: AKI in setting of liver failure + ascites; exclude hypovolemia (fluid challenge); urine sodium (<10 mEq/L); creatinine; liver function tests; renal Doppler (normal)	Standard of Care Emerging/Investigational: Liver transplant (definitive — resolves HRS); midodrine + octreotide as bridge where terlipressin unavailable
NEONATAL NEPHROLOGY					
154	Neonatal AKI (NAKI) — Prematurity-associated	AKI in premature neonates from immature nephron development, reduced GFR, and frequent exposure to nephrotoxins, hemodynamic compromise, and infection	US: ~30% of VLBW infants in US; AKI in ~18% of all NICU admissions; independent predictor of CKD in adulthood World: ~30% of VLBW infants globally; consistent rates across high-income country NICUs; adult CKD risk from neonatal nephron endowment deficit Sources: Jetton et al. NEJM 2017 (AWAKEN study); Askenazi et al. CJASN 2011; Selewski et al. Ped Nephrol 2019	Modified neonatal KDIGO staging (serum creatinine + urine output); serial creatinine trend (accounts for maternal creatinine at birth); urinalysis; renal Doppler ultrasound; novel biomarkers (NGAL, KIM-1)	Standard of Care - Diagnose using neonatal-specific criteria — serum creatinine ≥ 0.3 mg/dL rise in 48 hours, OR $\geq 1.5\times$ reference SCr rise, OR urine output <0.5 mL/kg/hr for >6 hours; cord blood creatinine as baseline for first 24–48 hr - Treat underlying cause — sepsis (antibiotics, vasopressors), asphyxia (therapeutic hypothermia), hemodynamic instability (fluids, inotropes), nephrotoxin removal - IV fluid optimization — normovolemia; avoid both dehydration and fluid overload; fluid overload independently worsens outcomes - Diuretics (furosemide) — for volume overload only; does NOT convert oliguric to non-oliguric AKI; does NOT improve outcomes; use for fluid management only

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Dopamine — low-dose "renal dose" dopamine NOT proven effective in neonates; not routinely recommended - Peritoneal dialysis — treatment of choice for NAKI requiring RRT; access and circuit; flexible volume removal; safe in small neonates - Continuous renal replacement therapy (CRRT) — increasingly available for NAKU in NICUs; higher-volume clearance; specialized circuits for neonates (<3 kg challenging) - Drug dosing adjustment — pharmacokinetics altered in neonatal AKI; dose all renally cleared drugs per eGFR - Long-term follow-up — NAKI survivors have elevated CKD and HTN risk; annual BP and urinalysis surveillance recommended per Neonatal Kidney Collaborative
155	Neonatal Hypertension	Blood pressure >95th percentile for corrected gestational age in neonate; often secondary to renovascular disease (UAC), CLD, PDA ligation, renal anomalies, or medications	US: ~0.2–3% of US NICU patients; higher in premature infants; UAC most common cause in preterm World: ~0.2–3% globally in NICU; consistent rates in Western neonatal registries; echocardiographic assessment recommended Sources: Dionne et al. Pediatrics 2012; Blowey et al. Ped Nephrol 2011; Neonatal HTN Consensus 2017	Accurate blood pressure measurement (oscillometric + Doppler confirmation); renal Doppler ultrasound; renal function; catecholamines (exclude pheochromocytoma); drug history; echocardiogram	Standard of Care - Identify and treat underlying cause — renovascular (UAC-associated RAT, renal artery stenosis), CKD, renal parenchymal disease, coarctation, volume overload, pheochromocytoma, hyperaldosteronism, medication-related (steroids, pressors, caffeine) - Antihypertensive pharmacotherapy — infants have limited medication options; first-line agents: - Isradipine — calcium channel blocker; oral or NG; first-line for most neonatal HTN; 0.05–0.15 mg/kg/dose q6–8h - Amlodipine — CCB; longer duration; oral; convenient once/twice daily - Hydralazine IV — acute management; vasodilator; IV bolus - Esmolol / labetalol IV — for hypertensive urgency; titratable - Captopril / enalaprilat — ACEi; USE WITH EXTREME CAUTION in neonates; risk of acute oliguric AKI; neonatal kidneys highly RAS-dependent; avoid in first month of life or bilateral RAS - Remove umbilical artery catheter — if RAT/renovascular cause - Serial BP monitoring — oscillometric per limb; confirm with Doppler; target BP <95th percentile for gestational and postnatal age - Echocardiogram — assess LVH, PDA, coarctation; cardiomegaly as marker of severity
156	 Oligohydramnios — Renal Cause	Reduced amniotic fluid from impaired fetal urine production; caused by bilateral renal agenesis, severe bilateral obstruction (PUV), or ARPKD; leads to pulmonary hypoplasia	US: ~0.5–1% of US pregnancies have oligohydramnios; renal causes ~15–20% of all oligohydramnios World: ~0.5–1% of pregnancies globally; renal agenesis and LUTO predominant renal causes; prognosis based on residual renal function Sources: Chamberlain et al. Am J Obstet Gynecol 1984; Shipp et al. AJOG 1996	Prenatal ultrasound (AFI, renal evaluation); fetal MRI; amniocentesis (therapeutic and diagnostic); genetic evaluation; postnatal renal ultrasound and function assessment	Standard of Care Emerging/Investigational: Serial amniocentesis — experimental (may improve pulmonary development)

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
METABOLIC / SYSTEMIC DISEASE WITH RENAL INVOLVEMENT					
157	 Fabry Disease — Renal (GLA)	X-linked lysosomal storage disorder (alpha-galactosidase A deficiency) causing glycosphingolipid accumulation; proteinuria, CKD, and cardiomyopathy in males; later onset in females	US: ~1/40,000–117,000 in US; X-linked; classic males severely affected; females increasingly recognized World: ~1/40,000–117,000 globally; higher in Sardinia (1/1,400); X-linked; female carriers ~50% symptomatic Sources: Eng et al. NEJM 2001; Schiffmann JASN 2009; Fabry Registry International data	Alpha-galactosidase A enzyme activity (low in males); GLA genetic testing; lyso-Gb3 (sensitive biomarker); kidney biopsy (zebra bodies on EM, podocyte inclusions); echocardiogram; MRI brain	Standard of Care - Enzyme replacement therapy (ERT): - Agalsidase beta (Fabrazyme, Sanofi) — FDA approved 2003; 1.0 mg/kg IV every 2 weeks; approved for adults and children ≥2 years - Agalsidase alfa (Replagal, Takeda) — 0.2 mg/kg IV every 2 weeks; approved in Europe/internationally; NOT FDA-approved in US; available via compassionate use - Pegunigalsidase alfa (Elfabrio, Chiesi/Protalix) — PEGylated ERT; FDA approved May 2023 for adults; reduced immunogenicity; prolonged half-life; BALANCE Phase 3 trial vs. agalsidase beta - Migalastat (Galafold, Amicus) — oral pharmacological chaperone; FDA approved Aug 2018 for adults (≥18 yr) with amenable GLA variants; stabilizes endogenous α-Gal A; ASPIRE Phase 3b trial (2025) confirms safety/efficacy in adolescents 12–17 yr with amenable variants — pediatric regulatory approval expected - RAAS blockade (ACEi/ARB) — for proteinuria; standard regardless of ERT/chaperone status - Pain management (gabapentin, carbamazepine, pregabalin) — for neuropathic pain/acroparesthesias - Renal transplant — for ESKD; does not cure systemic disease Emerging/Investigational - Gene therapy (4D-310, ST-920, FLT190) — AAV-mediated in vivo GLA gene delivery; Phase 1/2 trials ongoing - Substrate reduction therapy (venglustat, lucerastat) — reduces GL-3 synthesis; Phase 2/3 trials; oral administration advantage - Chaperone + ERT combination — investigational for non-amenable or partially amenable variants - Next-gen ERT biosimilars — in development globally
158	 Glycogen Storage Disease Type Ia — Renal (G6PC)	G6PC mutations causing hepatic glycogen accumulation; renal involvement from glycogen deposition, FSGS, and progressive CKD in second decade	US: ~1/100,000 in US; GSD Ia most common GSD; renal involvement in >75%; hypercalciuria and proteinuria by adolescence World: ~1/100,000 globally; G6PC mutations; GSD Ia distribution worldwide; renal FSGS-like lesions; dietary treatment cornerstone Sources: Weinstein et al. Nat Genet 2000; Chen JASN 1991; GSD Ia Registry data	Genetic testing (G6PC); kidney biopsy (glycogen deposition, FSGS); urinalysis (proteinuria); urine calcium; eGFR; hepatic imaging; liver biopsy	Standard of Care - Continuous glucose supplementation — cornerstone; uncooked cornstarch (oral); continuous nasogastric feeds overnight in infants; glycemic control prevents secondary complications including renal disease - RAAS blockade (ACEi/ARB) — for proteinuria; slows nephropathy progression - Allopurinol — for hyperuricemia (secondary to glucose-6-phosphate accumulation) - Low-fat diet — reduces hyperlipidemia; reduces hepatic steatosis

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- Citrate supplementation — if renal stone risk (lactic aciduria and hypocitraturia) - Liver transplant — corrects metabolic defect; secondary prevention of renal complications; used for liver failure, HCC, or extreme metabolic instability Emerging/Investigational - SGLT2 inhibitors — emerging evidence for renal protection in GSD Ia; reduces glomerular hyperfiltration and proteinuria; off-label use in adolescents being studied - Gene therapy (AAV-G6PC) — Phase 3 STEADFAST trial (AT-307, Ultragenyx); monthly SC injection showed normalization of glucose in fasted patients; potential to eliminate cornstarch dependence; most advanced genetic therapy for GSD Ia - mRNA therapy (mRNA-3745, Moderna) — Phase 1/2; hepatic G6PC delivery - Hepatocyte transplantation — bridge to liver transplant; limited centers
159	 Methylmalonic Acidemia — Renal (MMUT)	MUT/MMAB/MMAA mutations causing methylmalonic acid accumulation; CKD from tubulointerstitial injury is major long-term complication even after hepatic-renal transplant	US: ~1/80,000–100,000 in US; MMA; renal failure in >60% by adulthood; mut0 genotype worst prognosis World: ~1/80,000–100,000 globally; mut0 (MMUT deficiency) and CblA/CblB types; ESKD major morbidity; liver/kidney transplant improves metabolic control Sources: Hörster et al. JIMD 2007; Baumgartner et al. Ped Nephrol 2014; MMA Natural History Study	Urine methylmalonic acid (markedly elevated); plasma amino acids; ammonia; renal function (progressive CKD); kidney biopsy (interstitial fibrosis); MMUT genetic testing	Standard of Care - Low protein diet — limits propionate precursor amino acids (isoleucine, valine, threonine, methionine) - L-carnitine supplementation — conjugates toxic propionylcarnitine; facilitates urinary excretion - RAAS blockade — for proteinuria/CKD progression - Vitamin B12 (hydroxocobalamin, high-dose IM) — for B12-responsive forms (cblA, cblB subtypes); limited effect in isolated MMUT deficiency - Metabolic crisis management — IV glucose, carnitine; protein restriction during illness; avoid catabolism - Dialysis — for severe metabolic crises (removes MMA acutely) - Combined liver-kidney transplant — liver transplant reduces MMA production significantly; kidney transplant for ESKD; combined approach addresses both organ failure and partially corrects metabolic defect Emerging/Investigational - mRNA therapy (mRNA-3705, Moderna) — encoding MMUT; Phase 2 trials; monthly IV infusion; reduces plasma MMA; potential to prevent long-term organ complications - Propionate scavenging approaches (NKTT320) — reduces systemic propionate load; early stage - Gene therapy (AAV-MMUT) — Phase 1/2 studies; selective hepatic expression; preclinical data promising - Renal-specific approaches — mitochondrial targeted antioxidants for tubular protection

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
160	 Mitochondrial Nephropathy (MELAS / Pearson / CoQ10 deficiency)	Mitochondrial DNA or nuclear gene mutations causing renal tubular dysfunction (Fanconi, RTA) or glomerulosclerosis as part of multisystem mitochondrial disease	US: ~1/5,000–10,000 for all mitochondrial disease in US; renal in ~20–30%; CoQ10 deficiency treatable World: ~1/5,000–10,000 globally for mitochondrial disease; renal involvement variable by syndrome; MELAS, Pearson, MERRF each affect kidneys differently Sources: Menezes et al. Ped Nephrol 2012; MELAS Consortium; Rötig et al. Nat Genet 1990	Urinalysis (tubular dysfunction markers); plasma lactate; muscle biopsy (mitochondrial morphology); mitochondrial DNA analysis; nuclear gene panel; respiratory chain enzyme activity; renal biopsy	Standard of Care - CoQ10 supplementation — dramatically effective for primary CoQ10 biosynthesis defects (ADCK4/COQ6/COQ2/PDSS2 mutations); less effective for respiratory chain complex disorders - L-carnitine — for secondary carnitine deficiency; mitochondrial energy support - Thiamine (vitamin B1) — for PDHC deficiency subtype; complex I-related - Riboflavin (vitamin B2) — for complex I/II deficiency; riboflavin-responsive MADD - Avoid mitochondrial toxins — metformin (complex I inhibitor), valproate, aminoglycosides, NSAIDs - Renal transplant — for ESKD; metabolic disease continues in transplanted kidney if not CoQ10-responsive; select carefully - Supportive management — electrolytes, Fanconi syndrome management (phosphate, bicarbonate, potassium) Emerging/Investigational - KL1333 — NAD⁺ precursor; Phase 2 trials in mitochondrial myopathy - Bezafibrate — PPAR-γ agonist; promotes mitochondrial biogenesis; clinical trials ongoing in MELAS - Elamipretide (SS-31) — mitochondrial membrane peptide; Phase 2/3 trials in Barth syndrome and cardiac mitochondrial disease - Gene therapy for specific respiratory chain defects — highly early stage; limited by heteroplasmy - mRNA-based mitochondrial gene delivery — investigational
161	 Wilson Disease — Renal Tubular (ATP7B)	ATP7B mutations causing copper accumulation in liver, brain, and kidneys; renal manifestations include Fanconi syndrome, hematuria, and nephrolithiasis	US: ~1/30,000 in US; AR; renal tubular disease common with liver disease; Kayser-Fleischer rings pathognomonic World: ~1/30,000 globally; ATP7B mutations; most common in Sardinia (1/7,000) and some Asian populations; copper chelation treatment Sources: Bull et al. Nat Genet 1993; European Wilson Disease Study Group 2012; EAU/EASL Guidelines	Serum ceruloplasmin (low); 24-hour urine copper (elevated); slit-lamp (Kayser-Fleischer rings); liver biopsy (copper quantification); ATP7B genetic testing; urine amino acids if Fanconi suspected	Standard of Care - D-Penicillamine — copper chelating agent; traditional first-line treatment; removes copper; urinary copper monitoring confirms response; maintenance 10–20 mg/kg/day; significant adverse effects (proteinuria, lupus-like, aplastic anemia — rare); most frequently initiated treatment historically - Trientine (triethylenetetramine) — copper chelator; first-line alternative to D-penicillamine; better tolerated; fewer autoimmune adverse effects; FDA approved; increasingly preferred over D-penicillamine - Trientine tetrahydrochloride (Cuvrior) — FDA approved 2022 for Wilson disease in adults; tablet formulation; more stable than trientine HCl; expanding to pediatric use - Zinc acetate (Galzin) — blocks intestinal copper absorption; approved for maintenance phase and

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					presymptomatic patients; first-line for presymptomatic or maintenance after initial chelation; generally safest - Combination therapy — chelator + zinc for severe initial disease; then zinc alone for maintenance - Liver transplant — for fulminant hepatic failure, decompensated cirrhosis unresponsive to chelation; normalizes copper metabolism; renal Fanconi syndrome resolves post-transplant - Renal Fanconi syndrome management — bicarbonate, phosphate, potassium supplementation during active copper-driven tubular injury; resolves with chelation therapy - Dietary copper restriction — avoid shellfish, nuts, chocolate, liver; cooking in copper pots Emerging - Bis-choline tetrathiomolybdate (WTX101/ALXN1840) — hepatic copper chelator; Phase 3 trials for neurological Wilson; reduces copper rapidly; promising for neurological disease - Gene therapy (AAV8-ATP7B) — hepatocyte-targeted; corrects copper metabolism; Phase 1/2 trials (GATEWAY trial, Passage Bio); early results
162	 Sickle Cell Nephropathy	Renal complications of sickle cell disease including hyposthenuria, hematuria, papillary necrosis, FSGS, and progressive CKD from repeated sickling-ischemia cycles	US: ~1/500 African Americans; ~8% of US Black children carry HbS gene; renal involvement universal in SCD World: ~1/1,000–1,500 in some African populations; highest burden in Sub-Saharan Africa; microalbuminuria from early childhood Sources: Powars et al. NEJM 1991; Ataga et al. JASN 2019; SCD Nephropathy Consensus 2015	Urinalysis (microhematuria, proteinuria); urine osmolality (low — hyposthenuria); eGFR; kidney biopsy (FSGS in proteinuric cases); urine protein:creatinine; renal ultrasound	Standard of Care - Hydroxyurea — increases HbF; reduces sickling; reduces proteinuria and hyperfiltration; slows nephropathy progression; recommended for all moderate-severe SCD including SS and Sβ0 - RAAS blockade (ACEi/ARB) — for proteinuria and microalbuminuria; initiated when UACR ≥30 mg/g - L-glutamine (Endari) — FDA approved Jul 2017 for SCD ≥5 yr; reduces oxidative stress; reduces acute pain crises - Renal transplant — for ESKD; acceptable outcomes; sickle cell in allograft can occur in non-SS donor kidneys (screen donors) - Crizanlizumab (Adakveo, Novartis) — anti-P-selectin; FDA approved Nov 2019; reduces vaso-occlusive crises; renal microvascular benefit theoretical <i>FDA-Approved Gene Therapies (December 2023):</i> - Exagamglogene autotemcel / exa-cel (Casgevy, Vertex/CRISPR Therapeutics) — CRISPR-Cas9 gene editing; FDA approved Dec 8, 2023 for SCD (SS and Sβ0) ≥12 yr with recurrent VOCs; reactivates HbF via BCL11A erythroid enhancer disruption; ~90%+ free of severe VOC in trials; potentially curative; one-time treatment - Lovotibeglogene autotemcel / lovo-cel (Lyfgenia, bluebird bio) — lentiviral gene therapy; FDA approved Dec 8, 2023 for SCD ≥12 yr with VOC history; encodes anti-sickling HbAT87Q; 88% complete resolution of VOC events in trials; potentially curative

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					Note: Gene therapies require myeloablative conditioning (busulfan); significant infrastructure needed; risks include infertility and secondary malignancy monitoring; accessible at select specialized centers Emerging/Investigational ▶ Voxelotor (Oxbryta) — HbS polymerization inhibitor; FDA approved Nov 2019 (VOLUNTARY WITHDRAWAL by Pfizer Sept 2024 due to mortality signal in Phase 3 trial — no longer available); removed from market ▶ Hematopoietic stem cell transplant (allogeneic) — curative for matched sibling donors; increasing use of haploidentical protocols ▶ Mitapivat — pyruvate kinase activator; reduces sickling; Phase 3 trials ▶ SGLT2 inhibitors — renal hemodynamic effects; early trials in SCD nephropathy ▶ Omidubicel — enhanced cord blood transplant; Phase 3 for SCD
163	 Diabetic Nephropathy — Type 1, Pediatric Onset	Progressive CKD from glomerular hyperfiltration, mesangial expansion, and glomerulosclerosis in children/adolescents with long-standing T1DM; microalbuminuria is early marker	US: ~1/300–500 US youth have T1DM; microalbuminuria in ~8–25% after 10 years; macroproteinuria in minority during childhood World: ~1/300–500 in high-income countries; lower in developing countries; microalbuminuria risk factors include HbA1c and duration Sources: DCCT/EDIC Research Group NEJM 1993; T1D Exchange Registry; IDF Diabetes Atlas 2021	Annual urine albumin:creatinine ratio (from 5 years diabetes duration or age 11); eGFR; blood pressure; HbA1c; lipids; ophthalmic screening (retinopathy correlates with nephropathy)	Standard of Care • Optimal glycemic control — HbA1c <7% (ISPAD target); continuous glucose monitoring; insulin pump; most powerful intervention for nephropathy prevention and progression • RAAS blockade — ACEi/ARB first-line when UACR ≥30 mg/g (microalbuminuria) or elevated blood pressure; slows progression • Blood pressure control — target <90th percentile for age; RAAS blockade preferred • Lipid management — statins for adolescents with dyslipidemia • Lifestyle modification — low-sodium diet, weight management, smoking avoidance Emerging/Investigational ▶ SGLT2 inhibitors — growing evidence in adolescents; empagliflozin (Jardiance) FDA approved for adult CKD Sept 2023; dapagliflozin approved for adults; pediatric trials (BUTTERFLY, DERIVE-KIDS) underway; off-label use in motivated adolescents with T1D microalbuminuria at some centers ▶ Finerenone (Kerendia, Bayer) — non-steroidal MRA; Phase 3 FIDELITY-DM1 trial for T1D nephropathy; cardiorenal benefits in T2D (FIDELIO-DKD, FIGARO-DKD); T1D-specific data pending ▶ GLP-1 receptor agonists — semaglutide injectable FDA approved Jan 2025 for CKD in adults with T2D (not T1D); T1D renal data limited but mechanisms of renal protection relevant

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					▶ Atrasentan (endothelin A antagonist) — Phase 3 trials in diabetic kidney disease ▶ Sparsentan — DUPLEX trial includes T2D; T1D interest growing
164	 Obesity-related Glomerulopathy — Pediatric	Glomerular hyperfiltration and glomerulomegaly from obesity-driven hemodynamic changes; presents with proteinuria and mild microhematuria; can progress to FSGS	US: Increasing incidence in US with childhood obesity epidemic; ~2% of obese children have proteinuria World: Increasing globally with obesity; ~2% of obese children have proteinuria; distinct FSGS-like glomerulomegaly on biopsy Sources: D'Agati et al. JASN 2016; Praga et al. JASN 2001; Obesity-related glomerulopathy review KI 2008	Urinalysis (proteinuria, minimal hematuria); urine protein:creatinine; eGFR (often supranormal initially — hyperfiltration); BMI; kidney biopsy (glomerulomegaly, secondary FSGS)	Standard of Care - Weight loss — primary and most powerful intervention; sustained 5–10% weight loss reduces proteinuria significantly; lifestyle, dietary, and behavioral interventions - RAAS blockade (ACEi/ARB) — for proteinuria reduction; mainstay regardless of weight loss progress Emerging/Investigational ▶ SGLT2 inhibitors — empagliflozin (Jardiance) FDA approved for adult CKD Sept 2023; reduces glomerular hyperfiltration; emerging evidence for renal protection in obese/diabetic adolescents; trials ongoing ▶ GLP-1 receptor agonists: - Semaglutide SC (Ozempic/Wegovy, Novo Nordisk) — FDA approved Jan 2025 for CKD in adults with T2D; profound weight loss + renal protection (FLOW trial: 24% reduction in kidney disease composite endpoint); increasing off-label interest in obese adolescents with glomerulopathy - Liraglutide (Saxenda) — FDA approved for obesity in adolescents ≥12 yr; indirect renal benefit via weight loss - Tirzepatide (Mounjaro/Zepbound) — GLP-1/GIP dual agonist; SURMOUNT trials; emerging renal protection data; adolescent obesity trials underway ▶ Bariatric surgery — for severe obesity (BMI ≥35 with comorbidities) in adolescents; most effective weight reduction; sustained remission of glomerulopathy documented
165	 Amyloidosis — Renal (AA in autoinflammatory / JIA / familial)	AA amyloid deposition from chronic inflammation (JIA, FMF, autoinflammatory conditions) or AL amyloid (rare pediatric plasma cell dyscrasia); nephrotic syndrome and CKD	US: Rare in US children; AA amyloidosis in ~5–10% of long-standing JIA; FMF-associated in untreated Mediterranean patients World: Rare globally in children; AA amyloidosis predominant form; FMF-associated highest in Middle East without colchicine treatment Sources: Lachmann et al. NEJM 2007; Berntson & Fasth Acta Paediatr 2001; FMF International Registry	Kidney biopsy (Congo red — apple green birefringence, SAA immunostaining for AA); serum amyloid A (elevated in AA); SAA gene (FMF: MEFV); urinalysis (nephrotic range proteinuria)	Standard of Care Emerging/Investigational: IL-1 blockade (anakinra, canakinumab) for autoinflammatory amyloidosis; eprodisate (anti-fibrillogenesis) — not approved
166	 Oxalosis — Systemic (primary hyperoxaluria end-stage)	Systemic oxalate deposition in multiple organs (bone, heart, retina, peripheral nerves) occurring when GFR falls <30 and renal clearance of oxalate is	US: Develops in PH1 patients with advanced CKD/ESKD in US; systemic oxalate deposition in bones, heart, eyes World: Globally; systemic oxalosis develops in PH1 ESKD; combined	Plasma oxalate (markedly elevated); bone biopsy; echocardiography (cardiac oxalate); retinal exam; plasma oxalate >50 μmol/L indicates systemic	Standard of Care

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		impaired; complication of PH1/2	liver-kidney transplant indicated to correct enzymatic defect Sources: Cochat & Rumsby NEJM 2013; Milliner et al. NEJM 1994; OHO Registry data	deposition; genetic confirmation of PH	
TRANSPLANT NEPHROLOGY					
167	Pediatric Kidney Transplant — Acute Cellular Rejection	T cell-mediated immune response against allograft; presents with rising creatinine, oliguria, graft tenderness in days to weeks post-transplant; Banff grade I–III	US: ~10–15% of US pediatric kidney transplants; highly sensitized recipients higher risk; treatable if caught early World: ~10–15% of pediatric transplants globally; T-cell mediated; steroid pulses effective; European Pediatric Transplant Registry Sources: Dharnidharka et al. JASN 2015; NAPRTCS Transplant Report; ERA-EDTA pediatric transplant registry	Allograft biopsy (Banff classification: tubulitis, interstitial infiltration, intimal arteritis); serum creatinine; DSA testing; tacrolimus levels; urinalysis	Standard of Care - Pulse IV methylprednisolone — first-line for all acute cellular rejection (ACR); 3–5 days; effective for Banff IA/IB - Rabbit antithymocyte globulin (rATG / Thymoglobulin) — for steroid-resistant ACR or Banff ≥IIA (vascular rejection); 7–14 days; depletes T cells - Optimize maintenance immunosuppression — ensure therapeutic tacrolimus levels; review adherence; consider conversion if CNI toxicity - Adjust mycophenolate and steroids — dose optimization as adjunct - Biopsy surveillance — protocol biopsies at 1, 3, 12 months at many pediatric centers; subclinical rejection detected in 15–30% - Donor-derived cell-free DNA (dd-cfDNA) — emerging non-invasive biomarker for rejection surveillance; increasingly used in pediatric transplant programs Emerging/Investigational - Extended-release tacrolimus (LCPT / Envarsus XR) — once-daily formulation; improving adherence in adolescents; growing pediatric evidence (Tönshoff 2025 Pediatric Transplantation review) - Tegoprobart (AT-1501) — anti-CD40L monoclonal antibody; Phase 2 BESTOW trial for kidney transplant completed enrollment Sept 2024; topline results expected 2025; novel CNI-free maintenance strategy - Costimulation blockade optimization — belatacept increasingly used as CNI-sparing maintenance in select pediatric patients (EBV-positive recipients); FDA approval study in pediatric kidney Tx underway - Precision biopsy scoring tools — AI-augmented Banff scoring under development
168	Pediatric Kidney Transplant — Antibody-Mediated Rejection (AMR)	Donor-specific antibody (DSA)-mediated allograft injury; acute or chronic; characterized by microvascular inflammation and C4d deposition; leading cause of late graft loss	US: ~10–20% of US pediatric transplants; increasing recognition; re-transplants and sensitized patients highest risk World: ~10–20% globally of pediatric transplants; chronic AMR major cause of late graft loss; DSA monitoring essential	DSA by Luminex single antigen bead assay; allograft biopsy (Banff: microvascular inflammation, C4d, arterial intimal fibrosis); eGFR trend; urinalysis	Standard of Care - IVIg (high dose, 1–2 g/kg) — immunomodulation; neutralizes DSA; reduces complement activation; cornerstone of AMR therapy - Plasma exchange / plasmapheresis — rapid DSA removal; typically 5–10 exchanges; combined with IVIg

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
			Sources: NAPRTCS Transplant Report 2011; Dharnidharka et al. JASN 2015; Solez et al. Am J Transplant 2007		• Rituximab (anti-CD20) — B-cell depletion; adjunct to PE/IVIG for acute AMR with significant DSA burden; reduces de novo DSA production • Eculizumab (anti-C5) — for severe acute AMR with active complement-mediated microvascular injury (positive C4d, g/ptc ≥ 2); case series and cohort data support use • Bortezomib (proteasome inhibitor) — targets antibody-secreting plasma cells; used in refractory AMR; significant toxicity (neuropathy); most pediatric evidence in refractory cases • Corticosteroids — high-dose pulse as adjunct <i>Important Note: As of 2025, NO treatment has received FDA approval specifically for AMR. All therapies are off-label. Evidence base consists of center-specific protocols and cohort studies. PNRC multicenter study (2024): ~28% of pediatric AMR episodes progressed to graft failure/severe dysfunction at 12 months regardless of treatment modality.</i> Emerging/Investigational ▢ Daratumumab (anti-CD38, J&J) — depletes CD38+ plasma cells and NK cells; FDA approved for multiple myeloma; case series and a pediatric case report (2025) demonstrate DSA reduction and creatinine improvement in ABMR; Phase 2 trials in kidney Tx AMR ongoing ▢ Felzartamab (anti-CD38, MorphoSys/HI-Bio) — Phase 2 trial in late AMR showed microvascular inflammation resolution, rejection transcript downregulation, and dd-cfDNA decrease; Phase 3 planning underway ▢ Imlifidase (Idefixir, Hansa Biopharma) — IgG-cleaving enzyme; EU approved 2020 for desensitization; Phase 2 RCT vs. plasmapheresis published 2024 (Halleck et al.); rapid IgG depletion; DSA rebound risk limits durability; used in highly sensitized desensitization protocols and compassionate-use AMR ▢ Tocilizumab (anti-IL-6R) — most evidence for chronic active AMR (cAMR); Phase 2/3 experience; may stabilize graft function in cAMR; less effective for acute AMR ▢ Riliprubart (BIVV020, Sanofi) — anti-C1s; Phase 2 trials in AMR; targets classical complement pathway ▢ Clazakizumab (anti-IL-6) — Phase 3 IMAGINE study for cAMR failed primary endpoint; Phase 2 data showed graft function stabilization ▢ Tegoprubart — CD40L blockade; enrollment complete; data 2025 ▢ dd-cfDNA (Allosure/AlloSure) — validated non-invasive monitoring tool; used for AMR treatment response assessment
169	Chronic Allograft Nephropathy	Progressive allograft fibrosis and dysfunction from multiple immune (chronic rejection, subclinical AMR)	US: ~50% of allografts show chronic histological changes by 10 years in US; late graft failure ~3–5%/year	Allograft biopsy (Banff: interstitial fibrosis, tubular atrophy (IFTA), arterial intimal thickening); eGFR	 Standard of Care

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		and non-immune (CNI toxicity, recurrent disease) causes; leading cause of late graft failure	World: ~50% globally show chronic allograft changes by 10 years; consistent across transplant registries; Banff criteria used Sources: Nankivell et al. NEJM 2003; Solez et al. Am J Transplant 2007; ERA-EDTA transplant outcomes	trend; DSA; tacrolimus levels; proteinuria; urinalysis	- Optimize tacrolimus exposure — minimize trough variability (high variability = worse outcomes); consider extended-release tacrolimus for adherence - Treat subclinical rejection — protocol biopsies identify subclinical AMR/TCR; early treatment slows progression - Minimize nephrotoxins — tacrolimus dose reduction where possible; avoid nephrotoxic medications (NSAIDs, contrast without hydration, aminoglycosides) - RAAS blockade — for proteinuria and blood pressure; slows CKD progression in allograft - Manage modifiable CKD risks — dyslipidemia, anemia, CKD-MBD, blood pressure, obesity - Conversion from CNI to mTOR inhibitor (everolimus/sirolimus) — reduces CNI nephrotoxicity component; beneficial in some patients; risk of rejection with conversion - Convert to belatacept — CNI-free maintenance; requires EBV-positive recipient (risk of PTLD if EBV-negative); growing evidence for long-term renal function preservation Emerging/Investigational - Rituximab / obinutuzumab — for antibody-mediated chronic allograft nephropathy (transplant glomerulopathy driven by DSA) - Belatacept — industry-sponsored Phase 3 pediatric trial (BENEFIT-PEDS) ongoing as of 2025; CNI-free protocol - Felzartamab (anti-CD38) — Phase 2 data for late AMR; potential application in cAMR-driven CAN - Tocilizumab — cAMR stabilization; ongoing clinical use at some centers - SGLT2 inhibitors — emerging evidence for renoprotection in allografts; reducing hyperfiltration in functioning grafts; off-label use growing - Non-invasive biomarkers (dd-cfDNA, urinary CXCL9/CXCL10) — for surveillance of subclinical rejection in stable allografts
170	 Recurrent FSGS Post-transplant	Return of circulating permeability factor causing immediate post-transplant proteinuria in recipients with primary FSGS; risk ~30–50% for first transplant, >80% for second	US: ~30–50% of US pediatric primary FSGS recipients develop recurrence; rapid ESKD and young age risk factors World: ~30–50% globally; circulating permeability factor pathogenic; plasmapheresis partially effective; European data consistent Sources: Senggutuvan et al. Ped Nephrol 1990; Dall'Amico et al. JASN 1999; FSGS-Recurrence Registry	Heavy proteinuria (often within hours-days post-transplant); allograft biopsy (FSGS on LM, foot process effacement on EM); urine protein:creatinine ratio; eGFR	 Standard of Care - Prophylactic plasma exchange (plasmapheresis) — most effective intervention; 8–10 sessions peri-transplant, continuing post-operatively; reduces recurrence risk from ~50% to ~25%; especially important for prior allograft loss to recurrent FSGS - Rituximab (peri-transplant) — depletes B cells; reduces circulating permeability factor; used with or without PE; typical protocol: 375 mg/m² pre-transplant ± 1–2 doses post-transplant - High-dose cyclosporine — used at some centers for receptor-blockade / lipid raft stabilization; less common since rituximab data

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- RAAS blockade — antiproteinuric maintenance - LDL apheresis — for refractory proteinuria; reduces circulating lipid-bound permeability factor Near-Approval (as of March 2026) - Sparsentan (Filspari, Travers) — dual endothelin A/AT1R antagonist; PDUFA date April 13, 2026 for FSGS (FDA extended from January 13, 2026 requesting additional clinical benefit data); Phase 3 DUPLEX trial; DUET Phase 2 included podocyte gene mutation patients; if approved, first FDA-approved FSGS-specific therapy; already approved for IgAN Emerging/Investigational - Inaxaplin (VX-147, Vertex) — oral APOL1 channel inhibitor; FDA Breakthrough Therapy + Rare Pediatric Disease Designation for APOL1-mediated FSGS; Phase 3 AMPLITUDE trial (NCT05312879) includes adolescents 10–17; relevant for APOL1 high-risk genotype (G1/G2) recurrent FSGS - Obinutuzumab — for rituximab-refractory recurrent FSGS; improved B-cell depletion; case reports emerging - Abatacept / CTLA4-Ig — Phase 2 data (B7-1/CD80 podocyte expression mechanism); NCT02078895; variable results - Anti-CD20 alternatives (ofatumumab, ublituximab) — rituximab-resistant cases - Povetacicept (BAFF/APRIL dual inhibitor) — early investigation
171	 Recurrent C3 Glomerulopathy / aHUS Post-transplant	Recurrence of complement-mediated disease in allograft; C3GN and DDD recur in ~50–80%; aHUS recur without anti-complement prophylaxis	US: C3GN: ~50% recurrence; DDD: ~80–90% recurrence; aHUS: near 100% without eculizumab prophylaxis World: Consistent rates globally; C3G and aHUS recurrence data from European and North American registries; eculizumab dramatically reduces aHUS recurrence Sources: Medjeral-Thomas et al. JASN 2014; Zuber et al. NEJM 2008; aHUS Post-transplant Registry	Allograft biopsy; complement studies; DSA; anti-CFH antibodies; genetic testing; urinalysis; eGFR	Standard of Care - Eculizumab (anti-C5) — CRITICAL for aHUS prevention; prophylactic eculizumab started peri-transplant and continued post-transplant; prevents complement-mediated allograft loss; lifelong in most patients with high-risk complement mutations (CFH, CFI, C3, CFB) - Ravulizumab — extended-dosing anti-C5; FDA approved for aHUS; monthly dosing; increasingly preferred over eculizumab in maintenance phase for adherence - Mycophenolate mofetil ± low-dose steroids — for recurrent C3G without identified complement dysregulation; evidence base limited - Rituximab — for anti-CFH antibody-mediated recurrent disease; combined with immunosuppression (cyclophosphamide, mycophenolate) - Monitor complement biomarkers — C3, C4, CH50, AH50, sC5b-9, complement genetics pre-transplant; guides prophylaxis intensity - Meningococcal + encapsulated bacteria vaccination — mandatory before complement inhibition Emerging/Investigational

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					▶ Pegcetacoplan (Empaveli, Apellis) — FDA approved July 2025 for C3G (≥ 12 yr); proximal C3 inhibitor; highly relevant for recurrent C3G post-transplant; upstream blockade may provide more complete complement suppression than eculizumab for C3G subtype ▶ Iptacopan (Factor B inhibitor) — Phase 2/3 trials ongoing in C3G including post-transplant recurrence ▶ Danicopan (Factor D inhibitor) — Phase 2 trials for C3G ▶ Targeted complement strategy — matching inhibitor to specific complement pathway defect (e.g., proximal C3 inhibitor for C3G; terminal C5 inhibitor for aHUS with CFH mutation)
172	Post-transplant Lymphoproliferative Disease (PTLD) — Renal	EBV-driven lymphoproliferative disorder in immunosuppressed transplant recipients; ranges from benign hyperplasia to aggressive lymphoma; can involve renal allograft	US: ~1–5% of US pediatric kidney transplant recipients; EBV-seronegative recipients at highest risk World: ~1–5% globally of pediatric kidney transplants; PTLD rates higher in pediatric vs. adult transplant; EBV drive paramount Sources: NAPRTCS Transplant Report; Dharnidharka et al. Transplantation 2002; European PTLD Registry	EBV viral load monitoring; CT scan (lymphadenopathy, organ involvement); PET scan; tissue biopsy (EBER staining, CD20); bone marrow biopsy if systemic	Standard of Care • Reduction of immunosuppression (RIS) — cornerstone; first intervention for all PTLD; reduce or stop MMF first; reduce tacrolimus target; may induce remission in 20–40% of early/low-grade PTLD; balance against rejection risk • EBV PCR monitoring — routine surveillance; pre-emptive rituximab in EBV-seronegative recipients with rising EBV viremia (evidence: Pollack et al. 2025; Ashoor et al. PNRC 2024); prevents PTLD development • Rituximab (anti-CD20) — for CD20+ B-cell PTLD (>90% of pediatric cases are EBV+ CD20+); monotherapy after RIS fails; 75–90% response in polymorphic PTLD; escalate to chemoimmunotherapy if incomplete response • R-CHOP (rituximab + cyclophosphamide + doxorubicin + vincristine + prednisone) — for monomorphic PTLD (DLBCL histology) or rituximab-refractory disease; standard chemoimmunotherapy regimen • R-EPOCH — alternative to R-CHOP for high-grade/CNS-involved disease • Surgical resection — for localized, resectable disease (e.g., GI PTLD) • WHO/ICC 2022 classification — guides treatment decisions: nondestructive vs. polymorphic vs. monomorphic PTLD require different approaches; histological subtyping essential <i>New Emerging Data (2024–2025):</i> • Polatuzumab vedotin (Polivy, Genentech) — anti-CD79b antibody-drug conjugate; FDA approved for DLBCL IPI ≥ 2; case report of Pola-R-CHP for EBV+ PTLD DLBCL (Paranji 2024); non-GCB subtype (majority of EBV+ PTLD) shows greater benefit from pola-R-CHP vs R-CHOP; increasing off-label use in monomorphic PTLD • Brentuximab vedotin (anti-CD30, Seagen) — ~70–85% of PTLD express CD30; Phase 1/2 trial (rituximab + brentuximab): ORR 75%, CR 60%; significant neutropenia/infection toxicity; option for refractory/relapsing PTLD

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					- EBV-specific T-cell adoptive immunotherapy — transfers EBV-specific cytotoxic T lymphocytes (CTLs); Phase 2 data for refractory EBV+ PTLD; centers with off-the-shelf EBV-CTL products increasing - CAR-T cell therapy — 76.5% success rate in 17 patients with refractory PTLD across real-world data; anti-CD19 CAR-T used in pediatric Burkitt-PTLD (case series); increasing use at specialized centers - Nanatinostat + valganciclovir — EBV lytic reactivation strategy for refractory EBV+ PTLD; Phase 1b/2 final results (VT3996-201) published; novel mechanism - Pre-emptive rituximab prophylaxis for EBV-seronegative recipients — growing adoption per 2022 IPTA Nashville pediatric PTLD consensus guidelines
173	Calcineurin Inhibitor Nephrotoxicity	Tacrolimus or cyclosporine-induced renal injury from afferent arteriolar vasoconstriction (acute, reversible) and tubular toxicity/interstitial fibrosis (chronic, potentially irreversible)	US: Near-universal histological CNI changes with long-term use; clinically significant CKD in ~10–30% of US pediatric recipients World: Near-universal globally with chronic CNI exposure; ~10–30% develop significant CKD; CNI-sparing protocols emerging Sources: Nankivell et al. NEJM 2003; Solez et al. Am J Transplant 2007; CNI Toxicity Consensus Report 2018	Allograft biopsy (arteriolar hyalinosis, tubular vacuolization, striped fibrosis); drug levels (tacrolimus/cyclosporine); eGFR trend; exclude rejection	Standard of Care - Minimize CNI exposure — reduce tacrolimus doses to lowest effective level; low-exposure protocols using C0 trough $\leq 5\text{--}7$ ng/mL in maintenance phase - Extended-release tacrolimus (LCPT / Envarsus XR) — once-daily dosing; lower peak-to-trough variability; improved CNI-nephrotoxicity profile in adults; growing pediatric use; approved US 2015; pediatric PK/PD data maturing - Therapeutic drug monitoring (TDM) — regular tacrolimus and cyclosporine levels; area under curve (AUC)-based dosing more accurate than trough-only - Avoid combined nephrotoxins — NSAIDs, contrast, aminoglycosides, antifungals (fluconazole/voriconazole elevate CNI levels) - RAAS blockade — for proteinuria and hypertension related to CNI arteriopathy - Conversion to mTOR inhibitor (everolimus, sirolimus) — reduces CNI nephrotoxicity when conversion safe (post-12 months, stable allograft, no significant proteinuria); BENEFIT-EXT and Symphony trials data - Conversion to belatacept — CNI-free regimen; FDA approved 2011; requires EBV-seropositive recipient; BENEFIT trials: superior eGFR at 7 years vs. CNI; increasing use at pediatric centers for CNI-intolerant patients Emerging/Investigational - Tegoprubart (anti-CD40L) — Phase 2 BESTOW trial completed enrollment 2024; CNI-free induction/maintenance; topline results 2025 - Voclosporin — calcineurin inhibitor approved for LN; potentially more nephrotoxicity-favorable profile in transplant context; investigational

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
					▶ CYP3A5 pharmacogenomics testing — guides initial tacrolimus dosing to reduce early toxicity; increasing clinical adoption ▶ Kidney organoid models — preclinical tools for CNI nephrotoxicity mechanistic studies
174	BK Virus Nephropathy — Post-transplant	BK polyomavirus reactivation causing tubulointerstitial nephritis in kidney transplant recipients; major cause of graft loss if untreated	US: ~5–10% of US kidney transplant recipients; most in first 2 years; children have higher primary BKV infection rates World: ~5–10% globally of kidney transplant recipients; children have higher rates than adults due to primary BKV infection peri-transplant Sources: Hirsch et al. NEJM 2002; Ramos et al. Transplantation 2002; BKV Nephropathy Consensus Report	Serum/plasma BK virus PCR (>10,000 copies/mL high-risk); urine decoy cells (screening); allograft biopsy (SV40 immunostaining for definitive diagnosis)	 Standard of Care • Reduce immunosuppression — primary and most effective intervention; reduce MMF dose first (by 25–50%), then tacrolimus target; balance against rejection risk; serial BK PCR monitoring guides titration • Switching tacrolimus to cyclosporine — reduces immunosuppressive burden via different CNI profile; used in some centers for persistent BKVN despite IS reduction • IVIG — immunomodulation; off-label; used in refractory BKVN with persistent viremia or declining graft function; limited evidence • Leflunomide — antiviral/immunomodulatory properties; used in refractory BKVN; active metabolite teriflunomide has anti-BKV activity in vitro; limited randomized trial evidence; some centers use as adjunct • Protocol biopsy / BK PCR surveillance — routine plasma BK PCR every 3 months for 2 years post-transplant per AST guidelines; allows pre-emptive IS reduction before established nephropathy Emerging/Investigational ▶ Brincidofovir (hexadecyloxypropyl cidofovir) — oral lipid conjugate of cidofovir; greater intracellular concentration; less nephrotoxicity than parenteral cidofovir; Phase 2 data for BK viremia; off-label use in severe/refractory BKVN at select centers ▶ Cidofovir (IV) — use with extreme caution; significant nephrotoxicity; reserved for life-threatening BKVN with no alternatives; dose and frequency reduced in transplant recipients ▶ Fluoroquinolone prophylaxis (ciprofloxacin, levofloxacin) — inhibit BK viral replication in vitro; prevention studies conflicting; not recommended as routine prophylaxis; debated ▶ Artesunate — antiparasitic with antiviral BK properties; isolated case reports; very limited evidence ▶ BKV-specific T-cell immunotherapy — adoptive transfer of BKV-specific cytotoxic T lymphocytes; emerging; clinical trials (e.g., NCT03267134); most relevant for patients unable to tolerate IS reduction ▶ Vaccines targeting BKV VP1 capsid protein — preclinical development ▶ CMX001 / brincidofovir real-world data accumulating — Phase 3 experience in hematopoietic SCT informing BK use

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
CHRONIC KIDNEY DISEASE					
175	 Chronic Kidney Disease (CKD) — Pediatric	Sustained reduction in kidney function or structural kidney damage for >3 months in children; etiologies include CAKUT, glomerulonephritis, hereditary nephropathies, and sequelae of AKI	US: ~15–75/million children in US (CKiD Study); 56% CAKUT, 14% FSGS, 5% SLE as leading causes World: ~15–75/million children globally; higher rates in lower-income countries due to post-infectious GN; CAKUT predominant in high-income countries Sources: Warady & Chadha Ped Nephrol 2007; ESPN/ERA-EDTA registry; CKiD Study CJASN 2008	eGFR calculation (Schwartz formula); urinalysis; serum electrolytes; hemoglobin; PTH; phosphate; 25-OH vitamin D; echocardiogram; growth chart; renal ultrasound	Standard of Care - RAAS blockade (ACEi/ARB) — cornerstone; antiproteinuric and antihypertensive; slows CKD progression - CKD-MBD management — phosphate binders; active vitamin D (calcitriol, alfacalcidol); calcimimetics (cinacalcet) for secondary HPT - Anemia — erythropoiesis-stimulating agents (EPO, darbepoetin alfa); iron supplementation; HIF-PHI agents (daprodustat, roxadustat) emerging for dialysis patients - Growth failure — recombinant human growth hormone (rhGH); nutritional optimization; early nutritional support - Nutrition — low-phosphate diet; adequate caloric/protein intake for growth; sodium restriction - Prepare for RRT — peritoneal dialysis (preferred in children); hemodialysis; pre-emptive transplant planning - Renal transplant — preferred RRT modality; improves growth, neurodevelopment, and QoL FDA-Approved (2023–2025) - Empagliflozin (Jardiance, BI/Lilly) — SGLT2 inhibitor; FDA approved Sept 2023 for adult CKD to reduce eGFR decline and kidney failure; EMPA-KIDNEY trial; pediatric CKD trials ongoing - Semaglutide SC (Ozempic, Novo Nordisk) — GLP-1RA; FDA approved Jan 2025 for CKD in adults with T2D; FLOW trial; pediatric applicability limited but relevant for adolescents with diabetic CKD Emerging/Investigational - SGLT2 inhibitors (dapagliflozin, canagliflozin) — growing pediatric trial data; DERIVE-KIDS and BUTTERFLY trials - Finerenone — non-steroidal MRA; Phase 3 trials across CKD spectrum; favorable cardiac and renal outcomes vs. spironolactone - GLP-1 agonists (tirzepatide, liraglutide) — for CKD with obesity/metabolic syndrome component in adolescents - Sparsentan — endothelin/AT1R antagonist; reduces proteinuria across proteinuric CKD beyond IgAN - Novel EPO alternatives (HIF-PHI) — daprodustat FDA approved Feb 2023 for dialysis patients; may offer advantages in pediatric anemia management
176	 CKD-Mineral Bone Disease (CKD-MBD)	Complex disorder of calcium, phosphate, PTH, and vitamin D metabolism in CKD; causes bone disease (renal osteodystrophy),	US: Universal complication of pediatric CKD stages 3–5 in US; stunting in ~40% of pediatric CKD; high PTH from early CKD World: Universal complication globally; FGF23 elevation earliest	PTH; phosphate; calcium; 25-OH vitamin D; ALP; bone density (DXA); hand X-ray (growth plates); urinary calcium; FGF23 (emerging biomarker)	Standard of Care Phosphate binders — sevelamer carbonate (Renvela) and lanthanum carbonate (Fosrenol) preferred in older children (calcium-free; avoid calcium loading); calcium

#	Condition Name	Description	Frequency / Epidemiology	Current Diagnostic Methods	Current Treatment (SOC vs. Emerging)
		vascular calcification, and growth failure in children	biomarker; most severe manifestations in children due to active bone growth Sources: KDIGO CKD-MBD Guidelines 2017; Salusky et al. Ped Nephrol 2011; ESPN/ERA-EDTA data		carbonate/acetate in younger children if phosphate very high - Active vitamin D — calcitriol (1,25-OH vitamin D); alfacalcidol; paricalcitol; dose adjusted to PTH target; avoid oversuppression - Cinacalcet (Sensipar) — calcimimetic; FDA approved for secondary HPT in CKD on dialysis ≥18 yr; off-label pediatric use; effective for PTH reduction; risk of hypocalcemia - Growth hormone — for growth failure in CKD; FGF23 elevation blunts GH/IGF axis - Renal transplant — normalizes CKD-MBD; post-transplant bone disease requires continued monitoring Emerging/Investigational ▶ Etelcalcetide (Parsabiv, Amgen) — next-generation IV calcimimetic; FDA approved for adult dialysis HPT; superior PTH reduction vs. cinacalcet; emerging pediatric dialysis use; IV administration in hemodialysis setting ▶ Burosumab (Crysvita) — anti-FGF23; approved for XLH; FGF23 elevated in CKD-MBD; off-label investigation for CKD-related FGF23 excess and phosphate dysregulation ▶ Evocalcet — oral calcimimetic; available in Japan; under global development ▶ Gene therapy for inherited mineral metabolism disorders (XLH, ARHR) — not directly CKD-MBD therapy but prevents its occurrence ▶ SGLT2 inhibitors — secondary benefit: reduce hyperphosphatemia and FGF23 through improved GFR in early/moderate CKD